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INSTANTON FLOER HOMOLOGY, SUTURES, AND EULER CHARACTERISTICS

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Abstract. This is a companion paper to an earlier work of the authors. In this paper, we provide an axiomatic definition of Floer homology for balanced sutured manifolds and prove that the graded Euler characteristic of this homology is fully determined by the axioms we proposed. As a result, we conclude that $\chi(SHI(M,\gamma)) = \chi(SFH(M,\gamma))$ for any balanced sutured manifold (M,γ). In particular, for any link L in S^3, the Euler characteristic $\chi(KHI(S^3,L))$ recovers the multi-variable Alexander polynomial of L, which generalizes the knot case. Combined with the authors' earlier work, we provide more examples of $(1,1)$-knots in lens spaces whose KHI and $\widehat{HFK}$ have the same dimension. Moreover, for a rationally null-homologous knot in a closed oriented 3-manifold Y, we construct a canonical $\mathbb{Z}_2$ on $KHI(Y,K)$, the decomposition of $I^\sharp(Y)$ discussed in the previous paper, and the minus version of instanton knot homology $\underline{\mathrm{KHI}}^-(Y,K)$ introduced by the first author.

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1. Introduction

Sutured manifold theory was introduced by Gabai [Gab83] and Floer theory was introduced by Floer [Flo88]. They are both powerful tools in the study of 3-manifolds and knots. The first combination of these theories, called sutured Floer homology, was introduced by Juhász [Juh06] based on Heegaard Floer theory, with some pioneering work done by Ghiggini [Ghi08] and Ni [Ni07]. Later, Kronheimer and Mrowka made parallel constructions in monopole (Seiberg-Witten) theory and instanton theory [KM10b]. Different versions of Floer theories have different merits. For example, Heegaard Floer theory is more computable, while instanton theory is closely related to representation varieties of fundamental groups. Hence it is important to understand the relationship between different versions of Floer theories. In this line, Lekili [Lek13], Baldwin and Sivek [BS20b]) proved that sutured (Heegaard) Floer homology is isomorphic to sutured monopole Floer homology, though the relation with sutured instanton Floer homology is still open.

Conjecture 1.1 ([KM10b])**.** For a balanced sutured manifold (M,γ), we have

$$SHI(M,\gamma) \cong SFH(M,\gamma) \otimes \mathbb{C}.$$

In particular, for a knot K in a closed oriented 3-manifold Y, there are isomorphisms

$$I^\sharp(Y) \cong \widehat{HF}(Y) \otimes \mathbb{C}, \text{ and } KHI(Y,K) \cong \widehat{HFK}(Y,K) \otimes \mathbb{C}.$$

Here SHI is sutured instanton Floer homology [KM10b], SFH is sutured (Heegaard) Floer homology [Juh06], $I^\sharp$ is framed instanton Floer homology [KM11], $\widehat{HF}$ is the hat version of Heegaard Floer homology [OS04d], KHI is instanton knot homology [KM10b], and $\widehat{HFK}$ is (Heegaard) knot Floer homology [OS04b, Ras03].

In this paper, instead of studying the full homologies, we study their Euler characteristics under some gradings, and obtain the following theorem.

Theorem 1.2. *Suppose (M,γ) is a balanced sutured manifold and $S_1,\dots,S_n$ are properly embedded admissible surfaces (c.f. Definition 2.21) generating $H_2(M,\partial M)/\mathrm{Tors}$. Then there exist $\mathbb{Z}^n$ gradings on $SHI(M,\gamma)$ and $SFH(M,\gamma)$ induced by these surfaces, respectively. Equivalently, we have*

$$SHI(M,\gamma) = \bigoplus_{(i_1,\dots,i_n)\in\mathbb{Z}^n} SHI(M,\gamma,(S_1,\dots,S_n),(i_1,\dots,i_n))$$

and the similar result for $SFH(M,\gamma)$. Moreover, there exists relative $\mathbb{Z}_2$ gradings $\mathbf{SH}(M,\gamma)$ and $SFH(M,\gamma)$, respecting the decompositions. Define

$$\chi(SHI(M,\gamma)) := \sum_{(i_1,\dots,i_n)\in\mathbb{Z}^n} \chi(SHI(M,\gamma,(S_1,\dots,S_n),(i_1,\dots,i_n)) \cdot t_1^{i_1}\cdots t_n^{i_n}. \tag{1.1}$$

The graded Euler characteristic $\chi(SFH(M,\gamma))$ is defined similarly to (1.1). Then we have

$$\chi(SHI(M,\gamma)) \sim \chi(SFH(M,\gamma)),$$

where $\sim$ means two polynomials equal up to multiplication by $\pm t_1^{j_1}\cdots t_n^{j_n}$ for some $(j_1,\dots,j_n)\in\mathbb{Z}^n$,

Remark 1.3. Suppose that $t_1,\dots,t_n$ represent generators of

$$H = H_1(M)/\mathrm{Tors} \cong H_2(M,\partial M)/\mathrm{Tors}.$$

Then $\sim$ means the equality holds for elements in $\mathbb{Z}[H]/\pm H$.

The graded Euler characteristic $\chi(SFH(M,\gamma))$ was studied by Friedl, Juhász, and Rasmussen [FJR09]. Applying their results, we can relate the graded Euler characteristics of links with classical invariants obtained from fundamental groups.

Consider a finitely generated group $\pi = \langle x_1, \dots, x_n | r_1, \dots, r_k \rangle$. Let $H = H_1(\pi)/\mathrm{Tors}$ be the abelianization of π modulo torsions. For a generator x_i and a word w, let $\partial w/\partial x_i$ be the Fox derivative of w with respect to x_i. Equivalently, it satisfies the following conditions.

(1) For any word $w = u \cdot v$, we have $\frac{\partial w}{\partial x_i} = \frac{\partial u}{\partial x} + u \cdot \frac{\partial v}{\partial x}$.
(2) $\frac{\partial x_i}{\partial x_i} = 1$ and $\frac{\partial x_j}{\partial x_i} = 0$ for any $j \neq i$.

Consider $A = \{\partial r_j/\partial x_i\}_{i,j}$ as a matrix with entries in $\mathbb{Z}[H]$ by the projection map $\mathbb{Z}[\pi] \to \mathbb{Z}[H]$. Let $E(\pi)$ be the ideal generated by the minor determinant of A of order $n-1$. Since $\mathbb{Z}[H]$ is a unique facterization domain, one can consider the greatest common divisor of the elements of $E(\pi)$, which is well-defined up to multiplication by a nonzero element in $\pm H$. This is denoted by $\Delta(\pi)$ and called the **Alexander polynomial** of π (*c.f.* [Tur02]). For a 3-manifold M, the Alexander polynomial of M is defined by $\Delta(M) := \Delta(\pi_1(M))$. For an n-component link L in S^3, we write $t_1, \dots, t_n$ for homology classes of meridians of components of L and define $\Delta_L(t_1, \dots, t_n) := \Delta(S^3 \backslash \mathrm{int} N(L))$ as the **multi-variable Alexander polynomial** of L. If $n = 1$ and $L = K$ is a knot, we can fix the ambiguity of $\pm H$ by assuming $\Delta_K(t_1) = \Delta_K(t_1^{-1})$ and $\Delta_K(1) = 1$. In this case, we call it the **symmetrized Alexander polynomial** of K.

Theorem 1.4. *Suppose M is a compact manifold whose boundary consists of tori $T_1, \dots, T_n$ with $b_1(M) \geqslant 2$. Suppose*

$$\gamma = \bigcup_{j=1}^{n} m_j \cup (-m_j)$$

consists of two simple closed curves with opposite orientations on each torus. Suppose $H = H_1(M)/\mathrm{Tors}$ and $[m_1], \dots, [m_n]$ are homology classes. Then we have

$$\chi(SHI(M,\gamma)) = \Delta(M) \cdot \prod_{j=1}^{n} ([m_j] - 1) \in \mathbb{Z}[H]/\pm H. \tag{1.2}$$

In particular, suppose $L \subset S^3$ is an n-component link with $n \geqslant 2$. Define

$$KHI(L) := SHI(S^3 \backslash \mathrm{int} N(L), \bigcup_{j=1}^{n} m_j \cup (-m_j)), \tag{1.3}$$

where $m_1, \dots, m_n$ are meridians of components of L. Let $(i_1, \dots, i_n)$ denote the $\mathbb{Z}^n$ grading on $KHI(L)$ induced by Seifert surfaces of components of L. Then we have

$$\chi(KHI(L)) := \sum_{(i_1,\dots,i_n) \in \mathbb{Z}^n} \chi(KHI(L, (S_1, \dots, S_n), (i_1, \dots, i_n))) \cdot t_1^{i_1} \cdots t_n^{i_n} \sim \Delta_L(t_1, \dots, t_n) \cdot \prod_{j=1}^{n} (t_j - 1),$$

where $\sim$ means the equality holds for elements in $\mathbb{Z}[H]/\pm H$.

Remark 1.5. The similar result had been proved for link (Heegaard) Floer homology by Ozsváth and Szabó [OS08]. For instanton theory, the case of single-variable Alexander polynomial for links were understood by Kronheimer and Mrowka [KM10a] and independently by Lim [Lim09], while the case of the multi-variable polynomial was unknown before.

For knots, the corresponding corollary is the following.

Theorem 1.6. *Suppose K is a knot in a closed oriented 3-manifold Y. Suppose $Y(K) := Y \backslash \mathrm{int} N(K)$ is the knot complement and $b_1(Y(K)) = 1$. Let $[m] \in H = H_1(Y(K))/\mathrm{Tors} \cong \mathbb{Z}\langle t \rangle$ be the homology class of the meridian of K. Define $KHI(Y, K)$ similarly to $KHI(L)$ as in (1.3). Then we have*

$$\chi(KHI(Y, K)) = \Delta(Y(K)) \cdot \frac{[m] - 1}{t - 1} \in \mathbb{Z}[H]/\pm H.$$

Remark 1.7. Analogue results in Heegaard Floer theory can be find in [RR17, Proposition 2.1] and [Ras07, Proposition 3.1].

An application of Theorem 1.6 is to compute the instanton knot homology of some special families of knots. In [LY20], the authors proved the following.

Theorem 1.8 ([LY20, Theorem 1.4])**.** *Suppose $K \subset Y$ is a $(1, 1)$-knot in a lens space (including S^3). Then we have*

$$\dim_{\mathbb{C}} KHI(Y, K) \leqslant \dim_{\mathbb{F}_2} \widehat{HFK}(Y, K).$$

Obviously, a lower bound of $\dim_{\mathbb{C}} KHI(Y, K)$ can be obtained from $\chi(KHI(Y, K))$. If this lower bound coincides with the upper bound from Theorem 1.8, then we figure out the precise dimension of $KHI(Y, K)$. This trick applies to $(1, 1)$-knots in S^3 which are either homologically thin knots or Heegaard Floer L-space knots. In [LY20] we worked with knots in S^3 because prior to the current paper, the Euler characteristics of instanton knot homology were only understood in that case. On the other hand, in [Ye20], the second author observed some families of $(1, 1)$-knots in general lens spaces whose $\dim_{\mathbb{F}_2} \widehat{HFK}(Y, K)$ is determined by $\chi(\widehat{HFK}(Y, K))$. Hence, with Theorem 1.6 and results from [Ye20], we conclude the following.

Corollary 1.9. *Suppose Y is a lens space, and $K \subset Y$ is a $(1, 1)$-knot such that*

(1) either K admits an L-space surgery (c.f. [RR17, Lemma 3.2] and [GLV18, Theorem 2.2]), or K is a constrained knot (c.f. [Ye20, Section 4]),
(2) we have $H_1(Y(K)) \cong \mathbb{Z}$, where $Y(K)$ is the knot complement of K.

Then we have

$$\dim_{\mathbb{C}} KHI(Y, K) = \dim_{\mathbb{Z}} \widehat{HFK}(Y, K).$$

Remark 1.10. Greene, Lewallen, and Vafaee [GLV18] provided a clear criterion to check if a $(1, 1)$-knot admits an L-space surgery.

Remark 1.11. The condition $H_1(Y(K)) \cong \mathbb{Z}$ is necessary since terms related Euler characteristics of torsion spin^c structures may cancel out when we consider the map induced by the projection $H_1(Y(K)) \to H_1(Y(K))/\mathrm{Tors}$.

Next, we roughly explain the idea to prove Theorem 1.2. For a closed 3-manifold Y, the Euler characteristic $\chi(I^\sharp(Y))$ was understood by Scaduto [Sca15]. The strategy is to build Y from Dehn surgeries along a link in an integral homology sphere. The behavior of the Euler characteristic under the surgery exact triangle was fully understood as in [KM07, Section 42.3] and it is known that the Floer homology of any integral homology sphere has Euler characteristic 1. Hence we can calculate $\chi(I^\sharp(Y))$ inductively.

However, things become more complicated when we take into account surfaces inside 3-manifolds. Suppose $R \subset Y$ is a closed homologically essential surface. Then R induces a $\mathbb{Z}$ grading $I^\sharp(Y)$ by $\mu(R)$ (**c.f.** [KM10b, Section 7]). When trying to understand the graded Euler characteristic, the previous strategy does not apply directly. The reason is that, the surgery curves may have nontrivial algebraic intersections with the surface R, so the maps in surgery exact triangles may not behave

well under the grading associated to R. We are faced with the same problem when proving Theorem 1.2.

Our strategy is the following. Suppose (M, γ) is a balanced sutured manifold and suppose $S_1, \ldots, S_n$ are properly embedded surfaces in M. After attaching 1-handles along ∂M, we can find a framed link in the interior of the resulting manifold such that the link is disjoint from all the surfaces. Moreover, surgeries along the link with all slopes chosen between in $\{0, 1\}$ produce only handlebodies. Since the surgery link is disjoint from the surfaces $S_1, \ldots, S_n$ that induce the gradings, the maps in surgery exact triangles preserve the gradings. Hence it suffices to understand the case for sutured handlebodies. In this case, we can further use bypass exact triangles to reduce sutured handlebodies to product sutured manifolds. It is known that product sutured manifolds have Floer homology of dimension one. Since the behavior of Euler characteristics under bypass exact triangles and surgery exact triangles are the same for both instanton theory and Heegaard Floer theory, we finally conclude that these two versions of Floer theories must have the same graded Euler characteristic.

In the above argument, it is not necessary to treat instanton theory and Heegaard Floer theory separately. Instead, we only use some formal properties that are shared by both theories and hence we can deal with them at the same time. This observation can be made more general. In Kronheimer and Mrowka's definition of sutured (monopole or instanton) Floer homology, they constructed a closed 3-manifold called the closure, out of a balanced sutured manifold in a topological manner, and defined the Floer homology for a balanced sutured manifold to be some direct summands of the Floer homology of its closure. Then they used the formal properties of monopole theory and instanton theory to show that the construction is independent of the choice of the closures. In the following series of work [BS15, BS16a, BS16b, Li19, GL19], most arguments were also carried out based on topological constructions and hence only depend on the formal properties of Floer theories.

In this paper, we summarize the necessary properties of Floer theory that are used to build a sutured homology for balanced sutured manifolds. In Subsection 2.1, we state three axioms for $(3+1)$-TQFTs (functors from cobordism categories to categories of vector spaces) called

- (A1) the adjunction inequality axiom,
- (A2) the surgery exact triangle axiom,
- (A3) the canonical $\mathbb{Z}_2$ (mod 2) grading axiom.

A $(3+1)$-TQFT satisfying these axioms is called a **Floer-type** theory. For any Floer-type theory $\mathbf{H}$ and any balanced sutured manifold (M, γ), we construct a vector space $\mathbf{SH}(M, \gamma)$ called the **formal sutured homology** of (M, γ). More precisely, we have the following theorem.

Theorem 1.12. *Suppose $\mathbf{H}$ is a $(3+1)$-TQFT and (M, γ) is a balanced sutured manifold. If $\mathbf{H}$ satisfies Axioms (A1) and (A2), then there is a vector space $\mathbf{SH}(M, \gamma)$ for any balanced sutured manifold (M, γ), well-defined up to multiplication by a nonzero element in the base field $\mathbb{F}$. Suppose $S_1, \ldots, S_n$ are properly embedded admissible surfaces generating $H_2(M, \partial M)/\mathrm{Tors}$. Then there exist a $\mathbb{Z}^n$ grading on $\mathbf{SH}(M, \gamma)$ induced by these surfaces. Equivalently, we have*

$$\mathbf{SH}(M, \gamma) = \bigoplus_{(i_1, \ldots, i_n) \in \mathbb{Z}^n} \mathbf{SH}(M, \gamma, (S_1, \ldots, S_n), (i_1, \ldots, i_n)). \tag{1.4}$$

Furthermore, if $\mathbf{H}$ satisfies Axiom (A3), then there exists a relative $\mathbb{Z}_2$ grading $\mathbf{SH}(M, \gamma)$, respecting the decomposition in (1.4). Moreover, the graded Euler characteristic $\chi(\mathbf{SH}(M, \gamma))$, defined similarly to (1.1) and determined up to multiplication by a nonzero element in $\pm H_1(M)/\mathrm{Tors}$, is independent of the choice of the Floer-type theory.

Remark 1.13. *A priori*, the definition fo formal sutured homology depends on a large and fixed integer g. See Convection 2.2.

Remark 1.14. The construction of **SH** is essentially due to work of Kronheimer and Mrowka [KM10b]. Note that instanton theory, monopole theory and Heegaard Floer theory all satisfy Axioms (A1), (A2), and (A3) with coefficients $\mathbb{C}$, $\mathbb{F}_2$ and $\mathbb{F}_2$, respectively, up to mild modifications (*c.f.* Subsection 2.1). Moreover, Axioms (A1), (A2), and (A3) are not limited by the scope of gauge-theoretic theories mentioned above and may hold for other more general $(3+1)$-TQFTs.

There is one further step to prove Theorem 1.2 from Theorem 1.12. For Heegaard Floer theory, the construction coming from Theorem 1.12 is different from the original version of sutured (Heegaard) Floer homology defined by Juhász [Juh06]. It had been shown by Lekili [Lek13], Baldwin and Sivek [BS20b] that these two constructions coincide with each other. Although not shown explicitly, their proofs also imply that the isomorphism between these two constructions respects gradings. Based on their work, we show the following proposition.

Proposition 1.15. *Suppose (M,γ) is a balanced sutured manifold and suppose $H = H_1(M)/\mathrm{Tors}$. Suppose* **SHF** *is the sutured homology for balanced sutured manifolds constructed in Theorem 1.12 for Heegaard Floer theory. Then we have*

$$\chi(\mathbf{SHF}(M,\gamma)) = \chi(SFH(M,\gamma)) \in \mathbb{Z}[H]/\pm H.$$

Other than Theorem 1.12, there are more results that can be induced from axioms and formal properties of the formal sutured homology **SH**. Since the proofs in the authors' previous papers [BLY20, LY20] are only based on formal properties, all results can be applied to **SH** without essential changes. In particular, the following theorem is just the main theorem of [BLY20], replacing SHI by **SH**.

Theorem 1.16 ([BLY20, Theorem 1.1])**.** *Suppose $\mathcal{H}$ is an admissible Heegaard diagram for a balanced sutured manifold (M,γ). Let $\mathfrak{S}(\mathcal{H})$ be the set of generators of $SFC(\mathcal{H})$. Suppose* **H** *is a Floer-type theory with coefficient $\mathbb{F}$. Then we have*

$$\dim_{\mathbb{F}} \mathbf{SH}(M,\gamma) \leqslant |\mathfrak{S}(H)|.$$

Combining the lower bound from Theorem 1.12 and the upper bound from Theorem 1.16, we obtain the following corollary.

Corollary 1.17. *Formal sutured homology* **SH** *is independent of the choice of the $(3+1)$-TQFT satisfying Axioms (A1), (A2) and (A3) in the following cases (c.f. Definition 5.1):*

- *alternating knots in S^3 (c.f. [OS03]),*
- *$(1,1)$-knots satisfying the assumption of Corollary 1.9, including all torus knots in S^3,*
- *strong L-spaces (including double branched covers of non-split alternating links) and knots induced by strong Heegaard diagrams (c.f. [Gre13, GL16]).*

Next, we deal with the $\mathbb{Z}_2$ grading on formal sutured homology $\mathbf{SH}(M,\gamma)$. Following [KM10b], to construct $\mathbf{SH}(M,\gamma)$, we first construct a closure Y from (M,γ). From a fixed balanced sutured manifold (M,γ), we can construct infinitely many different closures (with the same genus), and the Floer homology of each closure has its own $\mathbb{Z}_2$ grading. Although we can construct isomorphisms between the Floer homology of different closures, the maps do not necessarily respect the $\mathbb{Z}_2$ grading. See [KM10a, Section 2.6] for a concrete example. Thus, we cannot obtain a canonical $\mathbb{Z}_2$ grading on $\mathbf{SH}(M,\gamma)$ and the Euler characteristic can only be defined up to a sign (since we do know the maps between closures are homogenous with respect to the $\mathbb{Z}_2$ gradings).

However, if we focus on balanced sutured manifolds whose underlying 3-manifolds are knot complements, it is possible to obtain a canonical $\mathbb{Z}_2$ grading. The idea is based on the following observation. For knot complements, the unknot in S^3 becomes a good reference. We could compare closures of a general knot with closures of the unknot and then fix the relative $\mathbb{Z}_2$ grading. As an application, for a knot K in a closed oriented 3-manifold Y, we obtain a canonical $\mathbb{Z}_2$ grading on $KHI(Y,K)$. Moreover, this canonical $\mathbb{Z}_2$ grading also carries over to the decomposition of $I^\sharp(Y)$ introduced by the authors.

Theorem 1.18 ([LY20, Theorem 1.10])**.** *Suppose Y is a closed 3-manifold, and $K \subset Y$ is a null-homologous knot. Suppose $\widehat{Y}$ is obtained from Y by performing the q/p surgery along K with $q > 0$. Then there is a decomposition up to isomorphism*

$$I^\sharp(\widehat{Y}) \cong \bigoplus_{i=0}^{q-1} I^\sharp(\widehat{Y}, i),$$

associated to the knot K and the slope q/p.

Proposition 1.19. *Under the hypothesis and the statement of Theorem 1.18, there is a well-defined $\mathbb{Z}_2$ grading on $I^\sharp(\widehat{Y}, i)$. For $i = 0, \dots, q-1$, we have*

$$\chi(I^\sharp(\widehat{Y}, i)) = \chi(I^\sharp(Y)).$$

Corollary 1.20. *Suppose $K \subset S^3$ is a knot and the 3-manifold $S^3_r(K)$ obtained from S^3 by the r-surgery along K is an instanton L-space. Suppose further that $r = q/p$ with $q > 0$. Then for $i = 0, \dots, q-1$, we have*

$$I^\sharp(S^3_r(K), i) \cong \mathbb{C}.$$

Proof. Applying Proposition 1.19 to $Y = S^3$, we have

$$\dim_{\mathbb{C}} I^\sharp(S^3_r(K)) \geqslant |\chi(I^\sharp(S^3_r(K))| = |\sum_{i=0}^{q-1} \chi(I^\sharp(S^3_r(K), i))| = |\sum_{i=0}^{q-1} \chi(I^\sharp(S^3))| = q. \tag{1.5}$$

By assumption, $S^3_r(K)$ is an instanton L-space, *i.e.*,

$$\dim_{\mathbb{C}} I^\sharp(S^3_r(K)) = |H_1(S^3_r(K))| = q.$$

Hence the inequality in (1.5) is sharp, which implies $\dim_{\mathbb{C}} I^\sharp(S^3_r(K), i) = 1$. □

The techniques to prove Proposition 1.19 can also be applied to study the minus version of instanton knot homology $\underline{\mathrm{KHI}}^-$, which was introduced by the first author in [Li19, Section 5].

Proposition 1.21. *Suppose $K \subset S^3$ is a knot and $\Delta_K(t)$ is the symmetrized Alexander polynomial of K. Then there is a $\mathbb{Z}_2$ grading on $\underline{\mathrm{KHI}}^-(S^3, K)$. Furthermore, we have*

$$\sum_{i \in \mathbb{Z}} \chi(\underline{\mathrm{KHI}}^-(S^3, K, i)) \cdot t^i = -\Delta_K(t) \cdot \sum_{i=0}^{+\infty} t^{-i}.$$

Conventions. If it is not mentioned, homology groups and cohomology groups are with $\mathbb{Z}$ coefficients. A general field is denoted by $\mathbb{F}$ and the field with two elements is denoted by $\mathbb{F}_2$.

If it is not mentioned, all manifolds are smooth and oriented. Moreover, all manifolds are connected unless we indicate disconnected manifolds are also considered. This usually happens When we discussing cobordism maps from a $(3+1)$-TQFT.

Suppose M is an oriented manifold. Let $-M$ denote the same manifold with the reverse orientation, called the **mirror manifold** of M. If it is not mentioned, we do not consider orientations of knots. Suppose K is a knot in a 3-manifold M. Then $(-M, K)$ is the **mirror knot** in the mirror manifold.

For a manifold M, let $\mathrm{int}(M)$ denote its interior. For a submanifold A in a manifold Y, let $N(A)$ denote the tubular neighborhood. The knot complement of K in Y is denoted by $Y(K) = Y \backslash \mathrm{int}(N(K))$.

For a simple closed curve on a surface, we do not distinguish between its homology class and itself. The algebraic intersection number of two curves α and β on a surface is denoted by $\alpha \cdot \beta$, while the number of intersection points between α and β is denoted by $|\alpha \cap \beta|$. A basis (m, l) of $H_1(T^2; \mathbb{Z})$ satisfies $m \cdot l = -1$. The **surgery** means the Dehn surgery and the slope q/p in the basis (m, l) corresponds to the curve $qm + pl$..

A knot $K \subset Y$ is called **null-homologous** if it represents the trivial homology class in $H_1(Y; \mathbb{Z})$, while it is called **rationally null-homologous** if it represents the trivial homology class in $H_1(Y; \mathbb{Q})$.

An argument holds for **large enough** or **sufficiently large** n if there exists a fixed $N \in \mathbb{Z}$ so that the argument holds for any integer $n > N$.

Organization. The paper is organized as follows. In Section 2, we introduce three axioms to define formal sutured homology for balanced sutured manifolds and prove the first part of Theorem 1.12. Moreover, we state many useful properties for the proof of the second part of Theorem 1.12. In 3, we discuss the modification of Heegaard Floer theory to make it suitable to formal sutured homology and prove Proposition 1.15. In Section 4, we prove the second part of Theorem 1.12. In Section 5, we construct a canonical $\mathbb{Z}_2$ grading for balanced sutured manifolds obtained from knots in closed 3-manifolds and prove Proposition 1.19 and Proposition 1.21.

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2. Axioms and formal properties for sutured homology

In this section, we construct formal sutured homology and introduce many important properties.

2.1. Axioms of a Floer-type theory for closed 3-manifolds.

In this subsection, we set up axioms to construct formal sutured homology for balanced sutured manifolds.

Let $\mathbf{Cob}^{3+1}$ be the cobordism category whose objects are closed oriented (possibly disconnected) 3-manifolds, and whose morphisms are oriented 4-dimensional cobordisms between closed oriented 3-manifolds. Let $\mathbf{Vect}_{\mathbb{F}}$ be the category of $\mathbb{F}$-vector spaces, where $\mathbb{F}$ is a suitably chosen coefficient field. A $(3+1)$ **dimensional topological quantum field theory**, or in short $(3+1)$**-TQFT**, is a monoidal functor

$$\mathbf{H} : \mathbf{Cob}^{3+1} \to \mathbf{Vect}_{\mathbb{F}}.$$

For a closed oriented 3-manifold Y, we write $\mathbf{H}(Y)$ for the related vector space, called the **Floer homology** of Y. For an oriented cobordism W, we write $\mathbf{H}(W)$ for the induced map between Floer homologies, called the **cobordism map** associated to W. Note that by definition of the categories, we have

$$\mathbf{H}(Y_1 \sqcup Y_2) = \mathbf{H}(Y_1) \otimes_{\mathbb{F}} \mathbf{H}(Y_2), \text{ and } \mathbf{H}(-Y) \cong \mathrm{Hom}_{\mathbb{F}}(\mathbf{H}(Y), \mathbb{F}).$$

Note that Floer theories are special cases of $(3+1)$-TQFTs. Summarized from known Floer theories, we propose the following definition.

Definition 2.1. A $(3+1)$-TQFT $\mathbf{H}$ is called a **Floer-type** theory if it satisfies the following three axioms (A1), (A2), and (A3).

A1. For a closed oriented 3-manifold Y and a second homology class $\alpha \in H_2(Y)$, there is a $\mathbb{Z}$-grading of $\mathbf{H}(Y)$ associated to α, *i.e.*, we have

$$\mathbf{H}(Y) = \bigoplus_{i \in \mathbb{Z}} \mathbf{H}(Y, \alpha, i).$$

This grading satisfies the following properties.

A1-1. For any odd integer i, we have $\mathbf{H}(Y, \alpha, i) = 0$.

A1-2. For $i \in \mathbb{Z} \backslash \{0\}$, the summand $\mathbf{H}(Y, \alpha, i)$ is a finite dimensional vector space over $\mathbb{F}$.

A1-3. For $i \in \mathbb{Z}$, we have $\mathbf{H}(Y, \alpha, i) \cong \mathbf{H}(Y, \alpha, -i)$.

A1-4. (Adjunction inequality). Suppose Σ is a connected closed oriented surface embedded in Y with $g(\Sigma) \geqslant 1$. For $|i| > 2g(\Sigma) - 2$, we have $\mathbf{H}(Y, [\Sigma], i) = 0$.

A1-5. If Y is a surface bundle over S^1 such that the fibre Σ is a connected closed oriented surface with $g(\Sigma) \geqslant 2$, then

$$\mathbf{H}(Y, [\Sigma], 2g(\Sigma) - 2) \cong \mathbb{F}.$$

A1-6. The gradings coming from multiple homology classes are compatible with each other, *i.e.*, if we have $\alpha_1, \ldots, \alpha_n \in H_2(Y)$, then there is a $\mathbb{Z}^n$-grading on $\mathbf{H}(Y)$, denoted by

$$\mathbf{H}(Y) = \bigoplus_{(i_1, \ldots, i_n) \in \mathbb{Z}^n} \mathbf{H}(Y, (\alpha_1, \ldots, \alpha_n), (i_1, \ldots, i_n)).$$

Moreover, we have

$$\mathbf{H}(Y, \alpha_1 + \cdots + \alpha_n, i) = \bigoplus_{\substack{(i_1, \ldots, i_n) \in \mathbb{Z}^n \\ i_1 + \cdots + i_n = i}} \mathbf{H}(Y, (\alpha_1, \ldots, \alpha_n), (i_1, \ldots, i_n)).$$

A1-7. Suppose W is an oriented cobordism from Y_1 to Y_2. Suppose $\alpha_1, \dots, \alpha_n \in H_2(Y_1)$ and $\beta_1, \dots, \beta_n \in H_2(Y_2)$ are homology classes such that for $i = 1, \dots, n$, we have

$$\alpha_i = \beta_i \in H_2(W).$$

Then the cobordism map $\mathbf{H}(W)$ respects the grading associated to those homology classes:

$$\mathbf{H}(W) : \mathbf{H}(Y_1, (\alpha_1, \dots, \alpha_n), (i_1, \dots, i_n)) \to \mathbf{H}(Y_1, (\beta_1, \dots, \beta_n), (i_1, \dots, i_n)).$$

A2. (Surgery exact triangle) Suppose M is a connected compact oriented 3-manifold with toroidal boundary. Let γ_1, γ_2, γ_3 be three connected oriented simple closed curves on ∂M such that

$$\gamma_1 \cdot \gamma_2 = \gamma_2 \cdot \gamma_3 = \gamma_3 \cdot \gamma_1 = -1.$$

Let Y_1, Y_2, and Y_3 be the Dehn fillings of M along curves γ_1, γ_2, and γ_3, respectively. Then there is an exact triangle

$$\begin{tikzcd} \mathbf{H}(Y_1) \arrow[rr] & & \mathbf{H}(Y_2) \arrow[dl] \\ & \mathbf{H}(Y_3) \arrow[ul] & \end{tikzcd} \tag{2.1}$$

Moreover, maps in the above triangle are induced by the natural cobordisms associated to different Dehn fillings.

Remark 2.2. It is worth mentioning that Axioms (A1) and (A2) are enough for defining formal sutured homology for balanced sutured manifolds. The following Axiom (A3) is only involved when considering Euler characteristics.

A3. For any closed oriented 3-manifold Y, there is a canonical $\mathbb{Z}_2$ grading on $\mathbf{H}(Y)$, denoted by

$$\mathbf{H}(Y) = \mathbf{H}_0(Y) \oplus \mathbf{H}_1(Y).$$

This grading satisfies the following properties.

A3-1. The $\mathbb{Z}_2$ grading is compatible with the grading in Axiom (A1). More precisely, if we have $\alpha_1, \dots, \alpha_n \in H_2(Y)$, then there is a $\mathbb{Z}_2 \oplus \mathbb{Z}^n$ grading on $\mathbf{H}(Y)$:

$$\mathbf{H}(Y) = \bigoplus_{j \in \{0,1\}} \bigoplus_{(i_1, \dots, i_n) \in \mathbb{Z}^n} \mathbf{H}_j(Y, (\alpha_1, \dots, \alpha_n), (i_1, \dots, i_n)).$$

A3-2. Suppose Σ_g is a connected closed oriented surface of genus $g \geqslant 2$. Suppose $Y = S^1 \times \Sigma_g$ and $\Sigma = \{1\} \times \Sigma_g$. Then we have

$$\mathbf{H}(Y, [\Sigma_g], 2g-2) = \mathbf{H}_1(Y, [\Sigma_g], 2g-2) \cong \mathbb{F}.$$

A3-3. Suppose W is a cobordism from Y_1 to Y_2. Then $\mathbf{H}(W)$ is homogeneous with respect to the canonical $\mathbb{Z}_2$ grading. Its degree can be calculated by the following degree formula

$$\deg(\mathbf{H}(W)) \equiv \frac{1}{2}(\chi(W) + \sigma(W) + b_1(Y_2) - b_1(Y_1) + b_0(Y_2) - b_0(Y_1)) \pmod 2. \tag{2.2}$$

Remark 2.3. The canonical $\mathbb{Z}_2$ grading is essentially determined by Axioms (A3-2) and (A3-3) (*c.f.* [KM07, Section 25.4]). The normalization of the $\mathbb{Z}_2$ grading for the generator of $\mathbf{H}(Y, [\Sigma_g], 2g-2)$ is not essential. Assuming $\mathbf{H}(Y, [\Sigma_g], 2g-2) = \mathbf{H}_1(Y, [\Sigma_g], 2g-2)$ shifts the canonical $\mathbb{Z}_2$ grading for all 3-manifolds. Based on [LPCS20, Lemma 3.8], we may relax the assumption in Axiom (A3-2) by $g = 2$.

The degrees of the maps in Axiom (A2) were described explicitly by Kronheimer and Mrowka [KM07, Section 42.3]. For the convenience of later usage, we present the discussion here.

Proposition 2.4 ([KM07, Section 42.3])**.** *Suppose δ is a nonzero element in* $\ker(i_*)$ *for the map*

$$i_* : H_1(\partial M; \mathbb{Q}) \to H_1(M; \mathbb{Q}).$$

In the surgery exact triangle (2.1), we can determine the parities of the maps f_1, f_2, and f_3 as follows.

(1) If there is an $i = 1, 2, 3$ so that $\gamma_i \cdot \delta = 0$, then f_{i-1} is odd and the other two are even.

(2) If $\gamma_i \cdot \delta \neq 0$ for any $i = 1, 2, 3$, then there is a unique $j \in \{1, 2, 3\}$ so that $\gamma_j \cdot \delta$ and $\gamma_{j+1} \cdot \delta$ are of the same sign. Then the map f_j is odd and the other two are even.

Here the indices are taken mod 3.

With Proposition 2.4, the following lemma is straightforward.

Lemma 2.5. *In the surgery exact triangle (2.1), after arbitrary shifts on the canonical $\mathbb{Z}_2$-grading on $I^{\omega_i}(Y_i)$ for all $i = 1, 2, 3$, exactly one of the following two cases happens.*

(1) If all three maps f_i are odd, then we have

$$\chi(\mathbf{H}(Y_1)) + \chi(\mathbf{H}(Y_2)) + \chi(\mathbf{H}(Y_3)) = 0.$$

(2) If there exists $i = 1, 2, 3$ so that f_i is odd and the other two are even, then

$$\chi(\mathbf{H}(Y_{i-1})) = \chi(\mathbf{H}(Y_i)) + \chi(\mathbf{H}(Y_{i+1})).$$

Here the indices are taken mod 3.

Remark 2.6. If there are no shifts, then case (2) in Lemma 2.5 happens due to Proposition 2.4.

In this paper we discuss three Floer theories, namely instanton theory, monopole (Seiberg-Witten) theory and Heegaard Floer theory. However, for any of these theories, we need some modifications as follows. Suppoe Y is an object of $\mathbf{Cob}^{3+1}$ and W is a morphism of $\mathbf{Cob}^{3+1}$.

Instanton theory. We should consider the decorated cobordism category $\mathbf{Cob}^{3+1}_{\omega}$ rather than $\mathbf{Cob}^{3+1}$. The objects should be admissible pairs (Y, ω), where $\omega \subset Y$ is a 1-cycle such that any component of Y contains at least one component of ω. The admissible condition means that for any component Y_0 of Y, there exists a closed oriented surface $\Sigma \subset Y_0$ such that $g(\Sigma) \geqslant 1$ and the algebraic intersection number $\omega \cdot \Sigma$ is odd. Morphisms are pairs (W, ν), where ν is a 2-cycle restricting to the given 1-cycles on ∂W. The Floer homology and the cobordism map are $I^{\omega}(Y)$ and $I(W, \nu)$ (*c.f.* [Flo90]), respectively.

The coefficient field is $\mathbb{F} = \mathbb{C}$. The decorations ω and ν do not influence Axiom (A1), where the $\mathbb{Z}$ grading is induced by the generalized eigenspaces of $(\mu(\alpha), \mu(\mathrm{pt}))$ actions for $\alpha \in H_2(Y)$ (*c.f.* [KM10b, Section 7]).

In the original statement of the surgery exact triangle in [Flo90], different 3-manifolds in the surgery exact triangle may have different choices of ω. However, by the argument in [BS20a, Section 2.2], we can assume that, in Axiom (A2), the 1-cycle ω is unchanged in all manifolds involved in the triangle.

The canonical $\mathbb{Z}_2$ grading for instanton theory was discussed by Kronheimer and Mrowka [KM10a, Section 2.6]. Indeed, the degree formula (2.2) is from their discussion.

Monopole theory. The Floer homology and the cobordism map are $\widetilde{HM}_\bullet(Y)$ and $\widetilde{HM}_\bullet(W)$ (*c.f.* [KM07]), respectively. Although we use $\widetilde{HM}_\bullet$ for monopole theory, $(3+1)$-TQFTs associated to other versions of monopole Floer homology $\widehat{HM}_\bullet$ and $\overline{HM}_\bullet$ are also implicitly used in the proof

of the Floer's excision theorem (*c.f.* [KM10b, Section 3]), which is important to show the sutured homology for balanced sutured manifolds is well-defined.

The $\mathbb{Z}$ grading in Axiom (A1) is induced by $\langle c_1(\mathfrak{s}), \alpha\rangle$ for $\mathfrak{s} \in \mathrm{Spin}^c(Y)$ and $\alpha \in H_2(Y)$ (*c.f.* [KM10b, Section 2.4]).

The coefficient field is $\mathbb{F} = \mathbb{F}_2$. This is because originally the surgery exact triangle is only proved in characteristics two (*c.f.* [KM07, Section 42]). However, it is worth mentioning that recently the surgery exact triangle in characteristic zero was study by Freeman [Fre], so it might be possible to extend the discussion to $\mathbb{F} = \mathbb{Q}$ or $\mathbb{C}$ for monopole theory. Here we do not discuss the construction using local coefficients as in [KM10b, Section 2.2], so we only use closures of fixed genus (*c.f.* Definition 2.8).

The canonical $\mathbb{Z}_2$ grading for monopole theory was discussed by Kronheimer and Mrowka [KM07, Section 25.4]. When considering cobordisms of connected 3-manifolds, the degree formula (2.2) is the same as the formula in [KM07, Definition 25.4.1].

Heegaard Floer theory. The Floer homology and the cobordism map are $HF^+(Y)$ and $HF^+(W)$ (*c.f.* [OS04d]). Similar to monopole theory, we will use other versions of Heegaard Floer homology $HF^-, HF^\infty, \mathbf{HF}^-, \mathbf{HF}^\infty$ for the proof of the Floer's excision theorem. See Section 3 for details.

The coefficient field is $\mathbb{F} = \mathbb{F}_2$. This is because we have to use the naturality results in [JTZ18], which works only for $\mathbb{F}_2$. Originally, to obtain a $(3+1)$-TQFT, we should consider the graph cobordism category $\mathbf{Cob}^{3+1}_\Gamma$ (*c.f.* [Zem19]) rather than $\mathbf{Cob}^{3+1}$. However, after modifying the naturality results in Section 3, we can show the Floer homology and and the cobordism map are independent of the choice of basepoints and graphs.

For characteristic zero, the naturality results for closed 3-manifolds were obtained by Gartner in [Gar19]. However, the naturality results for cobordisms are still under working. Hence we choose to focus on characteristics two.

Similar to monopole theory, the $\mathbb{Z}$ grading in Axiom (A1) is induced by $\langle c_1(\mathfrak{s}), \alpha\rangle$ for $\mathfrak{s} \in \mathrm{Spin}^c(Y)$ and $\alpha \in H_2(Y)$.

There are many ways to fix the $\mathbb{Z}_2$ grading for Heegaard Floer theory. See [OS04c, Section 10.4] and [FJR09, Section 2.4]. However, we can arrange the canonical $\mathbb{Z}_2$ grading to be the same as those for instanton theory and monopole theory. This is possible because the degree formula (2.2) only depends on algebraic-topological information of cobordisms and 3-manifolds.

2.2. **Formal sutured homology of balanced sutured manifold.**

In [KM10b], Kronheimer and Mrowka constructed sutured monopole Floer homology SHM and sutured instanton Floer homology SHI by closures of balanced sutured manifolds. The construction of the closure is purely topological. Hence we can adapt their construction to construct a formal version of sutured homology for balanced sutured manifolds. Here we briefly present the construction from [KM10b].

Definition 2.7 ([Juh06, KM10b])**.** A **balanced sutured manifold** (M, γ) consists of a compact oriented 3-manifold M with non-empty boundary together with a closed 1-submanifold γ on ∂M. Let $A(\gamma) = [-1, 1] \times \gamma$ be an annular neighborhood of $\gamma \subset \partial M$ and let $R(\gamma) = \partial M \backslash \mathrm{int}(A(\gamma))$. They satisfy the following properties.

(1) Neither M nor $R(\gamma)$ has a closed component.
(2) If $\partial A(\gamma) = -\partial R(\gamma)$ is oriented in the same way as γ, then we require this orientation of $\partial R(\gamma)$ induces one on $R(\gamma)$. The induced orientation on $R(\gamma)$ is called the **canonical orientation**.

(3) Let $R_+(\gamma)$ be the part of $R(\gamma)$ so that the canonical orientation coincides with the induced orientation on ∂M, and let $R_-(\gamma) = R(\gamma)\backslash R_+(\gamma)$. We require that $\chi(R_+(\gamma)) = \chi(R_-(\gamma))$. If γ is clear in the contents, we simply write $R_\pm = R_\pm(\gamma)$.

Definition 2.8. Suppose (M, γ) is a balanced sutured manifold. Let T be a connected compact oriented surface such that the numbers of components of ∂T and γ are the same. Let the **preclosure** $\widetilde{M}$ of (M, γ) be

$$\widetilde{M} := M \underset{\gamma = -\partial T}{\cup} [-1, 1] \times T.$$

The boundary of $\widetilde{M}$ consists of two components

$$\widetilde{R}_+ = R_+(\gamma) \cup \{1\} \times T, \text{ and } \widetilde{R}_- = R_-(\gamma) \cup \{-1\} \times T.$$

Let $h : \widetilde{R}_+ \xrightarrow{\cong} \widetilde{R}_-$ be a diffeomorphism which reverses the boundary orientations (*i.e.* preserves the canonical orientations). Let Y be the 3-manifold obtained from $\widetilde{M}$ by gluing $\widetilde{R}_+$ to $\widetilde{R}_-$ by h and let R be the image of $\widetilde{R}_+$ and $\widetilde{R}_-$ in Y. The pair (Y, R) is called a **closure** of (M, γ). The genus of R is called the **genus** of the closure (Y, R). For a closure (Y, R) with $g(R) \geqslant 2$, define

$$\mathbf{H}(Y|R) := \mathbf{H}(Y, [R], 2g(R) - 2).$$

Remark 2.9. For instanton theory, we should choose a point p on T and choose the diffeomorphism h such that $h(\{1\} \times p) = \{-1\} \times p$. The image of $[-1, 1] \times p$ in Y becomes a 1-cycle ω and we have $\omega \cdot R = 1$. We should use (Y, R, ω) for the definition of a closure. We do not mention this subtlety later since everything works well under this modification.

Suppose (M, γ) is a balanced sutured manifold. In [BS15], Baldwin and Sivek studied the naturality of $SHM(M, \gamma)$ and $SHI(M, \gamma)$. To relate different closures of (M, γ), they constructed canonical maps between corresponding Floer homologies of closures. Here we briefly present their construction in [BS15, Section 5].

Suppose (Y_1, R_1) and (Y_2, R_2) are two closures of (M, γ) of the same genus. We will construct a canonical map

$$\Phi_{12} : \mathbf{H}(Y_1|R_1) \to \mathbf{H}(Y_2|R_2).$$

Note that Y_2 can be thought of as obtained from Y_1 as follows. There exists an orientation preserving diffeomorphism

$$h_{12} : R_1 \to R_1$$

such that if we cut Y_1 open along R_1 and reglue using h_{12}, then we obtain a new 3-manifold Y_1' together with the surface $R_1' \subset Y_1'$. Furthermore, there exists a diffeomorphism

$$\phi : Y_1' \to Y_2$$

such that

$$\phi|_M = \mathrm{id}_M \text{ and } \phi(R_1') = R_2.$$

Let X_ϕ be a cobordism from Y_1' to Y_2 induced by ϕ. It is straightforward to check

$$\mathbf{H}(X_\phi) : \mathbf{H}(Y_1'|R_1') \to \mathbf{H}(Y_2|R_2)$$

is an isomorphism.

We can regard h_{12} as a composition of Dehn twists along curves on R_1:

$$h_{12} = D_{\alpha_1}^{e_1} \circ \cdots \circ D_{\alpha_n}^{e_n}.$$

Here $e_i \in \{\pm 1\}$, where $e_1 = 1$ means a positive Dehn twist, and $e_1 = -1$ means a negative Dehn twist. Suppose

$$N = \{i \mid e_i = -1\} \text{ and } P = \{i \mid e_i = 1\}.$$

Note that the resulting 3-manifold of cutting Y_1 open along R_1 and regluing by $D_{\alpha_i}^{e_i}$ is the same as the resulting 3-manifold of performing a $(-e_i)$-surgery along $\alpha_i \subset R_1 \subset Y_1$. We take a neighborhood $N(R_1)$ of $R_1 \subset Y_1$, and choose an identification

$$N(R_1) = [-1, 1] \times R_1.$$

Pick $-1 < t_1 < \cdots < t_n < 1$ so that $t_i \neq 0$ for $i = 1, \ldots, n$, and isotope α_i to the level $\{t_i\} \times R_1 \subset N(R_1) \subset Y_1$. Let Y_P be the 3-manifold obtained from Y_1 by performing (-1)-surgeries along α_i for all $i \in P$. There is a natural cobordism X_P from Y_1 to Y_P by attaching framed 4-dimensional 2-handles to the product $[0, 1] \times Y_1$ along $\alpha_i \times \{1\}$. Furthermore, the manifold Y_P can also be obtained from Y_1' by performing (-1)-surgeries along α_i for all $i \in N$. Hence there is a similar cobordism X_N from Y_1' to Y_P. Since $t_i \neq 0$, the surface $R_1 = \{0\} \times R_1$ survives in all surgeries. Let $R_P \subset Y_P$ be the corresponding surface.

Definition 2.10 ([BS15])**.** Define

$$\Phi_{12} = \mathbf{H}(X_\phi) \circ \mathbf{H}(X_N)^{-1} \circ \mathbf{H}(X_P) : \mathbf{H}(Y_1|R_1) \to \mathbf{H}(Y_2|R_2).$$

Proposition 2.11. *The maps*

$$\mathbf{H}(X_P) : \mathbf{H}(Y_1|R_1) \to \mathbf{H}(Y_P|R_N) \text{ and } \mathbf{H}(X_N) : \mathbf{H}(Y_1'|R_1') \to \mathbf{H}(Y_P|R_P)$$

are both isomorphisms.

Remark 2.12. Proposition 2.11 restates [BS15, Lemma 4.9]. However, the proof in that paper involves a non-vanishing result for minimal Lefschetz fibrations. See [BS15, Proposition B.1]. Yet this non-vanishing result is not covered by Axioms (A1), (A2), and (A3), so we present an alternative proof of Proposition 2.11 based on surgery exact triangles from Axiom (A2). Also, it is worth mentioning that Baldwin and Sivek worked with $\mathbb{Z}$ coefficients for monopole theory in [BS15], while we work with $\mathbb{Z}_2$ coefficients. This makes a difference as the existing proof of the surgery exact triangle in monopole theory is only proved with $\mathbb{Z}_2$ coefficients.

Proof of Proposition 2.11. The cobordisms X_P and X_N are constructed similarly, so we only prove X_P is an isomorphism. Furthermore, we can assume that P has only one element α_1. If it has more elements, then X_P is simply the composition of cobordisms associated to single Dehn surgeries. With this assumption, the manifold Y_P is obtained from Y_1 by performing a (-1)-surgery along α_1. Let Y_0 be obtained from Y_1 by performing a 0-surgery along α_1, and R_1 survives to become $R_0 \subset Y_0$. Then we have an exact triangle by Axioms (A1-7) and (A2):

$$\begin{tikzcd} \mathbf{H}(Y_1|R_1) \arrow[rr, "\mathbf{H}(X_P)"] & & \mathbf{H}(Y_P|R_P) \arrow[dl] \\ & \mathbf{H}(Y_0|R_0) \arrow[ul] & \end{tikzcd} \tag{2.3}$$

To show that $\mathbf{H}(X_P)$ is an isomorphism, it suffices to show that $\mathbf{H}(Y_0|R_0) = 0$. Indeed, since Y_0 is obtained from a 0-surgery along α_0, and α_0 can be isotoped to be a simple closed curve on R_1, the surface $R_0 \subset Y_0$ is compressible. Hence $\mathbf{H}(Y_0|R_0) = 0$ by the the adjunction inequality in Axiom (A1-4). $\square$

With Proposition 2.11 being settled down, the rest of the argument in [BS15] can be applied to our setup verbatim, and we have the following theorem.

Theorem 2.13 ([BS15])**.** *Suppose (M,γ) is a balanced sutured manifold and (Y_1,R_1) and (Y_2,R_2) are two closures of it of the same genus. Then the isomorphism*

$$\Phi_{12}:\mathbf{H}(Y_1|R_1)\to\mathbf{H}(Y_2|R_2)$$

defined in Definition 2.10 satisfies the following properties.

(1) The map Φ_{12} is well-defined up to multiplication by a non-zero element in $\mathbb{F}$.
(2) If $(Y_1,R_1)=(Y_2,R_2)$, then

$$\Phi_{12}\doteq \mathrm{id},$$

where $\doteq$ means the equation holds up to multiplication by a non-zero element in $\mathbb{F}$.
(3) If there is a third closure (Y_3,R_3) of the same genus, then we have

$$\Phi_{13}\doteq\Phi_{23}\circ\Phi_{12}.$$

Remark 2.14. In Baldwin and Sivek's original work, the requirement that the two closures have the same genus could be dropped, at the cost of involving local coefficient systems. However, up to the authors' knowledge, the discussion for the naturality of Heegaard Floer theory has not been carried out with local coefficients. Since it is enough to work with closures of a large and fixed closure in the current paper, we choose not to discuss on the local coefficients.

Definition 2.15 ([JTZ18, BS15])**.** A **projectively transitive system** of vector spaces over a field $\mathbb{F}$ consists of

(1) a set A and collection of vector spaces $\{V_\alpha\}_{\alpha\in A}$ over $\mathbb{F}$,
(2) a collection of linear maps $\{g^\alpha_\beta\}_{\alpha,\beta\in A}$ well-defined up to multiplication by a nonzero element in $\mathbb{F}$ such that
  (a) g^α_β is an isomorphism from V_α to V_β for any $\alpha,\beta\in A$, called a **canonical map**,
  (b) $g^\alpha_\alpha\doteq \mathrm{id}_{V_\alpha}$ for any $\alpha\in A$,
  (c) $g^\beta_\gamma\circ g^\alpha_\beta\doteq g^\alpha_\gamma$ for any $\alpha,\beta,\gamma\in A$.

A morphism of projectively transitive systems of vector spaces over a field $\mathbb{F}$ from $(A,\{V_\alpha\},\{g^\alpha_\beta\})$ to $(B,\{U_\gamma\},\{h^\gamma_\delta\})$ is a collection of maps $\{f^\alpha_\gamma\}_{\alpha\in A,\gamma\in B}$ such that

(1) f^α_γ is a linear map from V_α to U_γ well-defined up to multiplication by a nonzero element in $\mathbb{F}$ for any $\alpha\in A$ and $\gamma\in B$,
(2) $f^\beta_\delta\circ g^\alpha_\beta\doteq h^\gamma_\delta\circ f^\alpha_\gamma$ for any $\alpha,\beta\in A$ and $\gamma,\delta\in B$.

A **transitive system** of vector spaces over a field $\mathbb{F}$ if it is a projectively transitive system and all equations with $\doteq$ are replaced by ones with $=$. A morphism of transitive systems of vector spaces over a field $\mathbb{F}$ is defined similarly.

We can replace vector spaces by groups or chain complexs of vector spaces and define the projectively transitive system and the transitive system similarly.

Remark 2.16. A transitive system of vector spaces $(A,\{V_\alpha\},\{g^\alpha_\beta\})$ over a field $\mathbb{F}$ canonically defines an actual vector space over $\mathbb{F}$

$$V:=\coprod_{\alpha\in A}V_\alpha/\sim,$$

where $v_\alpha\sim v_\beta$ if and only if $g^\alpha_\beta(v_\alpha)=v_\beta$ for any $v_\alpha\in V_\alpha$ and $v_\beta\in V_\beta$. A morphism of transitive systems of vector spaces canonically defines an linear map between corresponding actual vector spaces.

Convention. If $\mathbb{F} = \mathbb{F}_2$, a projectively transitive system over $\mathbb{F}$ is also a transitive system since $\mathbb{F}_2$ has only one nonzero element. In this case, we do not distinguish the projectively transitive system, the transitive system and the corresponding actual vector space. For a general field $\mathbb{F}$, the morphisms between projectively transitive systems are also called maps.

Definition 2.17. Suppose $\mathbf{H}$ is a $(3+1)$-TQFT satisfying Axioms (A1) and (A2), and (M, γ) is a balanced sutured manifold, the **formal sutured homology** $\mathbf{SH}^g(M, \gamma)$ is the projectively transitive system consisting of

(1) the Floer homology $\mathbf{H}(Y|R)$ for closures (Y, R) of (M, γ) with a fixed and large enough genus g.
(2) the canonical maps Φ between Floer homologies as in Definition 2.10.

Convention. Through out the paper, when discussing formal sutured homology, we will pre-fix a large enough genus. So we omit it from the notation and write simply $\mathbf{SH}(M, \gamma)$ instead of $\mathbf{SH}^g(M, \gamma)$.

Remark 2.18. When $\mathbf{H}$ also satisfies Axiom (A3), since Φ is constructed by the composition of cobordism maps and their inverses, it is homogeneous with respect to the $\mathbb{Z}_2$ grading from Axiom (A3). Then there exists an induced relative $\mathbb{Z}_2$ grading on $\mathbf{SH}(M, \gamma)$.

In [BS16a, BS18], Baldwin and Sivek proved the bypass exact triangle for sutured monopole Floer homology and sutured instanton Floer homology. Their proof can be exported to our setup.

Theorem 2.19 ([BS16a, Theorem 5.2] and [BS18, Theorem 1.20])**.** *Suppose (M, γ_1), (M, γ_2), (M, γ_3) are three balanced sutured manifold such that the underlying 3-manifold is the same, and the sutures γ_1, γ_2, and γ_3 only differ in a disk as depicted in Figure 1. Then there exists an exact triangle*

$$\begin{tikzcd} \mathbf{SH}(-M, -\gamma_1) \arrow[rr, "\psi_1"] & & \mathbf{SH}(-M, -\gamma_2) \arrow[dl, "\psi_2"] \\ & \mathbf{SH}(-M, -\gamma_3) \arrow[ul, "\psi_3"] & \end{tikzcd}$$

Moreover, the maps ψ_i are induced by cobordisms, hence is homogeneous with respect to the relative $\mathbb{Z}_2$ grading on $\mathbf{SH}(M, \gamma_i)$.

We un-package the proof of Theorem 2.19 for later convenience.

Proposition 2.20 ([BS16a, Section 5] and [BS18, Section 4])**.** *Consider (M, γ_i) for $i = 1, 2, 3$ in Theorem 2.19, there is a closure (Y_1, R) of $(-M, -\gamma_1)$ with the following significance.*

(1) The genus $g(R)$ is large enough.
(2) There are pairwise disjoint curves $\zeta_1, \zeta_2, \zeta_3 \subset Y_1$ so that the following is true.
  (a) For $i = 1, 2, 3$, we have $\zeta_i \cap \mathrm{int}(M) = \varnothing$ and ζ_i can be isotoped to be disjoint from R.
  (b) If we perform a suitable Dehn surgery along ζ_1, then we obtain a closure (Y_2, R) of $(-M, -\gamma_2)$. If we perform a suitable Dehn surgery in Y_2 along ζ_2, then we obtain a closure (Y_3, R) of $(-M, -\gamma_3)$. If we perform a suitable Dehn surgery in Y_3 along ζ_3, then we obtain the closure (Y_1, R) of $(-M, -\gamma_1)$ again.
  (c) The maps ψ_1, ψ_2, and ψ_3 are induced the cobordism associated to Dehn surgeries along ζ_1, ζ_2, and ζ_3, respectively.
(3) There are two curves η_1 and η_2 on R, so that if we perform (-1)-surgeries on both of them, with respect to the surface framings from R, then the surgeries along ζ_1, ζ_2, and ζ_3 form an exact triangle as in Axiom (A2).

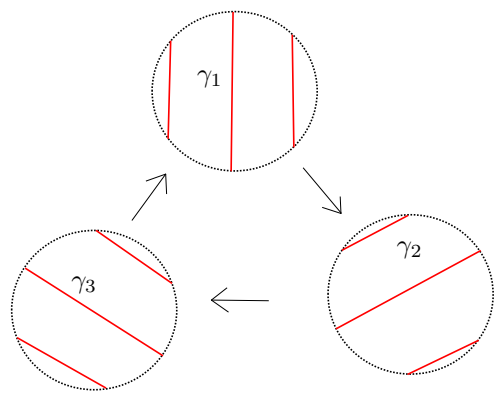

FIGURE 1. The bypass triangle.

2.3. **Gradings on formal sutured homology.**

Suppose (M,γ) is a balanced sutured manifold and $S \subset (M,\gamma)$ is a properly embedded surface in M. If S satisfies some admissible conditions, the first author constructed a grading on $SHM(M,\gamma)$ and $SHI(M,\gamma)$ in [Li19]. In this subsection, we adapt the construction to formal sutured homology $\mathbf{SH}(M,\gamma)$.

Definition 2.21 ([GL19])**.** Suppose (M,γ) is a balanced sutured manifold and $S \subset M$ is a properly embedded surface. The surface S is called an **admissible surface** if the followings hold.

(1) Every boundary component of S intersects γ transversely and non-trivially.
(2) $\frac{1}{2}|S \cap \gamma| - \chi(S)$ is an even integer.

Recall the construction of a closure of (M,γ) in Definition 2.8. Let T be a connected compact oriented surface of large enough genus and $\partial T \cong -\gamma$. Then we take

$$\widetilde{M} = M \cup [-1,1] \times T, \text{ with } \partial\widetilde{M} = \widetilde{R}_+ \sqcup (-\widetilde{R}_-).$$

Suppose $n = \frac{1}{2}|\partial S \cap \gamma|$ and

$$\partial S \cap \gamma = \{p_1, \dots, p_{2n}\}.$$

Definition 2.22 ([Li19])**.** A **pairing** $\mathcal{P}$ of size n is a collection of n couples

$$\mathcal{P} = \{(i_1, j_1), \dots, (i_n, j_n)\}$$

such that the followings hold.

(1) $\{1, \dots, 2n\} = \{i_1, j_1, \dots, i_n, j_n\}$.
(2) For any $k \in \{1, \dots, n\}$, the points p_{i_k} and p_{j_k} are positive and negative, respectively, as intersection points of ∂S and γ on ∂M.

Given a pairing $\mathcal{P}$ of size n, and assuming that $g(T)$ is large enough, we can extend S to a properly embedded surface in $\widetilde{M}$ as follows. Let $\alpha_1, \dots, \alpha_n$ be pairwise disjoint properly embedded arcs on T such that the followings hold.

(1) The arcs $\alpha_1, \dots, \alpha_n$ represent linearly independent homology classes in $H_1(T, \partial T)$.
(2) For any $k \in \{1, \dots, n\}$, we have $\partial\alpha_i = \{p_{i_k}, p_{j_k}\}$.

Givn $\alpha_1, \dots, \alpha_n$, we construct a surface properly embedded in $\widetilde{M}$

$$\widetilde{S}_{\mathcal{P}} := S \cup [-1,1] \times (\alpha_1 \cup \dots \cup \alpha_n).$$

Definition 2.23. A pairing $\mathcal{P}$ is called **balanced** if $\widetilde{S}_{\mathcal{P}} \cap \widetilde{R}_+$ and $\widetilde{S}_{\mathcal{P}} \cap \widetilde{R}_-$ have the same number of components.

For any balanced pairing $\mathcal{P}$, we can pick an orientation preserving diffeomorphism

$$h : \widetilde{R}_+ \xrightarrow{\cong} \widetilde{R}_-$$

so that

$$h(\widetilde{S}_{\mathcal{P}} \cap \widetilde{R}_+) = \widetilde{S}_{\mathcal{P}} \cap \widetilde{R}_-.$$

Thus, we obtain a closure (Y, R) of (M, γ) as well as a closed oriented surface $\bar{S}_{\mathcal{P}} \subset Y$. Then we define

$$\mathbf{SH}(M, \gamma, S, i) := \mathbf{H}(Y, ([R], [\bar{S}_{\mathcal{P}}]), (2g(R) - 2, 2i)).$$

Theorem 2.24. *Given an admissible surface S in a balanced sutured manifold (M, γ), the decomposition*

$$\mathbf{SH}(M, \gamma) = \bigoplus_{i \in \mathbb{Z}} \mathbf{SH}(M, \gamma, S, i)$$

is independent of all the choices made in the construction and hence is well-defined.

Proof. The decomposition follows from Axioms (A1-1) and (A1-7). This gives a $\mathbb{Z}$ grading on $\mathbf{SH}(M, \gamma)$. To show that this grading is well-defined, we need to show that it is independent of the following three types of choices:

(1) the choice of the balanced pairing $\mathcal{P}$,
(2) the choice of arcs $\alpha_1, \dots, \alpha_n$ with fixed endpoints,
(3) the choice of the diffeomorphism h.

In [Li19], the grading had been shown to be independent of the choices of type (2) and (3). The proof involves only Axioms (A1) and (A2) and hence can be applied to our current setup. However, the original argument for choices of type (1) in [Li19] involves closures of different genus, which is beyond the scope of our current paper as mentioned in Remark 2.14. Hence, we provide an alternative proof here. For the moment let us write the grading as

$$\mathbf{SH}(M, \gamma, S, \mathcal{P}, i)$$

to emphasize that the grading *a priori* depends on the choice of the balanced pairing. Theorem 2.24 then follows from the following proposition. □

Proposition 2.25. *Suppose $\mathcal{P}$ and $\mathcal{P}'$ are two balanced pairings, then for any $i \in \mathbb{Z}$, we have*

$$\mathbf{SH}(M, \gamma, S, \mathcal{P}, i) = \mathbf{SH}(M, \gamma, S, \mathcal{P}', i).$$

To relate two different pairings, in [Li19], the author introduced the following operation.

Definition 2.26. Suppose $\mathcal{P}$ is a pairing of size n and $\alpha_1, \dots, \alpha_n$ are related arcs. Suppose $k, l \in \{1, \dots, n\}$ are two indices so that the followings hold.

(1) The arcs $\{1\} \times \alpha_{i_k}$ and $\{1\} \times \alpha_{i_l}$ belong to different components of $\widetilde{S}_{\mathcal{P}} \cap R_+$.
(2) The arcs $\{-1\} \times \alpha_{i_k}$ and $\{-1\} \times \alpha_{i_l}$ belong to different components of $\widetilde{S}_{\mathcal{P}} \cap R_-$.

Then we can construct another pairing

$$\mathcal{P}' = (\mathcal{P}\backslash\{(i_k, j_k), (i_l, j_l)\}) \cup \{(i_k, j_l), (i_l, j_k)\}.$$

The operation of replacing $\mathcal{P}$ by $\mathcal{P}'$ is called a **cut and glue operation**.

Theorem 2.27 ([Kav19])**.** *Balanced pairings always exist. Moreover, any two balanced pairings are related by a finite sequence of cut and glue operations and their inverses.*

Lemma 2.28. *Suppose $\mathcal{P}$ and $\mathcal{P}'$ are two balanced pairings that are related by a cut and glue operation, then for any $i \in \mathbb{Z}$, we have*

$$\mathbf{SH}(M, \gamma, S, \mathcal{P}, i) = \mathbf{SH}(M, \gamma, S, \mathcal{P}', i).$$

Proof. Suppose k and l are the indices involved in the operation. From the first part of the proof of Theorem 2.24, we can freely make choices of type (2) and (3). Hence we can assume that there is a disk $D \subset \mathrm{int}(T)$ so that α_k and α_l intersects D in two arcs as depicted in Figure 2. Suppose

$$D_+ = \{1\} \times D \subset R_+, \text{ and } D_- = \{-1\} \times D \subset R_-.$$

We can choose an orientation preserving diffeomorphism

$$h : \widetilde{R}_+ \xrightarrow{\cong} \widetilde{R}_-$$

such that

$$h(\widetilde{S}_{\mathcal{P}} \cap \widetilde{R}_+) = \widetilde{S}_{\mathcal{P}} \cap \widetilde{R}_- \text{ and } h(D_+) = D_-.$$

Let (Y, R) be the corresponding closure of (M, γ) and $\bar{S}_{\mathcal{P}}$ be the closed surface defining the grading $\mathbf{SH}(M, \gamma, S, \mathcal{P})$. Let $\beta_k = \alpha_k \cap D$ and $\beta_l = \alpha_l \cap D$. It is obvious that if we remove the two arcs β_k and β_l from $D \subset T$, and glue back two new arcs β'_k and β'_l as shown in the middle subfigure of Figure 2, then we obtain two new properly embedded arcs α'_k and α'_l on T so that

$$\partial\alpha'_k = \{p_{i_k}, p_{j_l}\} \text{ and } \partial\alpha'_l = \{p_{i_l}, p_{j_k}\}.$$

Hence we change from $\mathcal{P}$ to $\mathcal{P}'$. Inside Y, if we remove $S^1 \times (\beta_k \cup \beta_l) \subset S^1 \times D \subset Y$ and glue back $S^1 \times (\beta'_k \cup \beta'_l) \subset S^1 \times D$, then we obtain the surface $\bar{S}_{\mathcal{P}'} \subset Y$ that gives rise to the grading $\mathbf{SH}(M, \gamma, S, \mathcal{P}')$. The lemma then follows from the fact

$$[\bar{S}_{\mathcal{P}}] = [\bar{S}_{\mathcal{P}'}] \in H_2(Y).$$

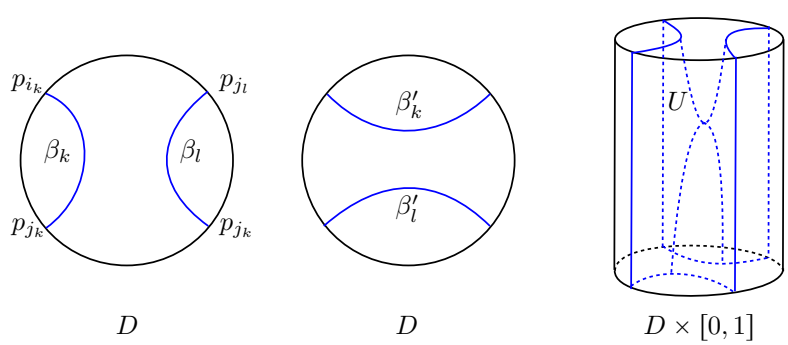

FIGURE 2. The disk D, the arcs $\beta_k, \beta_l, \beta'_k, \beta'_l$, and the surface U.

This fact (2.3) can be proved by constructing an explicit cobordism in $Y \times [0,1]$ from $\bar{S}_{\mathcal{P}} \subset Y \times \{0\}$ to $\bar{S}_{\mathcal{P}'} \subset Y \times \{1\}$: in the product $(Y \times [0,1], \bar{S}_{\mathcal{P}} \times [0,1])$, we can remove

$$S^1 \times (\beta_1 \cup \beta_2) \times [0,1] \subset S^1 \times D \times [0,1] \subset Y \times [0,1]$$

and glue back $S^1 \times U \subset D \times [0,1]$, where $U \subset D \times [0,1]$ is the surface shown in the right subfigure of Figure 2. □

Proof of Proposition 2.25. It follows immediately from Theorem 2.27 and Lemma 2.28. □

Having constructed the grading, the rest of the arguments in [Li19] can be applied to our current setup verbatim. Hence we have the following.

Theorem 2.29 ([Li19])**.** *Suppose (M,γ) is a balanced sutured manifold and $S \subset (M,\gamma)$ is an admissible surface. Then there is a $\mathbb{Z}$ grading on $\mathbf{SH}(M,\gamma)$ induced by S , which we write as*

$$\mathbf{SH}(M,\gamma) = \bigoplus_{i \in \mathbb{Z}} \mathbf{SH}(M,\gamma,S,i).$$

This decomposition satisfies the following properties.

(1) Suppose $n = \frac{1}{2}|\partial S \cap \gamma|$. If $|i| > \frac{1}{2}(n - \chi(S))$, then $\mathbf{SH}(M,\gamma,S,i) = 0$.

(2) If there is a sutured manifold decomposition $(M,\gamma) \overset{S}{\rightsquigarrow} (M',\gamma')$ in the sense of Gabai [Gab83], then we have

$$\mathbf{SH}(M,\gamma,S,\frac{1}{2}(n-\chi(S))) \cong \mathbf{SH}(M',\gamma').$$

(3) For any $i \in \mathbb{Z}$, we have

$$\mathbf{SH}(M,\gamma,S,i) = \mathbf{SH}(M,\gamma,-S,-i).$$

(4) For any $i \in \mathbb{Z}$, we have

$$\mathbf{SH}(M,\gamma,S,i) \cong \mathbf{SH}(M,\gamma,S,-i) \cong \mathbf{SH}(M,-\gamma,S,i) \cong \mathbf{SH}(M,\gamma,S,-i).$$

(5) For any $i \in \mathbb{Z}$, we have

$$\mathbf{SH}(-M,\gamma,S,i) \cong \mathrm{Hom}_{\mathbb{F}}(\mathbf{SH}(M,\gamma,S,-i),\mathbb{F}).$$

Proof. Term (1) comes from the adjunction inequality in (A1-4). Term (2) is a restatement of [KM10b, Proposition 7.11]. Term (3) is straighforward from the definition. Term (4) is from Axiom (A1-3). Term (5) is from the pairing (*c.f.* [Li18]):

$$\langle \cdot, \cdot \rangle : \mathbf{SH}(M,\gamma) \times \mathbf{SH}(-M,\gamma) \to \mathbb{F}$$

. □

Based on term (2) in Theorem 2.29, we can show formal sutured homology detects the tautness and the productness of balanced sutured manifolds.

Definition 2.30 ([Juh06])**.** A sutured manifold (M,γ) is called **taut** if M is irreducible and $R_+(\gamma)$ and $R_-(\gamma)$ are both incompressible and Thurston norm-minimizing in the homology class that they represent in $H_2(M,\gamma)$.

Theorem 2.31 ([Juh06, Juh08, KM10b])**.** *Suppose (M,γ) is a balanced sutured manifold so that M is irreducible. Then (M,γ) is taut if and only if $\mathbf{SH}(M,\gamma) \neq 0$.*

Definition 2.32. Suppose (M,γ) is a balanced sutured manifold. It is called a **homology product** if $H_1(M,R_+(\gamma))=0$ and $H_1(M,R_-(\gamma))=0$. It is called a **product sutured manifold** if

$$(M,\gamma)\cong([-1,1]\times\Sigma,\{0\}\times\partial\Sigma),$$

where Σ is a compact surface with boundary.

Theorem 2.33 ([Ni07, Juh08, KM10b])**.** *Suppose (M,γ) is a sutured manifold and a homology product. Then (M,γ) is a product sutured manifold if and only if $\mathbf{SH}(M,\gamma)\cong\mathbb{F}$.*

If $S\subset(M,\gamma)$ is not admissible, then we can perform an isotopy on S to make it admissible.

Definition 2.34. Suppose (M,γ) is a balanced sutured manifold, and S is a properly embedded surface. A **stabilization** of S is an isotopy of S to a surface S' in the following sense. This isotopy creates a new pair of intersection points

$$\partial S'\cap\gamma=(\partial S\cap\gamma)\cup\{p_+,p_-\}.$$

We require that there are arcs $\alpha\subset\partial S'$ and $\beta\subset\gamma$, oriented in the same way as $\partial S'$ and γ, respectively, so that the followings hold.

(1) We have $\partial\alpha=\partial\beta=\{p_+,p_-\}$.
(2) The curves α and β cobound a disk D so that $\mathrm{int}(D)\cap(\gamma\cup\partial S')=\varnothing$.

The stabilization is called **negative** if D can be oriented so that $\partial D=\alpha\cup\beta$ as oriented curves. It is called **positive** if $\partial D=(-\alpha)\cup\beta$. See Figure 3. We denote by $S^{\pm k}$ the result of performing k many positive or negative stabilizations of S.

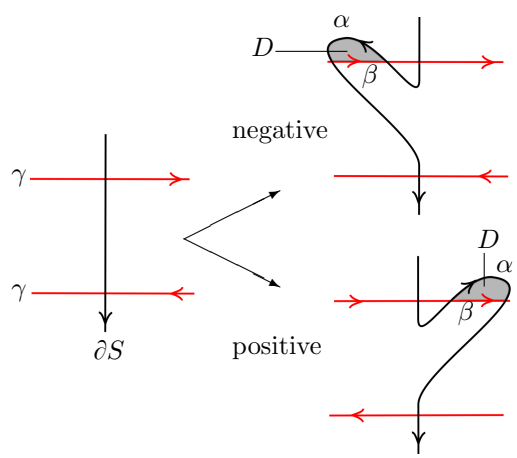

FIGURE 3. The positive and negative stabilizations of S.

The following lemma is straightforward.

Lemma 2.35. *Suppose (M,γ) is a balanced sutured manifold, and S is a properly embedded oriented surface. Suppose S^+ and S^- are the results of performing a positive and a negative stabilization on S, respectively. Then we have the following.*

(1) If we decompose (M,γ) along S or S^+, then the resulting two balanced sutured manifolds are diffeomorphic.

(2) If we decompose (M,γ) along S^-, then the resulting balanced sutured manifold (M',γ') is not taut, as $R_\pm(\gamma')$ would both become compressible.

Remark 2.36. Note that the definition of stabilizations of a surface depends on the orientations of the suture and the surface. If the orientation of the suture reverses, then positive and negative stabilizations switch between each other. Similarly, if we switch the orientation of the surface, then the positive and negative stabilizations also switch between each other.

One can also relate the gradings associated to different stabilizations of a fixed surface. The proof for SHM and SHI in [Li19] and [Wan20] can be adapted to our setup as well.

Theorem 2.37 ([Li19, Proposition 4.3] and [Wan20, Proposition 4.17])**.** *Suppose (M,γ) is a balanced sutured manifold and S is a properly embedded surface in M, which intersects the suture γ transversely. Suppose S has a distinguished boundary component so that all the stabilizations mentioned below are performed on this boundary component. Then, for any $p,k,l\in\mathbb{Z}$ so that the stabilized surfaces S^p and S^{p+2k} are both admissible, we have*

$$\mathbf{SH}(M,\gamma,S^p,l)=\mathbf{SH}(M,\gamma,S^{p+2k},l+k).$$

Note S^p is a stabilization of S as introduced in Definition 2.34, and, in particular, $S^0=S$.

If we have multiple admissible surfaces, then they together induce a multi-grading. This is proved for SHM and SHI by Ghosh and the first author in [GL19]. The proof can be adapted to our case without essential changes.

Theorem 2.38 ([GL19])**.** *Suppose (M,γ) is a balanced sutured manifold and $S_1,\dots,S_n$ are admissible surfaces in (M,γ). Then there exists a $\mathbb{Z}^n$ grading on $\mathbf{SH}(M,\gamma)$ induced by $S_1,\dots,S_n$, which we write as*

$$\mathbf{SH}(M,\gamma)=\bigoplus_{(i_1,\dots,i_n)\in\mathbb{Z}^n}\mathbf{SH}(M,\gamma,(S_1,\dots,S_n),(i_1,\dots,i_n)).$$

Theorem 2.39 ([GL19])**.** *Suppose (M,γ) is a balanced sutured manifold and $\alpha\in H_2(M,\partial M)$ is a nontrivial homology class. Suppose S_1 and S_2 are two admissible surfaces in (M,γ) such that*

$$[\partial S_1,\partial S_1]=[\partial S_2,\partial S_2]=\alpha\in H_2(M,\partial M).$$

Then, there exist a constant C so that

$$\mathbf{SH}(M,\gamma,S_1,l)=\mathbf{SH}(M,\gamma,S_2,l+C).$$

Based on the relative $\mathbb{Z}_2$ grading from Remark 2.18 and the $\mathbb{Z}^n$ grading from Theorem 2.38, we can define graded Euler characteristic of formal sutured homology.

Definition 2.40. Suppose (M,γ) is a balanced sutured manifold and $S_1,\dots,S_n$ are admissible surfaces in (M,γ). Consider the canonical isomorphisms

$$H_2(M,\partial M)\cong H^1(M)\cong H_1(M)/\mathrm{Tors}.$$

Let $\rho_1,\dots,\rho_n\in H=H_1(M)/\mathrm{Tors}$ be images of $[(S_1,\partial S_1)],\dots,[(S_n,\partial S_n)]$ respectively. The **graded Euler characteristic** of $\mathbf{SH}(M,\gamma)$ is

$$\chi(\mathbf{SH}(M,\gamma)):=\sum_{(i_1,\dots,i_n)\in\mathbb{Z}^n}\chi(\mathbf{SH}(M,\gamma,(S_1,\dots,S_n),(i_1,\dots,i_n)))\cdot(\rho_1^{i_1}\cdots\rho_n^{i_n})\in\mathbb{Z}[H]/\pm H.$$

Remark 2.41. Theorem 2.39 shows the definition of graded Euler characteristic is independent of the choice of $S_1,\dots,S_n$ if we regard it as an element in $\mathbb{Z}[H]/\pm H$. If the admissible surfaces $S_1,\dots,S_n$ and a particular closure of (M,γ) is fixed, then the ambiguity of $\pm H$ can be removed.

At last, from Theorem 2.19, Proposition 2.20, and Axiom (A1-7), the following proposition is straightforward.

Proposition 2.42. *Suppose (M,γ) is a balanced sutured manifold and $S \subset (M,\gamma)$ is an admissible surface. Suppose we perform the bypass change in a disk in Figure 1 that is disjoint from ∂S. Let γ_2 and γ_3 be the resulting two sutures. Then all the maps in the bypass exact triangle (2.19) are grading preserving, i.e., for any $i \in \mathbb{Z}$, we have an exact triangle*

$$\begin{tikzcd}
\mathbf{SH}(-M,-\gamma_1,S,i) \arrow[rr, "\psi_{1,i}"] & & \mathbf{SH}(-M,-\gamma_2,S,i) \arrow[dl, "\psi_{2,i}"] \\
& \mathbf{SH}(-M,-\gamma_3,S,i) \arrow[ul, "\psi_{3,i}"] &
\end{tikzcd}$$

where $\psi_{k,i}$ are the restriction of ψ_k in (2.19).

3. Heegaard Floer homology and the graph TQFT

In this section, we discuss the modification of Heegaard Floer theory to make it suitable to formal sutured homology.

3.1. Heegaard Floer homology for multi-pointed 3-manifolds.

In this subsection and the next subsection, we provide an overview of the graph TQFT for Heegaard Floer theory, constructed by Zemke [Zem19] (See also [HMZ18, Zem18]), and list some properties which are relevant to proofs in the third subsection about the Floer's excision theorem.

Definition 3.1. A **multi-pointed 3-manifold** is a pair $(Y,\mathbf{w})$ consisting of a closed, oriented 3-manifold Y (not necessarily connected), together with a finite collection of basepoints $\boldsymbol{w} \subset Y$, such that each component of Y contains at least one basepoint.

Given two multi-pointed 3-manifolds $(Y_0,\boldsymbol{w}_0)$ and $(Y_1,\boldsymbol{w}_1)$, a **ribbon graph cobordism** from $(Y_0,\boldsymbol{w}_0)$ to $(Y_1,\boldsymbol{w}_2)$ is a pair (W,Γ) satisfying the following:

(1) W is a cobordism from Y_0 to Y_1.
(2) Γ is an embedded graph in W such that $\Gamma \cap Y_i = \boldsymbol{w}_i$ for $i = 0,1$. Furthermore, each point of $\boldsymbol{w}_i$ has valence 1 in Γ.
(3) Γ has finitely many edges and vertices, and no vertices of valence 0.
(4) The embedding of Γ is smooth on each edge.
(5) Γ is decorated with a formal ribbon structure, *i.e.*, a formal choice of cyclic ordering of the edges adjacent to each vertex.

Definition 3.2. A **restricted graph** is a graph whose vertices have valence either 1 or 2. A ribbon graph cobordism (W,Γ) from $(Y_1,\boldsymbol{w}_1)$ to $(Y_2,\boldsymbol{w}_2)$ is called a **restricted graph cobordism** if Γ is restricted (so the cyclic ordering is unique) and any component of Γ does not connect two basepoints of the same manifold Y_i for $i = 1,2$.

Definition 3.3 ([OS08])**.** Suppose $(Y,\boldsymbol{w})$ is a connected multi-pointed 3-manifold. A **multi-pointed Heegaard diagram** $\mathcal{H} = (\Sigma,\alpha,\beta,\boldsymbol{w})$ for $(Y,\boldsymbol{w})$ is a tuple satisfying the following:

(1) Σ is a closed, oriented surface, embedded in Y , such that $\boldsymbol{w} \subset \Sigma\backslash(\alpha \cup \beta)$. Furthermore, Σ splits Y into two handlebodies U_α and U_β, oriented so that $\Sigma = \partial U_\alpha = -U_\beta$.
(2) $\alpha = \{\alpha_1,\dots,\alpha_n\}$ is a collection of $n = g(\Sigma) + |\boldsymbol{w}| - 1$ pairwise disjoint, simple, closed curves on Σ, bounding pairwise disjoint compressing disks in U_α. Each component of $\Sigma\backslash\alpha$ is planar and contains a single basepoint.

(3) $\beta = \{\beta_1, \dots, \beta_n\}$ is a collection of pairwise disjoint, simple, closed curves on Σ bounding pairwise disjoint compressing disks in U_β. Each component of $\Sigma \backslash \beta$ is planar and contains a single basepoint.

Suppose $\boldsymbol{w} = \{w_1, \dots, w_m\}$. Let the polynomial ring associated to $\boldsymbol{w}$ be

$$\mathbb{F}_2[U_{\boldsymbol{w}}] := \mathbb{F}_2[U_{w_1}, \dots, U_{w_m}].$$

Let $\mathbb{F}_2[U_{\boldsymbol{w}}, U_{\boldsymbol{w}}^{-1}]$ be the ring obtained by formally inverting each of the variables.

If $\boldsymbol{k} = (k_1, \dots, k_m)$ is an m-tuple, let

$$U_{\boldsymbol{w}}^{\boldsymbol{k}} := U_{w_1}^{k_1} \cdots U_{w_m}^{k_m}.$$

Suppose $\mathcal{H} = (\Sigma, \alpha, \beta, \boldsymbol{w})$ is a multi-pointed Heegaard diagram of a connected multi-pointed 3-manifold $(Y, \boldsymbol{w})$. Suppose $n = g(\Sigma) + |\boldsymbol{w}| - 1$. Consider two tori

$$\mathbb{T}_\alpha := \alpha_1 \times \cdots \times \alpha_n \text{ and } \mathbb{T}_\beta := \beta_1 \times \cdots \times \beta_n$$

in the symmetric product

$$\mathrm{Sym}^n \Sigma := (\prod_{i=1}^{n} \Sigma)/S_n.$$

The chain complex $CF^-(\mathcal{H})$ is a free $\mathbb{F}_2[U_{\boldsymbol{w}}]$-module generated by intersection points $\boldsymbol{x} \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta$. Define

$$CF^\infty(\mathcal{H}) := CF^-(\mathcal{H}) \otimes_{\mathbb{F}_2[U_{\boldsymbol{w}}]} \mathbb{F}_2[U_{\boldsymbol{w}}, U_{\boldsymbol{w}}^{-1}], \text{ and } CF^+(\mathcal{H}) := CF^\infty(\mathcal{H})/CF^-(\mathcal{H}).$$

To construct a differential on $CF^-(\mathcal{H})$, suppose $\mathcal{H}$ satisfies some extra admissibility conditions if $b_1(Y) > 0$ (*c.f.* [Zem19, Section 4.7]). Let $(J_s)_{s\in[0,1]}$ be an auxiliary path of almost complex structures on $\mathrm{Sym}^n\Sigma$ and let $\pi_2(\boldsymbol{x}, \boldsymbol{y})$ be the set of homology classes of Whitney disks connecting intersection points $\boldsymbol{x}$ and $\boldsymbol{y}$ (*c.f.* [OS08, Section 3.4]). For $\phi \in \pi_2(\boldsymbol{x}, \boldsymbol{y})$, let $\mathcal{M}_{J_s}(\phi)$ be the moduli space of J_s-holomorphic maps $u : [0, 1] \times \mathbb{R} \to \mathrm{Sym}^n\Sigma$ which represent ϕ. The moduli space $\mathcal{M}_{J_s}(\phi)$ has a natural action of $\mathbb{R}$, corresponding to reparametrization of the source. We write

$$\widehat{\mathcal{M}}_{J_s}(\phi) := \mathcal{M}_{J_s}(\psi)/\mathbb{R}.$$

For $\phi \in \pi_2(\boldsymbol{x}, \boldsymbol{y})$, let $\mu(\phi)$ be the expected dimension of $\mathcal{M}_{J_s}(\phi)$ for generic J_s and let $n_{w_i}(\phi)$ be the algebraic intersection number of $\{w_i\} \times \mathrm{Sym}^{n-1}\Sigma$ and any representative of ϕ. Define

$$n_{\boldsymbol{w}}(\phi) := (n_{w_1}(\phi), \dots, n_{w_m}(\phi)).$$

For a generic path J_s, define the differential on $CF^-(\mathcal{H})$ by

$$\partial_{J_s}(\boldsymbol{x}) = \sum_{\boldsymbol{y}\in\mathbb{T}_\alpha\cap\mathbb{T}_\beta} \sum_{\substack{\phi\in\pi_2(\boldsymbol{x},\boldsymbol{y}) \\ \mu(\phi)=1}} \#\widehat{\mathcal{M}}_{J_s}(\phi) U_{\boldsymbol{w}}^{n_{\boldsymbol{w}}(\phi)} \cdot \boldsymbol{y},$$

extended linearly over $\mathbb{F}_2[U_{\boldsymbol{w}}]$. The differential ∂_{J_s} can be extended on $CF^\infty(\mathcal{H})$ and $CF^+\infty(\mathcal{H})$ by tensoring with the identity map.

Lemma 3.4 ([OS08, Lemma 4.3])**.** *For a generic path J_s, the map ∂_{J_s} on $CF^\circ(\mathcal{H})$, where $\circ \in \{\infty, +, -\}$, satisfies*

$$\partial_{J_s} \circ \partial_{J_s} = 0.$$

For a disconnected multi-pointed 3-manifold $(Y, \boldsymbol{w}) = (Y_1, \boldsymbol{w_1}) \sqcup (Y_2, \boldsymbol{w_2})$, where Y_i is connected for $i = 1, 2$, suppose $\mathcal{H}_i$ is an admissible multi-pointed Heegaard diagram of Y_i and suppose J_{s_i} are corresponding generic paths of almost complex structures. Let the chain complex associated to $(Y, \boldsymbol{w})$ be

$$(CF^\circ(\mathcal{H}_1 \sqcup \mathcal{H}_2), \partial_{J_s}) := (CF^\circ(\mathcal{H}_1), \partial_{J_{s_1}}) \otimes_{\mathbb{F}_2} (CF^\circ(\mathcal{H}_2), \partial_{J_{s_2}}), \tag{3.1}$$

where $\circ \in \{\infty, +, -\}$.

Remark 3.5. In Zemke's original construction [Zem19], one should choose colors of basepoints and graphs to achieve the functoriality of TQFT. Here we implicitly choose different colors for all basepoints and we have the relation on the homology level

$$H(CF^\circ(\mathcal{H}_1 \sqcup \mathcal{H}_2), \partial_{J_s}) = H(CF^\circ(\mathcal{H}_1), \partial_{J_{s_1}}) \otimes_{\mathbb{F}_2} H(CF^\circ(\mathcal{H}_1), \partial_{J_{s_2}}). \tag{3.2}$$

Since we will only consider restricted graph cobordisms, the choice of colors satisfies the functoriality. Note that in the construction of [HMZ18, Zem18], the colors of all basepoints are the same and all U_{w_i} are identified as U, so (3.1) and (3.2) are not true in general.

The chain homotopy type of $(CF^\circ(\mathcal{H}), \partial_{J_s})$ is independent of the choice of the admissible diagram $\mathcal{H}$ and the generic path J_s. Indeed, we have the following theorem about naturality.

Theorem 3.6 ([Zem19, Proposition 4.6], see also [OS04d, JTZ18])**.** *Suppose that $(Y, \boldsymbol{w})$ is a multi-pointed 3-manifold. To each (admissible) pair $(\mathcal{H}, J)$ and $(\mathcal{H}', J')$, there is a well-defined map*

$$\Psi_{(\mathcal{H},J)\to(\mathcal{H}',J')} : (CF^-(\mathcal{H}), \partial_J) \to CF^-(\mathcal{H}'), \partial_{J'}),$$

which is well defined up to $\mathbb{F}_2[U_{\boldsymbol{w}}]$-equivariant chain homotopy. Furthermore, the following are satisfied:

(1) If $(\mathcal{H}, J)$, $(\mathcal{H}', J')$ and $(\mathcal{H}'', J'')$ are three (admissible) diagrams with (generic) paths of almost complex structures, then

$$\Psi_{(\mathcal{H},J)\to(\mathcal{H}'',J'')} \simeq \Psi_{(\mathcal{H}',J')\to(\mathcal{H}'',J'')} \circ \Psi_{(\mathcal{H},J)\to(\mathcal{H}',J')}.$$

(2) $\Psi_{(\mathcal{H},J)\to(\mathcal{H},J)} \simeq \mathrm{id}_{(CF^-(\mathcal{H}),\partial_J)}$.

Moreover, similar results hold for CF^∞ and CF^+.

Since all chain complexes discussed above can be decomposed by spin^c structures (*c.f.* [OS04d, Section 2.6]), we have the following definition.

Definition 3.7. Suppose $(Y, \boldsymbol{w})$ is a multi-pointed 3-manifold and $\mathfrak{s} \in \mathrm{Spin}^c(Y)$. For $\circ \in \{\infty, +, -\}$, define $CF^\circ(Y, \boldsymbol{w}, \mathfrak{s})$ to be the transitive system of chain complexes with canonical maps from Theorem 3.6, with respect to $\mathfrak{s}$, and define $HF^\circ(Y, \boldsymbol{w}, \mathfrak{s})$ to be the induced transitive system of homology groups.

For further using, we also define the completions of the chain complexes.

Definition 3.8. Let $\mathbb{F}_2[[U_{\boldsymbol{w}}]]$ be the ring of formal power series of $U_{\boldsymbol{w}}$. For $\circ \in \{\infty, +, -\}$, define

$$\mathbf{CF}^\circ(Y, \boldsymbol{w}, \mathfrak{s}) := \mathbf{CF}^\circ(Y, \boldsymbol{w}, \mathfrak{s}) \otimes_{\mathbb{F}_2[U_{\boldsymbol{w}}]} \mathbb{F}_2[[U_{\boldsymbol{w}}]].$$

Let $\mathbf{HF}^\circ(Y, \boldsymbol{w}, \mathfrak{s})$ be the induced homology groups.

Convention. When omitting the module structure, we have $\mathbf{CF}^+(Y, \boldsymbol{w}, \mathfrak{s}) = CF^+(Y, \boldsymbol{w}, \mathfrak{s})$. Hence we do not distinguish these groups.

The advantage of the completions is that we have the following proposition.

Proposition 3.9 ([MO17, Section 2], see also [OS04a, Lemma 2.3])**.** *If $(Y, \boldsymbol{w})$ is a multi-pointed 3-manifold and $\mathfrak{s} \in \mathrm{Spin}^c(Y)$ on any component is nontorsion, then $\mathbf{HF}^\infty(Y, \boldsymbol{w}, \mathfrak{s}) = 0$.*

Then the boundary map in the following long exact sequence induces a canonical isomorphism between $\mathbf{HF}^-(Y, \boldsymbol{w}, \mathfrak{s})$ and $HF^+(Y, \boldsymbol{w}, \mathfrak{s})$ for any nontorsion spin^c structure $\mathfrak{s}$.

Proposition 3.10. *From the short exact sequence*

$$0 \to \mathbf{CF}^-(Y, \boldsymbol{w}, \mathfrak{s}) \to \mathbf{CF}^\infty(Y, \boldsymbol{w}, \mathfrak{s}) \to CF^+(Y, \boldsymbol{w}, \mathfrak{s}) \to 0,$$

we have a long exact sequence

$$\cdots \to \mathbf{HF}^-(Y, \boldsymbol{w}, \mathfrak{s}) \to \mathbf{HF}^\infty(Y, \boldsymbol{w}, \mathfrak{s}) \to HF^+(Y, \boldsymbol{w}, \mathfrak{s}) \to \cdots$$

Definition 3.11. Suppose $(Y, \boldsymbol{w})$ is a multi-pointed 3-manifold and $\mathfrak{s} \in \mathrm{Spin}^c(Y)$ is a nontorsion spin^c structure. We write

$$HF(Y, \boldsymbol{w}, \mathfrak{s}) = HF_{\mathrm{red}}(Y, \boldsymbol{w}, \mathfrak{s}) := HF^+(Y, \boldsymbol{w}, \mathfrak{s}) \cong \mathbf{HF}^-(Y, \boldsymbol{w}, \mathfrak{s}).$$

3.2. Cobordism maps for restricted graph cobordisms.

Theorem 3.12 ([Zem19, Theorem A])**.** *Suppose $(W, \Gamma) : (Y_0, \boldsymbol{w}_0) \to (Y_1, \boldsymbol{w}_1)$ is a ribbon graph cobordism and $\mathfrak{s} \in \mathrm{Spin}^c(W)$. Then there are two chain maps*

$$F^A_{W,\Gamma,\mathfrak{s}}, F^B_{W,\Gamma,\mathfrak{s}} : CF^-(Y_0, \boldsymbol{w}_0, \mathfrak{s}|_{Y_0}) \to CF^-(Y_1, \boldsymbol{w}_1, \mathfrak{s}|_{Y_1}),$$

which is a diffeomorphism invariant of (W, Γ), up to $\mathbb{F}_2[U_{\boldsymbol{w}}]$-equivariant chain homotopy.

Proposition 3.13 ([Zem19, Theorem C])**.** *Suppose that (W, Γ) is ribbon graph cobordism which decomposes as a composition $(W, \Gamma) = (W_2, \Gamma_2) \cup (W_1, \Gamma_1)$. If $\mathfrak{s}_1$ and $\mathfrak{s}_2$ are spin^c structures on W_1 and W_2, respectively, then*

$$F^A_{W_2,\Gamma_2,\mathfrak{s}_2} \circ F^A_{W_1,\Gamma_1,\mathfrak{s}_1} = \sum_{\substack{\mathfrak{s} \in \mathrm{Spin}^c(W) \\ \mathfrak{s}|_{W_2} = \mathfrak{s}_2 \\ \mathfrak{s}|_{W_1} = \mathfrak{s}_1}} F^A_{W,\Gamma,\mathfrak{s}}.$$

The similar relation holds for $F^B_{W,\Gamma,\mathfrak{s}}$.

Convention. Since we will only consider restricted graph cobordisms, the map $F^A_{W,\Gamma,\mathfrak{s}}$ is chain homotopic to $F^B_{W,\Gamma,\mathfrak{s}}$. Hence we write $CF^-(W, \Gamma, \mathfrak{s})$ for the chain map and $HF^-(W, \Gamma, \mathfrak{s})$ for the induced map on the homology group. If Γ and $\mathfrak{s}$ are specified, we write $CF^-(W)$ and $HF^-(W)$ for simplicity, respectively. The chain maps on $CF^\infty, CF^+, \mathbf{CF}^-, \mathbf{CF}^\infty$ are obtained by tensoring with the identity maps, respectively. We use the similar notation for these chain maps and the induced maps on homology groups. All maps are called **cobordism maps**.

For CF^-, the cobordism map is defined by the composition of the following maps.

- For 4-dimensional 1-, 2-, and 3-handle attachments away from the basepoints, we use the maps defined by Ozsváth and Szabó [OS06].
- For 4-dimensional 0- and 4-handle attachments, or equivalently adding and removing a copy of S^3 with a single basepoint, respectively, we use the maps defined by the canonical isomorphism from the tensor product with $CF^-(S^3, w_0) \cong \mathbb{F}_2[U_0]$.
- For a ribbon graph cobordism $(Y \times [0, 1], \Gamma)$, we project the graph into Y and use the **graph action map** defined in [Zem19, Section 7].

Remark 3.14. For 4-dimensional 1-, 2-, and 3-handle attachments, Ozsváth and Szabó's original construction was for connected cobordisms of connected 3-manifolds. Zemke [Zem19, Section 8] extended the construction to cobordisms between disjoint 3-manifolds. For 4-dimensional 0- and 4-handle attachments, considering the coloring chain complex in [Zem19, Section 4.3], the isomorphism is indeed

$$CF^-(Y \sqcup S^3, \boldsymbol{w} \cup \{w_0\}) \cong CF^-(Y, \boldsymbol{w}) \otimes_{\mathbb{F}_2} CF^-(S^3, w_0) \cong CF^-(Y, \boldsymbol{w}) \otimes_{\mathbb{F}_2} \mathbb{F}_2[U_0].$$

The graph action map is obtained by the composition of maps associated to elementary graphs. The construction involves **free-stabilization maps** $S^{\pm}_w$ [Zem19, Section 6] and **relative homology maps** A_λ [Zem19, Section 5], where $S^{\pm}_w$ correspond to adding or removing a basepoint w and A_λ correspond to a path λ between two basepoints. When considering restricted graph cobordisms, we only need maps associated to 1-, 2-, 3-handle attachments and free-stabilizations.

Definition 3.15. Suppose $\mathcal{H} = (\Sigma, \alpha, \beta, \boldsymbol{w})$ is a multi-pointed Heegaard diagram for a multi-pointed 3-manifold $(Y, \boldsymbol{w})$. Let $D \subset \Sigma \backslash (\alpha \cup \beta)$ be a small disk containg a new basepoint $w_0 \in \Sigma \backslash (\alpha \cup \beta)$. Let α_0 and β_0 be two simple closed curves on Σ bounding a disk containing w_0 and $|\alpha_0 \cap \beta_0| = 2$. Suppose θ^+ and θ^- are the higher and the lower graded intersection points, respectively. Consider the Heegaard diagram $\mathcal{H}' = (\Sigma, \alpha \cup \{\alpha_0\}, \beta \cup \{\beta_0\}, \boldsymbol{w} \cup \{w_0\})$. See Figure 4.

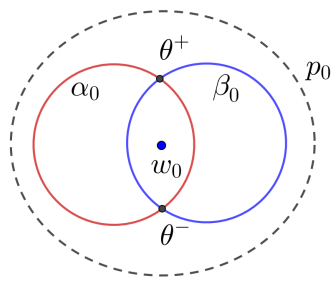

FIGURE 4. Free-stabilization in a small disk D.

For appropriately chosen almost complex structures, define the **free-stabilization** maps $S^{\pm}_{w_0}$ by

$$S^+_{w_0}(\boldsymbol{x}) = \boldsymbol{x} \times \theta^+,$$
$$S^-_{w_0}(\boldsymbol{x} \times \theta^-) = \boldsymbol{x},, \text{ and } S^-_{w_0}(\boldsymbol{x} \times \theta^+) = 0.$$

Remark 3.16. If we collapse ∂D to a point p_0, we obtain a doubly-pointed diagram on S^2 with two curves. Hence $\mathcal{H}'$ can be considered as the connected sum of $\mathcal{H}$ and $(S^2, \alpha_0, \beta_0, \{w_0, p_0\})$.

Proposition 3.17 ([Zem19, Section 6 and Lemma 8.13])**.** *The maps $S^{\pm}_{w_0}$ in Definition 3.15 determine well-defined chain maps on the level of transitive systems of chain complexes*

$$S^+_{w_0} : CF^-(Y, \boldsymbol{w}) \to CF^-(Y, \boldsymbol{w} \cup \{w_0\}),$$
$$S^-_{w_0} : CF^-(Y, \boldsymbol{w} \cup \{w_0\}) \to CF^-(Y, \boldsymbol{w}).$$

Moreover, they satisfy the following results.

(1) The maps $S^{\pm}_{w_0}$ commute with maps associated to 1-, 2-, and 3-handle attachments.
(2) For $\circ_1, \circ_2 \in \{+, -\}$, we have $S^{\circ_1}_{w_1} S^{\circ_2}_{w_2} \simeq S^{\circ_2}_{w_2} S^{\circ_1}_{w_1}$.

Remark 3.18. The free-stabilization maps can be regarded as restricted graph cobordisms with $W = Y \times [0,1]$. The graphs are shown in Figure 5. Alternatively, we can regard them as compositions of maps associated to handle attachements. The map $S^+_{w_2}$ is obtained by first attaching a 0-handle with an arc whose one endpoint is on the boundary and the other is in the interior, and then attaching a 1-handle away from basepoints. See Figure 5. The map $S^-_{w_2}$ is obtained by first attaching 3-handle and then 4-handle with an arc similarly.

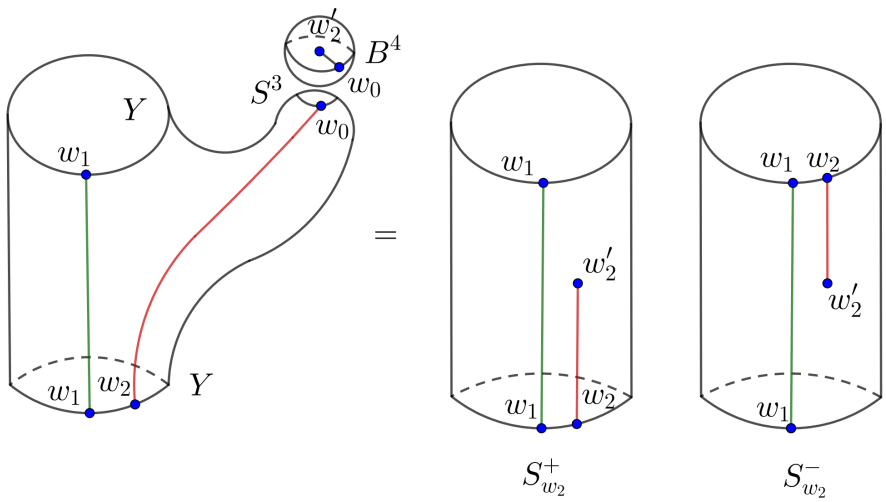

FIGURE 5. Restricted graph cobordisms related to free-stabilization maps.

Convention. All illustrations of cobordisms are from the top to the bottom.

We can calculate the effect of free-stabilization maps on the homology explicitly.

Proposition 3.19 ([OS08, Proposition 6.5])**.** *Consider the construction in Definition 3.15. For suitable choices of almost complex structures, the chain complex $CF^-(\mathcal{H}')$ is identified with the mapping cone of the following map*

$$CF^-(\mathcal{H}) \otimes_{\mathbb{F}_2} \mathbb{F}_2[U_0]\theta^- \xrightarrow{U_0 - U_1} CF^-(\mathcal{H}) \otimes_{\mathbb{F}_2} \mathbb{F}_2[U_0]\theta^+.$$

Corollary 3.20. *The map $S^+_{w_0}$ induces isomorphisms on HF^- and $\mathbf{HF}^-$ and induces the zero map on HF^+. The map $S^-_{w_0}$ induces an isomorphism on HF^+ and induces zero maps on HF^- and $\mathbf{HF}^-$.*

Proof. The arguments for HF^- and $\mathbf{HF}^-$ follows directly from Definition 3.15 and Proposition 3.19. For HF^+, consider the duality

$$CF^-(-Y, \boldsymbol{w}, \bar{\mathfrak{s}}) \cong \mathrm{Hom}_{\mathbb{F}_2}(CF^+(Y, \boldsymbol{w}, \mathfrak{s}), \mathbb{F}_2)$$

from [OS04d, Section 5.1] □

The following proposition implies the choice of the basepoints is not important.

Proposition 3.21 ([Zem19, Corollary 14.19 and Corollary F])**.** *Suppose $(Y, \boldsymbol{w})$ is a multi-pointed 3-manifold and $w_1 \in \boldsymbol{w}$. Then the $\pi_1(Y, w_1)$ action on $HF^-(Y, \boldsymbol{w})$ is always the identity map.*

Suppose $(Y_1, \boldsymbol{w}_1)$ and $(Y_2, \boldsymbol{w}_2)$ are two multi-pointed 3-manifolds with $|\boldsymbol{w}_1| = |\boldsymbol{w}_2|$. Suppose W is a cobordism from Y_1 to Y_2 and $\Gamma \subset W$ is a collection of paths connecting $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$. Then the cobordism map $HF^-(W, \Gamma)$ is independent of the choice of Γ. Moreover, if $Y = Y_1 = Y_2$, then $HF^-(W, \Gamma)$ is an isomorphism.

The similar results hold for $HF^\infty, HF^+, \mathbf{HF}^-, \mathbf{HF}^\infty$.

From Corollary 3.20 and Proposition 3.21, we can define a transitive system of groups based on different choices of basepoints.

Definition 3.22. Suppose Y is a closed, oriented 3-manifold and $\boldsymbol{w}_1, \boldsymbol{w}_2 \subset Y$ are two collections of basepoints in Y. Let $\boldsymbol{w}_1' = \boldsymbol{w}_1 \backslash \boldsymbol{w}_2$ and $\boldsymbol{w}_2' = \boldsymbol{w}_2 \backslash \boldsymbol{w}_1$. Define **transition maps** associated to $(\boldsymbol{w}_1, \boldsymbol{w}_2)$ as

$$\Psi^-_{\boldsymbol{w}_1 \to \boldsymbol{w}_2} := \prod_{w \in \boldsymbol{w}_1'} (S^+_w)^{-1} \circ \prod_{w \in \boldsymbol{w}_2'} S^+_w \quad \text{for } HF^- \text{ and } \mathbf{HF}^-$$

$$\Psi^+_{\boldsymbol{w}_1 \to \boldsymbol{w}_2} := \prod_{w \in \boldsymbol{w}_1'} (S^-_w)^{-1} \circ \prod_{w \in \boldsymbol{w}_2'} S^-_w \quad \text{for } HF^+,$$

where the products mean compositions.

Lemma 3.23. *Suppose Y is a closed, oriented 3-manifold and $\boldsymbol{w}_1, \boldsymbol{w}_2, \boldsymbol{w}_3 \subset Y$ are three collections of basepoints in Y. Suppose w is a basepoint in Y that is not in $\boldsymbol{w}_i$ for $i = 1, 2$. Then the followings hold for transition maps.*

(1) $\Psi^\pm_{\boldsymbol{w}_i \to \boldsymbol{w}_j}$ is well-defined for $i, j \in \{1, 2, 3\}$, i.e., the composition is independent of the order of maps.
(2) $\Psi^\pm_{\boldsymbol{w}_i \to \boldsymbol{w}_j}$ is an isomorphism for $i, j \in \{1, 2, 3\}$.
(3) $\Psi^\pm_{\boldsymbol{w}_i \to \boldsymbol{w}_i} = \mathrm{id}$ for $i = 1, 2, 3$.
(4) $\Psi^\pm_{\boldsymbol{w}_2 \to \boldsymbol{w}_3} \circ \Psi^\pm_{\boldsymbol{w}_1 \to \boldsymbol{w}_2} = \Psi^\pm_{\boldsymbol{w}_1 \to \boldsymbol{w}_3}$.
(5) $\Psi^\pm_{\boldsymbol{w}_1 \cup \{w\} \to \boldsymbol{w}_2 \cup \{w\}} \circ S^+_w = S^+_w \circ \Psi^\pm_{\boldsymbol{w}_1 \to \boldsymbol{w}_2}$.
(6) $\Psi^\pm_{\boldsymbol{w}_1 \to \boldsymbol{w}_2} \circ S^-_w = S^-_w \circ \Psi^\pm_{\boldsymbol{w}_1 \cup \{w\} \to \boldsymbol{w}_2 \cup \{w\}}$.

Proof. Terms (1), (4), (5) and (6) follow from term (2) of Proposition 3.17. Note that maps in terms (5) and (6) can be either isomorphisms or zero maps. Term (2) is trivial from the definition. Term (3) follows from Corollary 3.20. □

Lemma 3.24. *Suppose Y_1 and Y_2 are closed, oriented 3-manifolds and $\boldsymbol{w}_1, \boldsymbol{w}_2 \subset Y_1, \boldsymbol{w}_3, \boldsymbol{w}_4 \subset Y_2$ are collections of basepoints. Suppose W is a cobordism from Y_1 to Y_2 that is induced by a composition of 1-, 2-, 3-handle attachments away from all basepoints. Let Γ_1 and Γ_2 be induced graphs in W with*

$$\Gamma_1 \cap Y_1 = \boldsymbol{w}_1, \Gamma_1 \cap Y_2 = \boldsymbol{w}_3, \Gamma_2 \cap Y_1 = \boldsymbol{w}_2, \text{ and } \Gamma_2 \cap Y_2 = \boldsymbol{w}_4.$$

Then we have a commutative diagram

$$\begin{tikzcd}
HF^-(Y_1, \boldsymbol{w}_1) \arrow[r, "HF^-(W,\Gamma_1)"] \arrow[d, "\Psi^-_{\boldsymbol{w}_1 \to \boldsymbol{w}_2}"] & HF^-(Y_2, \boldsymbol{w}_3) \arrow[d, "\Psi^-_{\boldsymbol{w}_3 \to \boldsymbol{w}_4}"] \\
HF^-(Y_1, \boldsymbol{w}_2) \arrow[r, "HF^-(W,\Gamma_2)"] & HF^-(Y_2, \boldsymbol{w}_4)
\end{tikzcd}$$

The similar commutative diagrams hold for $\mathbf{HF}^-$ and HF^+.

Proof. This follows from term (1) of Proposition 3.17. □

Theorem 3.25. *Suppose Y is a closed, oriented 3-manifold. Then groups $HF^-(Y,\boldsymbol{w})$ for all $\boldsymbol{w}\subset Y$ and transition maps $\Psi^-_{\boldsymbol{w}_1\to\boldsymbol{w}_2}$ for all $\boldsymbol{w}_1,\boldsymbol{w}_2\subset Y$ form a transitive system, which is denoted by $HF^-(Y)$. Moreover, suppose (W,Γ) is a restricted graph cobordism from $(Y_1,\boldsymbol{w}_1)$ to $(Y_2,\boldsymbol{w}_2)$. Then $HF^-(W,\Gamma)$ induces a well-defined map from $HF^-(Y_1)$ to $HF^-(Y_2)$, which is independent of the choice of the restricted graph Γ and denoted by $HF^-(W)$.*

The similar arguments hold for $\mathbf{HF}^-$ and HF^+.

Proof. The well-definedness of $HF^-(Y)$ and $HF^-(W,\Gamma)$ follow from Lemma 3.23 and Lemma 3.24. The restricted graph cobordism is a composition of maps associated to 1-, 2-, 3-handle attachments and free-stabilizations. Note that free-stabilizations maps can be isomorphisms or zero maps on homology groups. Then the independence of Γ follows from Proposition 3.21 and above lemmas. The proofs for $\mathbf{HF}^-$ and HF^+ are similar. □

Remark 3.26. Groups and maps in Theorem 3.25 also split into spin^c structures. Suppose $\mathfrak{s}\in\mathrm{Spin}^c(W)$ is a nontorsion spin^c structure which restricts to nontorsion spin^c structure $\mathfrak{s}_i$ on Y_i for $i=1,2$. Then $\mathbf{HF}^-(Y_i,\mathfrak{s}_i)$ and $HF^+(Y_i,\mathfrak{s}_i)$ are canonically identified by the boundary map in Proposition 3.10. Moreover, the maps $\mathbf{HF}^-(W,\mathfrak{s})$ and $HF^+(W,\mathfrak{s})$ are the same under this identification. We write the map as

$$HF(W,\mathfrak{s}).$$

3.3. **Floer's excision theorem.**

In this subsection, we follow Kronheimer and Mrowka's idea in [KM10b, Section 3] to prove an excision theorem in Heegaard Floer theory.

Let Y be a closed, oriented 3-manifold, of either one or two components. In the latter case, let Y_1 and Y_2 be two components of Y. Let Σ_1 and Σ_2 be two closed, connected, oriented surfaces in Y with $g(\Sigma_1)=g(\Sigma_2)$. If Y has two components, suppose Σ_i is a non-separating surface in Y_i for $i=1,2$. If Y is connected, suppose Σ_1 and Σ_2 represent independent homology classes. In either case, let $F=\Sigma_1\cup\Sigma_2$. Let h be an orientation-preserving diffeomorphism from Σ_1 to Σ_2.

We construct a new manifold $\widetilde{Y}$ as follows. Let Y' be obtained from Y by cutting along Σ. We have

$$\partial Y'=\Sigma_1\cup(-\Sigma_1)\cup\Sigma_2\cup(-\Sigma_2).$$

If Y has two components, then we have $Y'=Y_1'\cup Y_2'$, where Y_i' is obtained from Y_i by cutting along Σ_i for $i=1,2$. Let $\widetilde{Y}$ be obtained by gluing the boundary component Σ_1 to the boundary component $-\Sigma_2$ and gluing Σ_2 to $-\Sigma_1$, using the diffeomorphism of h both times. See Figure 6 for the case that Y has two components.

In either case, $\widetilde{Y}$ is connected. Let $\widetilde{\Sigma}_i$ be the image of Σ_i in $\widetilde{Y}$ for $i=1,2$ and let $\widetilde{F}=\widetilde{\Sigma}_1\cup\widetilde{\Sigma}_2$.

Definition 3.27. Suppose Y is a closed, oriented 3-manifold and $F\subset Y$ is a closed, oriented surface. Let F_i for $i=1,\dots,m$ be components of F. Suppose further that $g(F_i)\geqslant 2$ and any component of Y contains at least one component of F. Let $\mathrm{Spin}^c(Y|F)$ denote the set of spin^c structures $\mathfrak{s}\in\mathrm{Spin}^c(Y)$ satisfying

$$\langle c_1(\mathfrak{s}),F_i\rangle=2g(F_i)-2>0 \text{ for any } F_i. \tag{3.3}$$

Define

$$HF(Y|F):=\bigoplus_{\mathfrak{s}\in\mathrm{Spin}^c(Y|F)}HF(Y,\mathfrak{s}).$$

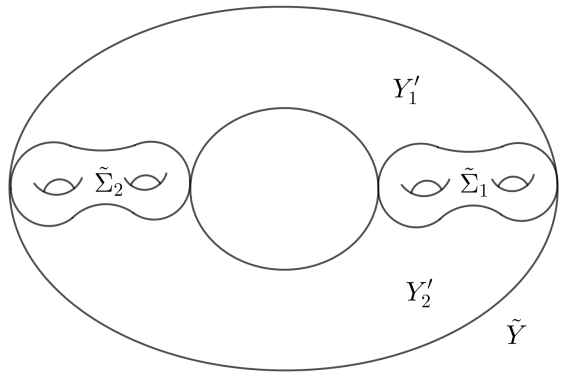

FIGURE 6. Construction of $\widetilde{Y}$.

Suppose (W, Γ) is a restricted graph cobordism and $G \subset W$ is a closed, oriented surface. Let G_i for $i = 1, \ldots, n$ be components of G. Suppose further that $g(G) \geqslant 2$ and any component of W contains at least one component of G. Let $\mathrm{Spin}^c(W|G)$ denote the set of spin^c structures $\mathfrak{s} \in \mathrm{Spin}^c(W)$ satisfying similar conditions in (3.3) by replacing F_i by G_i. Define

$$HF^-(W, \Gamma|G) := \sum_{\mathfrak{s} \in \mathrm{Spin}^c(W|G)} HF^-(W, \Gamma, \mathfrak{s}).$$

Let $HF^+(W, \Gamma|G)$, $\mathbf{HF}^-(W, \Gamma|G)$ and $HF(W, \Gamma|G)$ be defined similarly.

Remark 3.28. Note all spin^c structures in $\mathrm{Spin}^c(Y|F)$ are nontorsion, so $HF(Y, \mathfrak{s})$ is well-defined.

The following is the main theorem of this subsection.

Theorem 3.29 (Floer's excision theorem)**.** *Consider Y and $\widetilde{Y}$ constructed as above. If $g(\Sigma_1) = g(\Sigma_2) \geqslant 2$, then there is an isomorphism*

$$HF(Y|F) \cong HF(\widetilde{Y}|\widetilde{F}).$$

Moreover, this isomorphism and its inverse is induced by restricted graph cobordisms.

Before proving the main theorem, we introduce some lemmas parallel to results in monopole theory (*c.f.* [KM10b, Lemma 2.2, Proposition 2.5 and Lemma 4.7])

Lemma 3.30 ([Lek13, Corollary 17])**.** *Let $Y \to S^1$ be a fibered 3-manifold whose fiber F is a closed, connected, oriented surface with $g = g(F) \geqslant 2$. Then there is a unique $\mathfrak{s}_0 \in \mathrm{Spin}^c(Y|R)$ so that $HF(Y, \mathfrak{s}_0) \neq 0$. Moreover, we have*

$$HF(Y|F) = HF(Y, \mathfrak{s}_0) \cong \mathbb{F}_2.$$

Lemma 3.31. *Suppose $Y = \Sigma \times S^1$ such that $\Sigma = \Sigma \times \{1\} \subset Y$ is a closed, connected, oriented surface with $g(\Sigma) \geqslant 2$. Suppose $w_0 \in S^3$ and $w \in Y$ are basepoints. Let W be obtained from $\Sigma \times D^2$ by removing a 4-ball, considered as a cobordism from S^3 to Y. Let $\Gamma \subset W$ be any path connecting w_0 and w_1. Then the map*

$$\mathbf{HF}^-(W, \Gamma|\Sigma) : \mathbb{F}_2[[U_0]] \cong \mathbf{HF}^-(S^3, w_0) \to HF(Y|\Sigma) \cong \mathbb{F}_2$$

is nontrivial.

Consider $-W$ as a cobordism from $\Sigma \times S^1$ to S^3. Then the map

$$HF^+(-W, \Gamma|\Sigma) : \mathbb{F}_2 \cong HF(Y|\Sigma) \to HF^+(S^3, w_0) \cong \mathbb{F}_2[U_0, U_0^{-1}]/\mathbb{F}_2[U_0]$$

is also nontrivial.

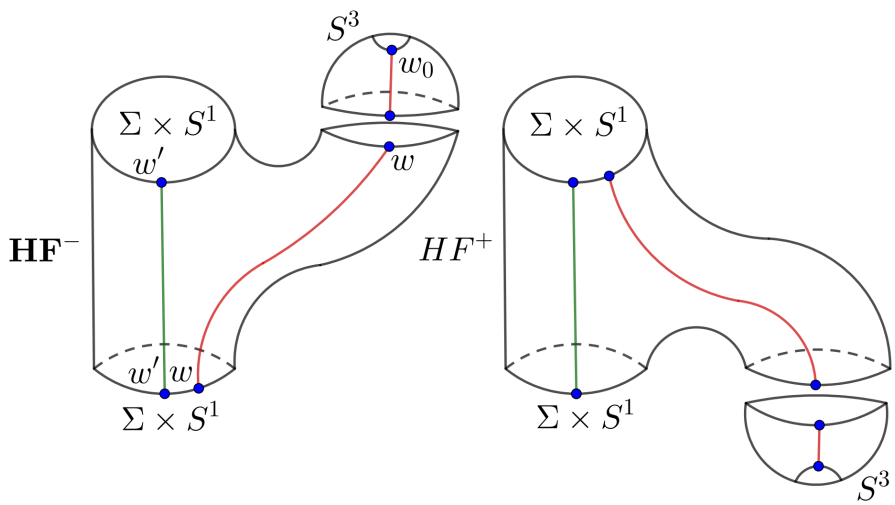

FIGURE 7. Nontrivial cobordism maps from compositions.

Proof. Suppose P is 2-dimensional pair of pants as shown in the left subfigure of Figure 7. Consider $W' = \Sigma \times P$ as a cobordism from $Y_1 \sqcup Y_2$ to Y_3, where $Y_i \cong Y$ for $i = 1, 2, 3$. Suppose w' is another basepoint in Y. Let w_i and w_i' be images of w and w' in Y_i for $i = 1, 2, 3$. Let $\Gamma' \subset W'$ be a collection of two paths γ_1 and γ_2, where γ_1 connects w_1' to w_3' and γ_2 connects w_2 to w_3.

Let $(W_1, \Gamma_1) = (Y_1 \times I, w' \times I)$ be the product cobordism. Suppose $\Sigma_i \subset Y_i$ is the image of $\Sigma \subset Y$ for $i = 1, 2, 3$. Consider the composition of the cobordism maps

$$\mathbf{HF}^-(W', \Gamma'|\Sigma_1 \cup \Sigma_2 \cup \Sigma_3) \circ \mathbf{HF}^-(W_1 \sqcup W|\Sigma_1 \cup \Sigma_2) : HF(Y_1|\Sigma_1) \otimes_{\mathbb{F}_2} \mathbf{HF}^-(S^3, w_0) \to HF(Y_3|\Sigma_3).$$

After filling the S^3 component by a 4-ball, or equivalently composing it with the map associated to a 0-handle attachment, we obtain the free-stabilization map S_w^+ (**c.f.** Remark 3.18). By Corollary 3.20, the result map is an isomorphism

$$HF(Y_1|\Sigma_1) \cong HF(Y_3|\Sigma_3).$$

Since

$$\mathbf{HF}^-(W_1 \sqcup W|\Sigma_1 \cup \Sigma_2) = \mathbf{HF}^-(W_1|\Sigma_1) \otimes_{\mathbb{F}_2} \mathbf{HF}^-(W|\Sigma_2),$$

we know $\mathbf{HF}^-(W|\Sigma_2)$ is a nontrivial map.

The proof about $HF^+(-W, \Gamma|\Sigma)$ is similar. We replace W' by $-W'$ and S_w^+ by S_w^- in the above proof. See the right subfigure of Figure 7 for the illustration of the composition of the cobordisms. □

Corollary 3.32. *Consider $Y = \Sigma \times S^1$ and (W, Γ) in Lemma 3.31. After filling two S^3 boundaries by 4-balls, we have*

$$\mathbf{HF}^-(-W, \Gamma|\Sigma) \circ HF^+(W, \Gamma|\Sigma) = \mathrm{id}_{HF(Y|\Sigma)}.$$

Proof. By Lemma 3.30, we have $HF(Y|\Sigma) \cong \mathbb{F}_2$, so any nontrivial map induces the identity map. By Lemma 3.31, the composition is a nontrivial map. □

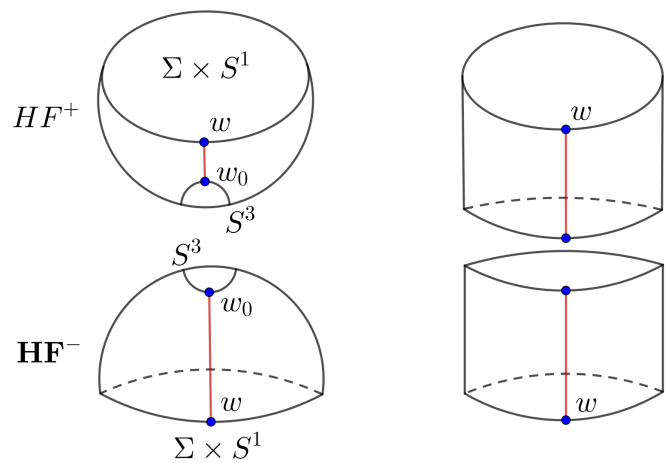

FIGURE 8. Restricted graph cobordisms inducing the same map.

Remark 3.33. The proof of [KM10b, Proposition 2.5] involves some arguments about the trace map, while the setting about trace cobordism in [Zem18] is different from what we use here (*c.f.* Remark 3.5). Corollary 3.32 implies that we can replace the cobordism in the left subfigure of Figure 8 by the right subfigure. This is not true for a general coloring in [Zem19]. To apply this replacement, we should decompose cobordisms in [KM10b, Section 3.2] into compositions of cobordisms.

Now we start to prove the main theorem of this subsection. Many constructions are similar to those in [KM10b, Section 3.2].

Proof of Theorem 3.29. **Step 1**. We construct a cobordism W from $\widetilde{Y}$ to Y and a cobordism $\bar{W}$ from Y to $\widetilde{Y}$.

Recall that Y' is obtained from Y by cutting along Σ_1 and Σ_2 and we have

$$\partial Y' = \Sigma_1 \cup (-\Sigma_1) \cup \Sigma_2 \cup (-\Sigma_2).$$

Suppose P_1 is a saddle surface, which can be regarded as a submanifold of a pair of pants with one boundary component on the top and two boundary components at the bottom. See Figure 9. Suppose

$$\partial P = \lambda_1 \cup \lambda_2 \cup \mu_1 \cup \mu_2 \cup \eta_{1,1} \cup \eta_{1,2} \cup \eta_{2,1} \cup \eta_{2,2},$$

where λ_1 and λ_2 are two arcs in the top boundary component of the pair of pants, μ_1 and μ_2 are two arcs in the bottom boundary components of the pair of pants, and $\eta_{i,j}$ is the arc connecting λ_i and μ_j for $i, j \in \{1, 2\}$.

Suppose $\Sigma \cong \Sigma_1 \cong \Sigma_2$. Remember that we have fixed a diffeomorphism h from Σ_1 to Σ_2. Suppose h' is an orientation-preserving diffeomorphism from Σ to Σ_1. Let W be the union

$$P_1 \times \Sigma \cup Y' \times I,$$

where $\eta_{1,1} \times \Sigma$ is glued to $\Sigma_1 \times I$, $\eta_{2,1} \times \Sigma$ is glued to $-\Sigma_1 \times I$, $\eta_{2,2} \times \Sigma$ is glued to $\Sigma_2 \times I$, and $\eta_{1,2} \times \Sigma$ is glued to $-\Sigma_2 \times I$, using h' and $h \circ h'$, respectively. Figure 9 illustrates the case that Y' has two components Y_1' and Y_2'. By the construction of $\widetilde{Y}$, the result manifold W is a cobordism from $\widetilde{Y}$ to Y.

The cobordism $\bar{W}$ is constructed similarly. Let P_2 be another saddle surface and let $\bar{W}$ be obtained by gluing $P_2 \times \Sigma$ and $Y' \times I$ as shown in Figure 9.

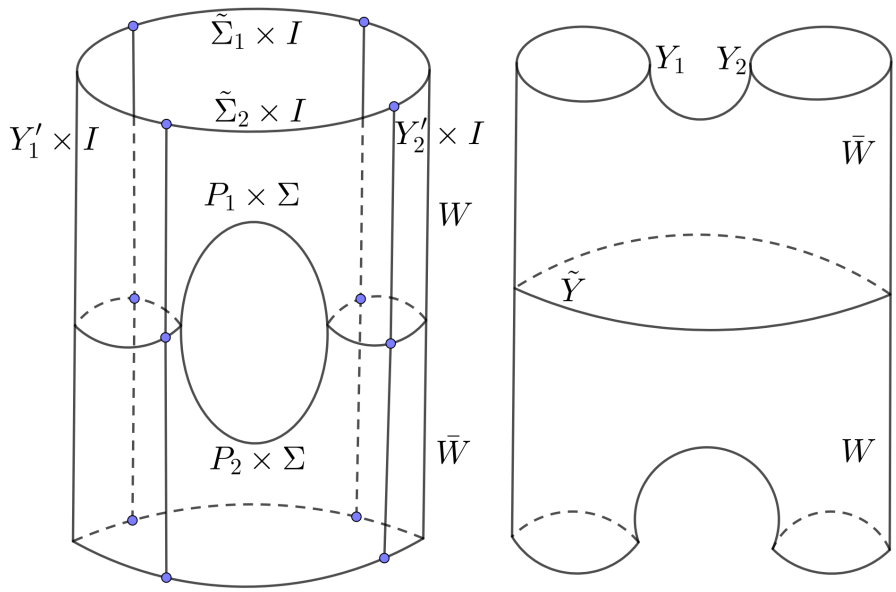

FIGURE 9. Compositions of W and $\bar{W}$.

Step 2. For some restricted graph Γ and some surface G in $W_A = \bar{W} \cup_{\widetilde{Y}} W$, we show the cobordism map

$$HF(W_A, \Gamma|G) := HF^+(W_A, \Gamma|G) = \mathbf{HF}^-(W_A, \Gamma|G)$$

induce the identity map on

$$HF(Y|F) := HF^+(Y|F) \cong \mathbf{HF}^-(Y|F).$$

We prove for the case that Y has two components Y_1 and Y_2. The proof for the case that Y is connected is similar. The cobordism W_A is shown in the right subfigure of Figure 9. Let w_1 and w_1' be two basepoints on Y_1 and let w_2 and w_2' be two basepoints on Y_2. Moreover, we can suppose $w_i \in Y_i'$ for $i = 1, 2$. For $i = 1, 2$, let γ_i be the union of images of $w_i \times I \subset Y_i' \times I$ in $\bar{W}$ and W. Let $\gamma_3 \subset W_A$ be a path in the union of images of $P_1 \times \Sigma$ and $P_2 \times \Sigma$ connecting w_1' to w_2', as shown in Figure 10. Let

$$\Gamma_A = \gamma_1 \cup \gamma_2 \cup \gamma_3.$$

It is straighforward to check by definition that Γ_A is restricted. Let w be a basepoint on $\Sigma \times S^3$. We decompose the restricted graph cobordism (W_A, Γ) into 4 pieces W_i as shown in Figure 10, where W_2 and W_3 are product cobordisms for $(Y_1 \sqcup \Sigma \times S^1 \sqcup Y_2, \{w_1, w, w_2\}))$. Let

$$G = F \cup \Sigma \times \{1\} = \Sigma_1 \cup \Sigma_2 \cup \Sigma \times \{1\}.$$

By Remark 3.26, we know $\mathbf{HF}^-(W_A, \Gamma_A|G)$ and $HF^+(W_A, \Gamma_A|G)$ induce the same map, so $\mathbf{HF}^-(W_i, \Gamma_A \cap W_i|G)$ and $HF^+(W_A, \Gamma_A \cap W_i|G)$ also induce the same map. We assign $\mathbf{HF}^-$ to W_1 and W_2 and HF^+ for W_3 and W_4.

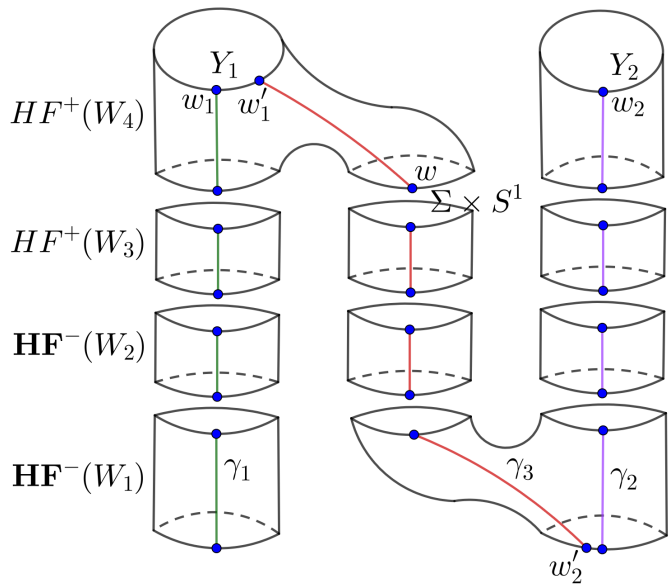

FIGURE 10. Decomposition of the cobordism W_A and the graph Γ.

Note that we have

$$HF(Y_1 \sqcup \Sigma \times S^1 \sqcup Y_2|G)) \cong HF(Y_1|\Sigma_1) \otimes_{\mathbb{F}_2} HF(\Sigma \times S^1|\Sigma \times \{1\}) \otimes_{\mathbb{F}_2} HF(Y_2|\Sigma_2).$$

By Corollary 3.32, we can replace the components of W_2 and W_3 corresponding to $\Sigma \times S^1$ by the cobordism in the left subfigure of Figure 8. After filling two S^3 boundaries by 4-balls, the resulting graph cobordism (W_A', Γ_A') is the composition of the graph cobordisms associated to $S^+_{w_2'}$ and $S^-_{w_1'}$. By Corollary 3.20 and our assignments of the cobordism maps, we know that

$$HF(W_A, \Gamma_A|G) = HF(W_A', \Gamma_A'|G)$$

is an isomorphism. By construction in Definition 3.22, we know that $HF(W_A, \Gamma_A|G)$ induces the identity map on $HF(Y|F)$.

Step 3. For some restricted graph Γ' and some surface G' in $W_B = W \cup_Y \bar{W}$, we show the cobordism map

$$HF(W_B, \Gamma'|G') := HF^+(W_B, \Gamma'|G') = \mathbf{HF}^-(W_B, \Gamma'|G')$$

induce the identity map on

$$HF(\widetilde{Y}|\widetilde{F}) := HF^+(\widetilde{Y}|\widetilde{F}) \cong \mathbf{HF}^-(\widetilde{Y}|\widetilde{F}).$$

We prove for the case that Y has two components Y_1 and Y_2. The proof for the case that Y is connected is similar. The cobordism W_B is shown in the left subfigure of Figure 9. The proof is essentially the same as that in Step 2. However, the decomposition of W_B is not obvious as before.

Let w_1, w_2, w_1'' and w_2'' be basepoints on $\widetilde{Y}$. Let γ_1' and γ_2' be defined similarly to γ_1 and γ_2, respectively. Let $\gamma_3' \subset W_B$ be a path in the union of images of $P_1 \times \Sigma$ and $P_2 \times \Sigma$ connecting w_1'' to w_2'', as shown in Figure 11. Let

$$\Gamma_B = \gamma_1' \cup \gamma_2' \cup \gamma_3'.$$

It is straighforward to check by definition that Γ_B is restricted.

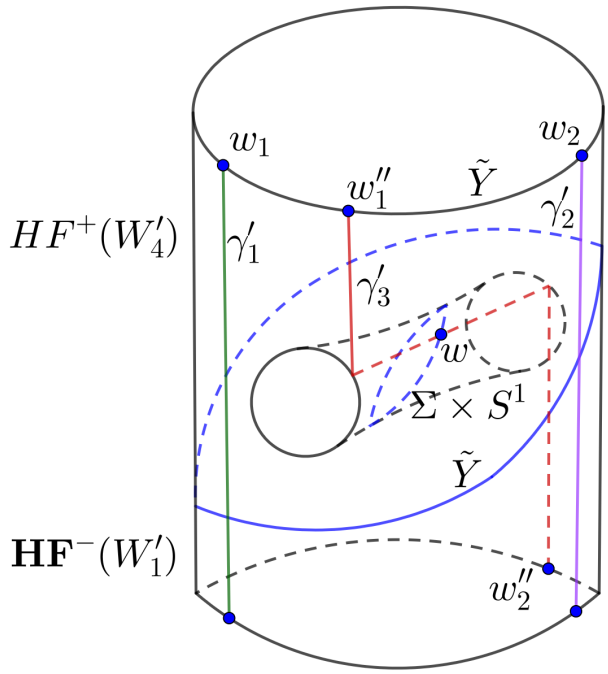

FIGURE 11. The cobordism W_B and the graph Γ'.

Similar to the proof for W_A, we decompose W_B into 4 pieces W_i', where W_2' and W_3' are product cobordisms for $(\widetilde{Y} \sqcup \Sigma \times S^1, \{w_1, w_2, w\})$. In Figure 11 we only draw W_1' and W_4'. Let

$$G' = \widetilde{F} \cup \Sigma \times \{1\} = \widetilde{\Sigma}_1 \cup \widetilde{\Sigma}_2 \cup \Sigma \times \{1\}.$$

Similarly, We assign $\mathbf{HF}^-$ to W_1' and W_2' and HF^+ for W_3' and W_4'. By similar argument to that in the proof for W_A, we know that $HF(W_B, \Gamma_B | G')$ induces the identity map on $HF(\widetilde{Y}|\widetilde{F})$.

Step 4. The results in Step 2 and Step 3 imply

$$\dim_{\mathbb{F}_2} HF(Y|F) \geqslant \dim_{\mathbb{F}_2} HF(\widetilde{Y}|\widetilde{F}) \geqslant \dim_{\mathbb{F}_2} HF(Y|F).$$

Then we know restricted graph cobordisms

$$(W, \Gamma_A \cap W), (\bar{W}, \Gamma_A \cap \bar{W}), (W, \Gamma_B \cap W), \text{ and } (\bar{W}, \Gamma_B \cap \bar{W})$$

with respect to G and G' induce isomorphisms $HF(W)$ and $HF(\bar{W})$ between $HF(Y|F)$ and $HF(\widetilde{Y}|\widetilde{F})$, respectively.

□

3.4. **Sutured Heegaard Floer homology.**

In this subsection, we introduce two equvalent definitions of sutured Heegaard Floer homology. The first one, is due to Juhász [Juh06], based on balanced diagrams of balanced sutured manifolds. The other follows from the construction in Section 2.2, which is essentially due to Kronheimer and Mrowka [KM10b]. These definitions are denoted by SFH and **SHF**, respectively. The equivalence of these definitions was shown by Lekili [Lek13], Baldwin and Sivek [BS20b]. We will focus on the equality for graded Euler characteristics of two homologies.

Definition 3.34 ([Juh06])**.** A **balanced diagram** $\mathcal{H} = (\Sigma, \alpha, \beta)$ is a tuple satisfying the following.

(1) Σ is a compact, oriented surface with boundary.
(2) $\alpha = \{\alpha_1, \dots, \alpha_n\}$ and $\beta = \{\beta_1, \dots, \beta_n\}$ are two sets of pairwise disjoint simple closed curves in the interior of Σ.
(3) The maps $\pi_0(\partial\Sigma) \to \pi_0(\Sigma\backslash\alpha)$ and $\pi_0(\partial\Sigma) \to \pi_0(\Sigma\backslash\beta)$ are surjective.

For such triple, let N be the 3-manifold obtained from $\Sigma \times [-1, 1]$ by attaching 3–dimensional 2–handles along $\alpha_i \times \{-1\}$ and $\beta_i \times \{1\}$ for $i = 1, \dots, n$ and let $\nu = \partial\Sigma \times \{0\}$. A balanced diagram (Σ, α, β) is called **compatible** with a balanced sutured manifold (M, γ) if the sutured manifold (N, ν) is diffeomorphic to (M, γ).

Suppose $\mathcal{H} = (\Sigma, \alpha, \beta)$ is a balanced diagram with $g = g(\Sigma)$ and $n = |\alpha| = |\beta|$. Consider two tori

$$\mathbb{T}_\alpha := \alpha_1 \times \cdots \times \alpha_n \text{ and } \mathbb{T}_\beta := \beta_1 \times \cdots \times \beta_n$$

in the symmetric product

$$\mathrm{Sym}^n\Sigma := (\prod_{i=1}^{n} \Sigma)/S_n.$$

The chain complex $SFC(\mathcal{H})$ is a free $\mathbb{F}_2$-module generated by intersection points $\boldsymbol{x} \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta$. Similar to the construction of CF^-, for a generic path of almost complex structures J_s on $\mathrm{Sym}^n\Sigma$, define the differential on $SFC(\mathcal{H})$ by

$$\partial_{J_s}(\boldsymbol{x}) = \sum_{\boldsymbol{y} \in \mathbb{T}_\alpha \cap \mathbb{T}_\beta} \sum_{\substack{\phi \in \pi_2(\boldsymbol{x}, \boldsymbol{y}) \\ \mu(\phi) = 1}} \#\widehat{\mathcal{M}}_{J_s}(\phi) \cdot \boldsymbol{y},$$

Theorem 3.35 ([Juh06, JTZ18])**.** *Suppose (M, γ) is a balanced sutured manifold. Then there is a balanced diagram $\mathcal{H}$ compatible with (M, γ). The groups $H(SFC(\mathcal{H}), \partial_{J_s})$ for different choices of $\mathcal{H}$ and J_s, together with some canonical maps, form a transitive system. Let $SFH(M, \gamma)$ denote this transitive system and also the associated actual group. Moreover, there is a decomposition*

$$SFH(M, \gamma) = \bigoplus_{\mathfrak{s} \in \mathrm{Spin}^c(M, \partial M)} SFH(M, \gamma, \mathfrak{s}).$$

Remark 3.36. The group $SFH(M, \gamma)$ generalizes Heegaard Floer homology [OS04d] and knot Flor homology [OS04b, Ras03]. Suppose Y is a closed 3-manifold and $K \subset Y$ is a knot. Let $Y(1)$ be obtained from Y by removing a 3-ball and let δ be a simple closed curve on $\partial Y(1)$. Let γ consist of two meridians of K. Then there are isomorphisms

$$SFH(Y(1), \delta) \cong \widehat{HF}(Y), \text{ and } SFH(Y(K), \gamma) \cong \widehat{HFK}(Y, K).$$

Definition 3.37. For a balanced sutured manifold (M, γ), let the $\mathbb{Z}_2$ grading of $SFH(M, \gamma)$ be induced by the sign of intersection points of $\mathbb{T}_\alpha$ and $\mathbb{T}_\beta$ for some compatible diagram $\mathcal{H} = (\Sigma, \alpha, \beta)$

(*c.f.* [FJR09, Section 3.4]). Suppose $H = H^2(M, \partial M)$ and choose $\mathfrak{s}_0 \in Spin^c(M, \partial M)$. The **graded Euler characteristic** of $SFH(M, \gamma)$ is

$$\chi(SFH(M,\gamma)) := \sum_{\substack{\mathfrak{s}\in \mathrm{Spin}^c(M,\partial M)\\ \mathfrak{s}-\mathfrak{s}_0=h\in H}} \chi(SFH(M,\gamma,\mathfrak{s}))\cdot h \in \mathbb{Z}[H]/\pm H.$$

Theorem 3.38 ([FJR09])**.** *Suppose (M, γ) is a balanced sutured manifold. Then*

$$\chi(SFH(M,\gamma)) = \tau(M,\gamma),$$

where $\tau(M, \gamma)$ is a torsion element computed from the map

$$\pi_1(R_-(\gamma), \mathrm{pt}) \to \pi_1(M, \mathrm{pt})$$

by Fox calculus.

Convention. The group $H^2(M, \partial M)$ may have torsions. To compare the graded Euler characteristic in Definition 3.37 with the one induced by admissible surfaces in Definition 2.40, we also regard $\chi(SFH(M, \gamma))$ as an element in $\mathbb{Z}[H']/\pm H'$, which is induced by the composition of maps

$$H = H^2(M, \partial M) \cong H_1(M) \to H_1(M)/\mathrm{Tors} = H'.$$

Then we define the second version of sutured Heegaard Floer homology.

Definition 3.39. Suppose (M, γ) is a balanced sutured manifold and (Y, R) is a closure of (M, γ) as in Definition 2.8. Define

$$SHF(M,\gamma) := HF(Y|R) = \bigoplus_{\mathfrak{s}\in \mathrm{Spin}^c(Y|R)} HF^+(Y,\mathfrak{s}).$$

Remark 3.40. By work of Kutluhan, Lee, and Taubes [KLT10], for any $\mathfrak{s} \in \mathrm{Spin}^c(Y)$, there is an isomorphism

$$HF^+(Y,\mathfrak{s}) \cong \widetilde{HM}_*(Y,\mathfrak{s}) = \widetilde{HM}_\bullet(Y,\mathfrak{s}).$$

The last group is used to define SHM in [KM10b].

Following the discussion in Section 2.2, we can prove the naturality of $SHF(M, \gamma)$ based on the Floer's excision theorem. Let $\mathbf{SHF}(M, \gamma)$ be the transitive system corresponding to $SHF(M, \gamma)$.

Theorem 3.41 ([Lek13, Theorem 24], see also [BS20b, Theorem 3.26])**.** *Suppose (M, γ) is a balanced sutured manifold and (Y, R) is a closure of (M, γ). Then there exists a balanced diagram $\mathcal{H} = (\Sigma, \alpha, \beta)$ compatible with (M, γ) and a singly-pointed Heegaard diagram $\mathcal{H}' = (\Sigma', \alpha', \beta')$ so that the followings hold.*

(1) Σ is a submanifold of Σ'.
(2) α and β are subsets of α' and β', respectively.
(3) Suppose $\alpha' = \alpha \cup \alpha''$ and $\beta' = \beta \cup \beta''$. There exists an intersection point $\boldsymbol{x}_1 \in \mathbb{T}_{\alpha''} \cap \mathbb{T}_{\beta''}$ so that the map

$$\begin{aligned} f : SFC(\mathcal{H}) &\to HF^+(\mathcal{H}') \\ \boldsymbol{c} &\mapsto \boldsymbol{c}\times \boldsymbol{x}_1 \end{aligned}$$

is a quasi-isomorphism.

Corollary 3.42 (Proposition 1.15)**.** *Suppose (M, γ) is a balanced sutured manifold and $H = H^2(M, \partial M)/\mathrm{Tors}$. Then we have*

$$\chi(SFH(M,\gamma)) = \chi(\mathbf{SHF}(M,\gamma)) \in \mathbb{Z}[H]/\pm H,$$

where $\chi(\mathbf{SHF}(M, \gamma))$ is defined as in Definition 2.40.

Proof. It suffices to show the quasi-isomorphism in Theorem 3.41 respects spin^c structures and $\mathbb{Z}_2$ gradings.

Consider the $\mathbb{Z}_2$ gradings at first. Suppose $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$ are two generators of $SFC(\mathcal{H})$. Note that the $\mathbb{Z}_2$ grading of $\boldsymbol{c}_i$ is defined by the sign of the corresponding intersection point in $\mathbb{T}_\alpha \cap \mathbb{T}_\beta$ for $i = 1, 2$. For $\boldsymbol{c}_i \times \boldsymbol{x}_1$, the $\mathbb{Z}_2$ grading is defined by mod 2 Maslov grading, which coincides with the sign of the corresponding intersection point in $\mathbb{T}_{\alpha'} \cap \mathbb{T}_{\beta'}$. Thus, we have

$$\mathrm{gr}_2(\boldsymbol{c}_1) - \mathrm{gr}_2(\boldsymbol{c}_2) = \mathrm{gr}_2(\boldsymbol{c}_1 \times \boldsymbol{x}_1) - \mathrm{gr}_2(\boldsymbol{c}_2 \times \boldsymbol{x}_1),$$

where gr_2 is the $\mathbb{Z}_2$ grading.

Then we consider spin^c structures. Consider $\boldsymbol{c}_i$ for $i = 1, 2$ as above. From [Juh06, Lemma 4.7], there is a one chain $\gamma_{\boldsymbol{c}_1} - \gamma_{\boldsymbol{c}_2}$ such that

$$\mathfrak{s}(\boldsymbol{c}_1) - \mathfrak{s}(\boldsymbol{c}_2) = \mathrm{PD}([\gamma_{\boldsymbol{c}_1} - \gamma_{\boldsymbol{c}_2}]),$$

where $\mathfrak{s}(\cdot) : \mathbb{T}_\alpha \cap \mathbb{T}_\beta \to \mathrm{Spin}^c(M, \partial M)$ is defined in [Juh06, Definition 4.5], and $\mathrm{PD} : H_1(M) \to H^2(M, \partial M)$ is the Poincaré duality map.

From [OS04d, Lemma 2.19], we have

$$\mathfrak{s}_z(\boldsymbol{c}_1 \times \boldsymbol{x}_1) - \mathfrak{s}_z(\boldsymbol{c}_2 \times \boldsymbol{x}_1) = \mathrm{PD}'(i_*([\gamma_{\boldsymbol{c}_1} - \gamma_{\boldsymbol{c}_2}])),$$

where $\mathfrak{s}_z(\cdot) : \mathbb{T}_{\alpha'} \cap \mathbb{T}_{\beta'} \to \mathrm{Spin}^c(Y)$ is defined in [OS04d, Section 2.6] and $\mathrm{PD}' : H_1(Y) \to H^2(Y)$ is the Poincaré duality map, and $i_* : H_1(M) \to H_1(Y)$ is the map induced by inclusion $i : M \to Y$.

Hence we have

$$c_1(\mathfrak{s}_z(\boldsymbol{c}_1 \times \boldsymbol{x}_1)) - c_1(\mathfrak{s}_z(\boldsymbol{c}_2 \times \boldsymbol{x}_1)) = 2\mathrm{PD}'(i_*([\gamma_{\boldsymbol{c}_1} - \gamma_{\boldsymbol{c}_2}])).$$

Finally, this corollary follows from definitions of graded Euler characteristics. □

4. Equivalence of graded Euler characteristics

In this section, we prove the graded Euler characteristic of formal sutured homology is independent of the choice of the Floer-type theory. Throughout the section, we assume that $\mathbf{H}$ is a Floer-type theory, *i.e.*, it satisfies all three axioms (A1), (A2), and (A3). For simplicity, we say 'a property is independent of $\mathbf{H}$' if a property is independent of the choice of the Floer-type theory. Suppose (M, γ) is a balanced sutured manifold and $H = H_1(M)/\mathrm{Tors}$. If the admissible surfaces and the closure of (M, γ) are fixed, then the graded Euler characteristic $\chi(\mathbf{SH}(M, \gamma))$ in Definition 2.40 is considered as a well-defined element in $\mathbb{Z}[H]$, rather than $\mathbb{Z}[H]/\pm H$. See Remark 2.41.

4.1. Balanced sutured handlebodies.

In this subsection, we deal with $\mathbb{Z}^n$ gradings for a balanced sutured handlebody. We start with the following lemma.

Lemma 4.1. *Suppose (M, γ) is a balanced sutured manifold, $S \subset (M, \gamma)$ is an admissible surface. Suppose (Y_1, R_1) and (Y_2, R_2) are two closures of (M, γ) of the same genus so that S extends to closed surfaces $\bar{S}_1$ and $\bar{S}_2$ as in Subsection 2.3. If the graded Euler characteristic of $\mathbf{H}(Y_1|R_1)$ has been known, then the graded Euler characteristic of $\mathbf{H}(Y_1|R_1)$ is determined from that of $\mathbf{H}(Y_1|R_1)$ and the topological data of (Y_1, R_1) and (Y_2, R_2).*

Proof. In Subsection 2.2, we construct a canonical map

$$\Phi_{12} : \mathbf{H}(Y_1|R_1) \to \mathbf{H}(Y_2|R_2).$$

From the proof of Theorem 2.24, the canonical map Φ_{12} necessarily preserves the grading induced by S, since that is the reason why the grading associated to S on $\mathbf{SH}(M, \gamma)$ is well-defined. From

the construction of Φ_{12} in Subsection 2.2, the canonical map is a composition of a few cobordism maps (or the inverse). Then the lemma follows from Axiom (A3-3). □

Next, we consider gradings associated to embedded disks.

Proposition 4.2. *Suppose H is a genus $g > 0$ handlebody and $\gamma \subset \partial H$ is a closed oriented 1-submanifold so that (H, γ) is a balanced sutured manifold. Pick $D_1, \dots, D_g$ a set of pairwise disjoint meridian disks in H so that $[D_1], \dots, [D_g]$ generate $H_2(H, \partial H)$. Then for any fixed multi-grading $\boldsymbol{i} = (i_1, \dots, i_g) \in \mathbb{Z}^g$ associated to $D_1, \dots, D_g$, the Euler characteristic*

$$\chi(\mathbf{SH}(-H, -\gamma, \boldsymbol{i})) \in \mathbb{Z}/\{\pm 1\}$$

depends only on (H, γ), $D_1, \dots, D_g$ and $\boldsymbol{i} \in \mathbb{Z}^g$, and is independent of $\mathbf{H}$. Furthermore, if a particular closure of $(-H, -\gamma)$ is fixed, then the sign ambiguity can be resolved.

Proof. We can think of H and $D_1, \dots, D_g \subset H$ as fixed while the suture γ may vary. For any suture γ, define

$$I(\gamma) = \min_{\gamma' \text{ is isotopic to } \gamma} \sum_{j=1}^{g} |D_j \cap \gamma'|,$$

where $|\cdot|$ denotes the number of points. We prove the proposition by induction on $I(\gamma)$. Since $[\gamma] = 0 \in H_1(\partial H)$, we know $|D_j \cap \gamma|$ is always even for $j = 1, \dots, g$.

Note that an isotopy of γ can be understood as combinations of positive and negative stabilizations in the sense of Definition 2.34, and the grading shifting behavior under such isotopy is described by Proposition 2.37, which is determined purely by topological data and is independent of $\mathbf{H}$. Hence we always assume that the suture γ has already realized $I(\gamma)$.

First, if $I(\gamma) < 2g$, then there exists a meridian disk D_j with $D_j \cap \gamma = \varnothing$. Then it follows from Theorem 2.31 that $\mathbf{SH}(-H, -\gamma) = 0$ since $-H$ is irreducible while $(-H, -\gamma)$ is not taut. Hence for any multi-grading $\boldsymbol{i} \in \mathbb{Z}^g$, we have

$$\chi(\mathbf{SH}(-H, -\gamma, \boldsymbol{i})) = 0.$$

If $I(\gamma) = 2g$, then either there exists some integer j so that

$$D_j \cap \gamma = \varnothing$$

or for $j = 1, \dots, g$, we have

$$|D_j \cap \gamma| = 2.$$

In the former case, we know that $\mathbf{SH}(-H, -\gamma) = 0$ and hence

$$\chi(\mathbf{SH}(-H, -\gamma, \boldsymbol{i})) = 0$$

for any multi-grading $\boldsymbol{i} \in \mathbb{Z}^g$. In the later case, we know that $(-H, -\gamma)$ is a product sutured manifold. It follows from Proposition 2.33 and Proposition 2.29 that

$$\mathbf{SH}(-H, -\gamma) = \mathbf{SH}(-H, -\gamma, \mathbf{0}) \cong \mathbb{F}.$$

Hence

$$\chi(\mathbf{SH}(-H, -\gamma, \boldsymbol{i})) = \begin{cases} \pm 1 & \boldsymbol{i} = \mathbf{0} = (0, \dots, 0) \\ 0 & \boldsymbol{i} \in \mathbb{Z}^g \backslash \{\mathbf{0}\} \end{cases}$$

Note the ambiguity ± 1 comes from the choice of the closure. If we look at a particular closure Y of $(-H, -\gamma)$, then the Euler characteristic has no sign ambiguity. Since (H, γ) is a product sutured manifold, there is a 'standard' closure $(S^1 \times \Sigma, \{1\} \times \Sigma)$ as in [KM10b]. By Axiom (A3-2), we have

$$\chi(\mathbf{H}(S^1 \times \Sigma | \{1\} \times \Sigma)) = -1.$$

Then for any other closure (Y, R), we can apply Lemma 4.1 and hence $\mathbf{SH}(Y, R)$ has a fixed Euler characteristic with no sign ambiguity.

Now suppose we have proved that, for all γ so that $I(\gamma) < 2n$, the Euler characteristic of $\mathbf{SH}(-H, -\gamma, \boldsymbol{i})$, viewed as an element in $\mathbb{Z}/\{\pm 1\}$, is independent of $\mathbf{H}$, and that when we look at any fixed closure of $(-H, -\gamma)$, the sign ambiguity can be resolved. Next we deal with the case when $I(\gamma) = 2n$.

Note we have dealt with the base case $I(\gamma) \leqslant 2g$, so we can assume that $n \geqslant g + 1$. Hence, without loss of generality, we can assume that $|D_1 \cap \gamma| \geqslant 4$. Within a neighborhood of ∂D_1, the suture γ can be depicted as in Figure 12. We can pick the bypass α as shown in the same figure. From Proposition 2.42, for any multi-grading $\mathbf{i} \in \mathbb{Z}^g$, we have an exact triangle

$$\begin{tikzcd} & \mathbf{SH}(-H, -\gamma, \boldsymbol{i}) \arrow[dr] & \\ \mathbf{SH}(-H, -\gamma'', \boldsymbol{i}) \arrow[ur] & & \mathbf{SH}(-H, -\gamma', \boldsymbol{i}) \arrow[ll] \end{tikzcd} \tag{4.1}$$

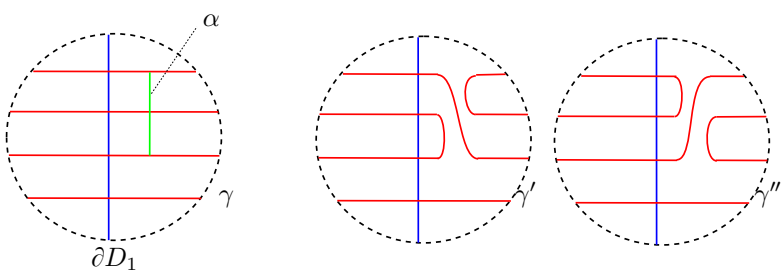

FIGURE 12. The bypass arc α that reduces the intersection function I.

Note that the suture γ' and γ'' are determined by the original suture γ and the bypass arc α, which are all topological data. From Figure 12, it is clear that

$$I(\gamma') \leqslant I(\gamma) - 2 \text{ and } I(\gamma'') \leqslant I(\gamma) - 2.$$

Hence the inductive hypothesis applies, and we know that the Euler characteristics of $\mathbf{SH}(-H, -\gamma'', \boldsymbol{i})$ and $\mathbf{SH}(-H, -\gamma', \boldsymbol{i})$ can be fixed independently of $\mathbf{H}$. Note the maps in the bypass exact triangle (4.1) are described by Proposition 2.20. Hence we conclude that the Euler characteristic of $\mathbf{SH}(-H, -\gamma, \boldsymbol{i})$ is also independent of $\mathbf{H}$. Thus, we finish the proof by induction. □

Next, we deal with gradings associated to general admissible surfaces.

Proposition 4.3. *Suppose H is a genus g handlebody, and S is a properly embedded surface in H. Suppose $\gamma \subset \partial H$ is a suture so that (H, γ) is a balanced sutured manifold and S is an admissible surface. Then the Euler characteristic*

$$\chi(\mathbf{SH}(-H, -\gamma, S, j)) \in \mathbb{Z}/\{\pm 1\}$$

depends only on (H, γ), S, and $j \in \mathbb{Z}$ and is independent of $\mathbf{H}$. Furthermore, if we fix a particular closure of $(-H, -\gamma)$, then the sign ambiguity can also be removed.

Before proving the theorem, we need the following lemma.

Lemma 4.4. *Suppose (M,γ) is a balanced sutured manifold and $S\subset(M,\gamma)$ is a properly embedded admissible surface. Suppose α is a boundary component of S so that α bounds a disk $D\subset\partial M$ and $|\alpha\cap\gamma|=2$. Let S' be the surface obtained by taking the union $S\cup D$ and then push D into the interior of M. Then for any $i\in\mathbb{Z}$, we have*

$$\mathbf{SH}(M,\gamma,S,i)=\mathbf{SH}(M,\gamma,S',i).$$

Proof. Push the interior of D into the interior of M and make $D\cap S'=\varnothing$. It is clear that

$$[S]=[S'\cup D]\in H_2(M,\partial M)\text{ and }\partial S=\partial(S'\cup D).$$

In Subsection 2.3, when constructing the grading associated to $S'\cup D$, we can pick a closure (Y,R) of (M,γ), so that S' and D extend to closed surfaces $\bar{S}'$ and $\bar{D}$ in Y, respectively. Since $|\partial D\cap\gamma|=2$, we know that $\bar{D}$ is a torus. Since $\partial S=\partial(S'\cup D)$, we know that S also extends to a closed surface $\bar{S}$ and from the fact that $[S]=[S'\cup D]$ we know that

$$[\bar{S}]=[\bar{S}'\cup\bar{D}]=[\bar{S}']+[\bar{D}].$$

Since $\bar{D}$ is a torus, from Axioms (A1-4) and (A1-6), we know that the decompositions of $\mathbf{H}(Y|R)$ with respect to $\bar{S}$ and $\bar{S}'$ are the same. Thus it follows that

$$\mathbf{SH}(M,\gamma,S,i)=\mathbf{SH}(M,\gamma,S',i).$$

□

Proof of Proposition 4.3. It is a basic fact that the map

$$\partial_*:H_2(H,\partial H)\to H_1(\partial H)$$

is injective, and $H_2(H,\partial H)$ is generated by g meridian disks, which we fix as $D_1,\dots,D_g$. Hence we assume that

$$[S]=a_1[D_1]+\cdots+a_g[D_g]\in H_2(H,\partial H).$$

Case 1. ∂S consists of only of ∂D_i, *i.e.*,

$$\partial S=\bigcup_{i=1}^{g}(\cup_{a_i}\partial D_i),$$

where $\cup_{a_i}\partial D_i$ means the union of a_i parallel copies of ∂D_i.

Then it follows immediately from the construction of the grading and Axiom (A1-6) that

$$\begin{aligned}\mathbf{SH}(-H,-\gamma,S,j)&=\mathbf{SH}(-H,-\gamma,\bigcup_{i=1}^{g}(\cup_{a_i}D_i),j)\\&=\bigoplus_{j_1+\cdots+j_g=j}\mathbf{SH}(-H,-\gamma,(D_1,\dots,D_g),(j_1,\dots,j_g)).\end{aligned}$$

Hence this case follows from Proposition 4.2.

Case 2. ∂S contains some component that is not parallel to ∂D_i for $j=1,\dots,g$.

Step 1. We modify S and show that it suffices to deal with the case when $S\cap D_j=\varnothing$ for $j=1,\dots,g$.

Note that $\mathrm{im}(\partial_*)\subset H_1(\partial H)$ is generated by $[\partial D_1],\dots,[\partial D_g]$, so we have $\partial S\cdot\partial D_i=0$ for $j=1,\dots,g$. Here $\cdot$ denotes the algebraic intersection number of two oriented curves on ∂H. This means that for $j=1,\dots,g$, the intersection points of ∂D_i with ∂S can be divided into pairs. Suppose two intersection points of ∂D_1 with ∂S of opposite signs are adjacent to each other on ∂D_1, as

depicted in Figure 13. We can perform a cut and paste surgery along D_1 and S to obtain a new surface S_1. From the same figure, it is clear that after isotopy, we can make

$$|\partial D_1 \cap \partial S_1| \leqslant |\partial D_1 \cap \partial S| - 2.$$

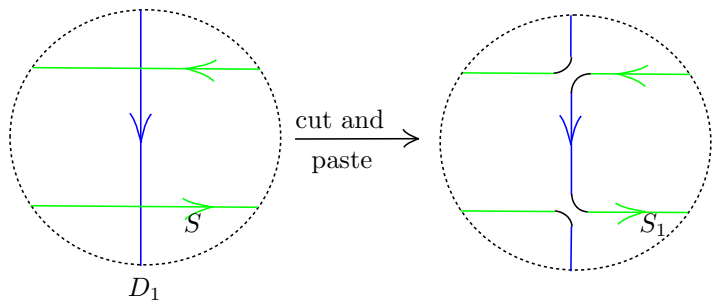

FIGURE 13. The cut and paste surgery on D_1 and S.

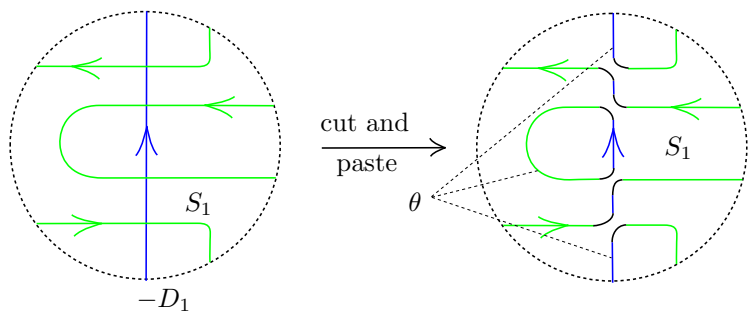

FIGURE 14. The cut and paste surgery on $-D_1$ and S_1.

Note also that if we perform a cut and paste surgery along S_1 and $-D_1$, we obtain another surface S_2. From Figure 14 it is clear that

$$\partial S_2 = \partial S \cup \theta,$$

where θ is the union of some null-homotopic closed curves on ∂H. We can isotope S_2 to make each component of θ intersects the suture twice. Let S_3 be the result of such an isotopy and S_4 be the surface obtained from S_3 by capping off every component of θ. Note we have

$$[S] = [S_4] \in H_2(H, \partial H) \text{ and } \partial S = \partial S_4.$$

Then from Lemma 4.4 we know that

$$\begin{aligned}
\mathbf{SH}(-H, -\gamma, S, j) &= \mathbf{SH}(-H, \gamma, S_4, j) \\
&= \mathbf{SH}(-H, \gamma, S_3, j) \\
&= \mathbf{SH}(-H, -\gamma, S_2, j + j(S_2, S_3)) \\
&= \bigoplus_{j_1 + j_2 = j + j(S_2, S_3)} \mathbf{SH}(-H, -\gamma, (D_1, S_1), (j_1, j_2))
\end{aligned}$$

Note the shift $j(S_2, S_3)$ depends on the isotopy from S_2 to S_3 and by Proposition 2.37, it is determined by the topological data and is independent of $\mathbf{H}$. Hence we reduce the problem to understanding the Euler characteristic of $\mathbf{SH}(-H, -\gamma)$ with multi-grading associated to (D_1, S_1), with

$$|\partial D_1 \cap \partial S_1| \leqslant |\partial D_1 \cap \partial S| - 2.$$

Repeating this argument, we finally reduce to the problem of understanding the Euler characteristic of $\mathbf{H}(-H, -\gamma)$ with multi-grading associated to $(D_1, \dots, D_g, S_g)$, with

$$\partial D_i \cap \partial S_g = \varnothing$$

for $j = 1, \dots, g$.

Step 2. We modify S further to reduce to Case 1.

If every component of ∂S_g is homotopically trivial, then we know that

$$[S_g] = 0 \in H_2(H, \partial H)$$

since the map $H_2(H, \partial H) \to H_1(\partial H)$ is injective. We isotope each component of ∂S_g by stabilization to make it intersect the suture γ twice and then cap it off by a disk. The resulting surface S_{g+1} is a homologically trivial closed surface in H, so $\mathbf{SH}$ is totally supported at grading 0 with respect to S_{g+1}. The grading shift between S_g and S_{g+1} can then be understood by Proposition 2.37, and is independent of $\mathbf{H}$.

Note that $\partial H \backslash (\partial D_1 \cup \dots \cup \partial D_g)$ is a $2g$-punctured sphere, so ∂S is homotopically trivial when removing punctures on the sphere. If some component C of ∂S_g is not null-homotopic, then C is obtained from some ∂D_j by performing handle slides (or equivalently, band sums) over $\partial D_1, \dots, \partial D_g$ for some times.

If we isotope C to make it intersect some ∂D_i twice and then apply the cut and paste surgery, the result curve is isotopic to the one obtained by performing a handle slide over ∂D_i. Explicitly, in Figure 13, suppose two right endpoints of arcs in ∂S (the green arcs) are connected, then the right part of ∂S_1 is a trivial circle and the left part of ∂S_1 is obtained from ∂S by performing a handle slide over ∂D_1. Thus, we can apply the cut and paste surgery for many times, which is equivalent to performing handle slides over $\partial D_1, \dots, \partial D_g$ for some times. Finally, we reduce C to the curve isotopic to ∂D_j. Then we reduce the problem to understanding the Euler characteristic of $\mathbf{SH}(-H, -\gamma)$ with multi-grading associated to $(D_1, \dots, D_g, S_{g+2})$, where S_{g+2} is a surface so that each component of ∂S_{g+2} is parallel to $\pm \partial D_i$ for some i. Case 1 applies to S_{g+2} and we finish the proof. □

Corollary 4.5. *Suppose H is a handlebody and γ is a suture on ∂H so that (H, γ) is a balanced sutured manifold. Suppose $S_1, \dots, S_n$ are properly embedded admissible surfaces in (H, γ). Then the Euler characteristic*

$$\chi(\mathbf{SH}(-H, -\gamma, (S_1, \dots, S_n), (i_1, \dots, i_n))) \in \mathbb{Z}/\{\pm 1\}$$

depends only on (H, γ), $S_1, \dots, S_n$, and $(i_1, \dots, i_n) \in \mathbb{Z}^n$, and is independent of $\mathbf{H}$. Furthermore, if we fix a particular closure of $(-H, -\gamma)$, then the sign ambiguity can also be removed.

Proof. The proof is similar to that for Proposition 4.3. □

4.2. **Gradings and 2-handle attachments.**

Suppose M is a compact oriented 3-manifold with boundary, and $S \subset M$ is a properly embedded surface. Suppose $\alpha \subset M$ is a properly embedded arc that intersects S transversely and $\partial \alpha \cap \partial S = \varnothing$.

Let $N = M \backslash \mathrm{int}(N(\alpha))$, $S_N = S \cap N$, and $\mu \subset \partial N$ be a meridian of α that is disjoint from S_N. Let γ_N be a suture on ∂N satisfies the following properties.

(1) (N, γ_N) is balanced, S is admissible, and $|\gamma_N \cap \mu| = 2$.
(2) If we attach a 2-handle along μ, then we obtain a balanced sutured manifold (M, γ_M).

From [BS16a, Section 4.2], there is a map

$$C_\mu : \mathbf{SHI}(-N, -\gamma_N) \to \mathbf{SHI}(-M, -\gamma_M)$$

constructed as follows.

Push μ into the interior of N to become μ'. Suppose $(N_0, \gamma_{N,0})$ is the manifold obtained from (N, γ_N) by a 0-surgery along μ' with respect to the framing from ∂N. Equivalently, $(N_0, \gamma_{N,0})$ can be obtained from (M, γ_M) by attaching a 1-handle. Since $\mu' \subset \mathrm{int}(N)$, the construction of the closure of (N, γ_N) does not affect μ'. Thus, we can construct a cobordism between closures of (N, γ_N) and $(N_0, \gamma_{N,0})$ by attaching a 4-dimensional 2-handle associated to the surgery on μ'. This cobordism induces a cobordism map

$$C_{\mu'} : \mathbf{SH}(-N, -\gamma_N) \to \mathbf{SH}(-N_0, -\gamma_{N,0}).$$

It is shown in [BS16a, Section 4.2] (or also [KM10b]) that attaching a 1-handle does not change the closure, so there is an identification

$$\iota : \mathbf{SH}(-M, -\gamma_M) \xrightarrow{=} \mathbf{SH}(-N_0, -\gamma_{N,0}).$$

Thus, we define

$$C_\mu = \iota^{-1} \circ C_{\mu'}.$$

In this subsection, we prove the following proposition.

Proposition 4.6. *For any $i \in \mathbb{Z}$, we have*

$$C_\mu(\mathbf{SH}(-N, -\gamma_N, S_N, i)) \subset \mathbf{SH}(-M, -\gamma_M, S, i).$$

Proof. **Step 1**. We consider how the map $C_{\mu'}$ changes the gradings associated to S_N and S_M.

Since μ is disjoint from S, so we can also make μ' disjoint from $S_N = S \cap N$. As a result, the surface S_N survives in $(N_0, \gamma_{N,0})$. From Axiom (A1-7), the cobordism map associated to the 0-surgery along μ' preserves the grading associated to S_N

$$C_{\mu'}(\mathbf{SH}(-N, -\gamma_N, S_N, i)) \subset \mathbf{SH}(-N_0, -\gamma_{N,0}, S_N, i).$$

Step 2. We show

$$\iota : \mathbf{SH}(-M, -\gamma_M, S, i) \xrightarrow{=} \mathbf{SH}(-N_0, -\gamma_{N,0}, S, i).$$

As discussed above, $(N_0, \gamma_{N,0})$ is obtained from (M, γ_M) by a 1-handle attachment. This one handle can be described explicitly as follows. In $(N_0, \gamma_{N,0})$, there is an annulus A bounded by μ and its push-off μ'. We can cap off μ' by the disk coming from the 0-surgery, and hence obtain a disk D with $\partial D = \mu$. By assumption, we know that $|\partial D \cap \gamma_{N,0}| = |\mu \cap \gamma_N| = 2$. Hence D is a compressing disk that intersects the suture twice. If we perform a sutured manifold decomposition on $(N_0, \gamma_{N,0})$ along D, it is straightforward to check the resulting balanced sutured manifold is (M, γ_M). However, in [Juh16], it is shown that decomposing along such a disk is the inverse operation of attaching a 1-handle, and the disk is precisely the co-core of the 1-handle. From this description, we can consider the 1-handle attached to (M, γ_M) as along two endpoints of α. Since $\partial\alpha \cap \partial S = \varnothing$, the surface S naturally becomes a properly embedded surface in $(N_0, \gamma_{N,0})$. From Axiom (A1-7), we know that the map ι preserves the gradings induced by S.

Step 3. We show that

$$\mathbf{SH}(-N_0, -\gamma_{N,0}, S, i) = \mathbf{SH}(-N_0, -\gamma_{N,0}, S_N, i).$$

If $S \cap \alpha = \varnothing$, then $S = S_N = S \cap N$ and the above argument is trivial. If $S \cap \alpha \neq \varnothing$, then S_N is obtained from S by removing disks containing intersection points in $\alpha \cap S$. Then $\partial S_N \backslash \partial S$ consists of a few copies of meridians of α. For simplicity, we assume that there is only one copy of the meridian of α, *i.e.*, $\partial S_N \backslash \partial S = \mu$. The general case is similar to prove.

After performing the 0-surgery along μ', we know that the surface $S_N \subset N_0$ is compressible. Indeed, we can pick $\mu'' \subset \mathrm{int}(S_N)$ parallel to $\mu \subset \partial S_N$. Then there is an annulus A' bounded by μ'' and μ', and we obtain a disk D' by capping μ' off by the disk coming from the 0-surgery. Performing a compression along the disk D', we know that S_N becomes the disjoint union of a disk D'' and the surface $S \subset N_0$. Note $\partial D''$ is parallel to the disk D discussed above. Since

$$\partial(D'' \cup S) = \partial S_N \text{ and } [D'' \cup S] = [S_N] \in H_2(N_0, \partial N_0),$$

From (A1-6), we know that

$$\begin{aligned}\mathbf{SH}(-N_0, -\gamma_{N,0}, S_N, i) &= \mathbf{SH}(-N_0, -\gamma_{N,0}, S \cup D'', i)\\ &= \sum_{i_1+i_2=i} \mathbf{SH}(-N_0, -\gamma_{N,0}, (S, D''), (i_1, i_2)).\end{aligned}$$

Since the disk D'' intersects γ'_N twice, from term (2) of Proposition 2.29, we know that

$$\mathbf{SH}(-N_0, -\gamma_{N,0}) = \mathbf{SH}(-N_0, -\gamma_{N,0}, D'', 0).$$

Hence we conclude that

$$\begin{aligned}\mathbf{SH}(-N_0, -\gamma_{N,0}, S_N, i) &= \sum_{i_1+i_2=i} \mathbf{SH}(-N_0, -\gamma_{N,0}, (S, D''), (i_1, i_2))\\ &= \mathbf{SH}(-N_0, -\gamma_{N,0}, S, i).\end{aligned}$$

□

Remark 4.7. Theorem 4.6 is a generalization of [BLY20, Lemma 2.2], where α is a tangle and S_N is an annulus.

4.3. **General balanced sutured manifolds.**

In this subsection, we prove our main theorem, which is a restatement of the second part of Theorem 1.12.

Theorem 4.8. *Suppose (M, γ) is a balanced sutured manifold and $\{S_1, \dots, S_n\}$ is a collection of properly embedded admissible surfaces. Then the Euler characteristic*

$$\chi(\mathbf{SH}(-M, -\gamma, (S_1, \dots, S_g), (i_1, \dots, i_n)))$$

depends only on (M, γ), $S_1, \dots, S_n$, and $(i_1, \dots, i_n) \in \mathbb{Z}^n$, and is independent of $\mathbf{H}$.

Corollary 4.9. *Suppose (M, γ) is a balanced sutured manifold and suppose $H = H_1(M)/\mathrm{Tors}$. Then the graded Euler characteristic*

$$\chi(\mathbf{SH}(M, \gamma)) = \chi(\mathbf{SH}^g(M, \gamma)) \in \mathbb{Z}[H]/\pm H$$

is independent of the choice of the fixed genus g of closures.

Proof. From Corollary 3.42 and Theorem 4.8, we know

$$\chi(\mathbf{SH}^g(M,\gamma)) = \chi(SFH(M,\gamma)) \in \mathbb{Z}[H]/\pm H,$$

where the right hand side is independent of the choice of the fixed genus g of closures. □

Proof of Theorem 4.8. First we can attach product 1-handles disjoint from $S_1,\dots,S_n$. From [BS16a, Section 4.2], attaching a 1-handle does not change the closure and hence does not make any difference to the multi-grading associated to $(S_1,\dots,S_n)$. Hence we can assume that γ is connected from now on. From [LY20, Section 3.1], we can pick a disjoint union of properly embedded arcs

$$\alpha = \alpha_1 \cup \cdots \cup \alpha_m$$

so that

(1) for $k = 1,\dots,m$, we have $\partial\alpha_k \cap R_+(\gamma) \neq \varnothing$ and $\partial\alpha_k \cap R_-(\gamma) \neq \varnothing$,
(2) $M\backslash\mathrm{int}(N(\alpha))$ is a handlebody.

Then we apply the arguments involved in [LY20, Section 3.2]: since γ is connected, we can pick pairwise disjoint arcs $\zeta_1,\dots,\zeta_m$ so that for any $k = 1,\dots,m$, we have

$$\partial\zeta_k = \partial\alpha_k \text{ and } |\zeta_k \cap \gamma| = 1.$$

For any $k = 1,\dots,g$, let $\beta_k \subset \zeta_k$ be a neighborhood of the intersection point $\zeta_k \cap \gamma$ and let

$$\zeta_k\backslash\beta_k = \zeta_{k,+} \cup \zeta_{k,-},$$

where $\zeta_{k,\pm} \subset R_\pm(\gamma)$. Push the interior of β_k into the interior of M to make it a properly embedded arc, which we still call β_k. Let

$$\beta = \beta_1 \cup \cdots \cup \beta_m.$$

Let $N = M\backslash\mathrm{int}(N(\beta))$, and let γ_N be the disjoint union of γ and a meridian for each component of β. It is explained in [LY20, Section 3.2] that (N,γ_N) can be obtained from (M,γ) by attaching 1-handles disjoint from $S_1,\dots,S_m$, so there is a canonical identification

$$\mathbf{SH}(-M,-\gamma,(S_1,\dots,S_n),(i_1,\dots,i_n)) = \mathbf{SH}(-N,-\gamma_N,(S_1,\dots,S_n),(i_1,\dots,i_n))$$

Let $H = M\backslash\mathrm{int}(N(\alpha\cup\beta))$. It is straightforward to check that H is a handlebody. Let Γ_μ be the disjoint union of γ and a meridian for each component of $\alpha\cup\beta$. Let the suture Γ_0 be obtained from Γ_μ by performing band sums along $\zeta_{k,+}$ and $\zeta_{k,-}$ for $k = 1,\dots,m$. See Figure 15. It is straightforward to check that (N,γ_N) can be obtained from (H,Γ_0) by attaching 2-handles along the meridians of all components of α.

We prove the theorem in the case when $m = 1$, while the general case follows from a straightforward induction. If $m = 1$, then α is connected. Suppose μ is the meridian of α. As explained in Subsection 4.2, attching a 2-handle along μ is the same as performing a 0-surgery along a push-off μ' of μ. There is an exact triangle associated to the surgeries along μ' that is discussed in [LY20, Section 3.2] (see also [GLW19, Section 3.1]):

(4.2)
$$\begin{tikzcd} & \mathbf{SH}(-N,-\gamma_N) \arrow[dr] & \\ \mathbf{SH}(-H,-\Gamma_0) \arrow[ur, "C_\mu"] & & \mathbf{SH}(-H,-\Gamma_1) \arrow[ll] \end{tikzcd}$$

The map C_μ is the map associated to the 2-handle attachment as discussed in Subsection 4.2. The suture Γ_1 is obtained from Γ_0 by twisting along $-\mu$ once. for $j = 1,\dots,n$, take $S_{j,H} = S_j \cap H$.

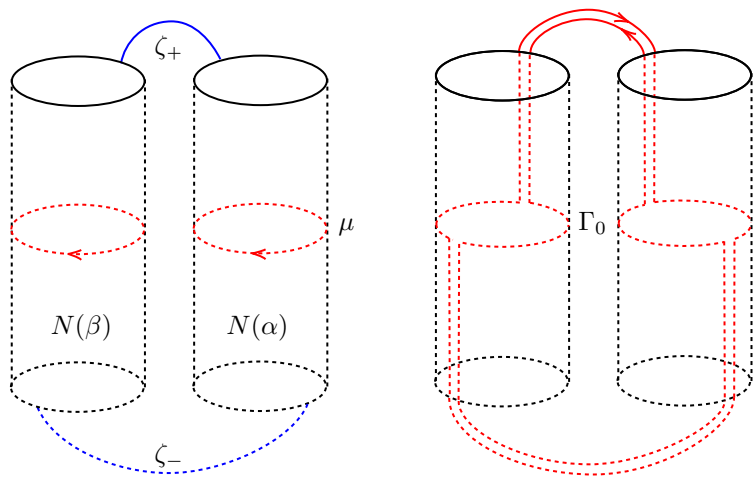

FIGURE 15. The suture Γ_0.

Since μ is disjoint from $S_{j,H}$ for $j = 1, \ldots, n$, the proof of Proposition 4.6 implies there is a graded version of the exact triangle (4.2):

(4.3)

$$\begin{tikzcd}
 & \mathbf{SH}(-N, -\gamma_N, (S_1, \ldots, S_n), (i_1, \ldots, i_n)) \arrow[dd] \\
\mathbf{SH}(-H, -\Gamma_0, (S_{1,H}, \ldots, S_{n,H}), ((i_1, \ldots, i_n))) \arrow[ur, "c_\mu"] & \\
 & \mathbf{SH}(-H, -\Gamma_1, (S_{1,H}, \ldots, S_{n,H}), ((i_1, \ldots, i_n))) \arrow[ul]
\end{tikzcd}$$

Then Theorem 4.8 follows from Proposition 2.4 and Corollary 4.5. □

5. The canonical mod 2 grading

Throughout this section, we focus on special cases of balanced sutured manifolds obtained from connected closed 3-manifolds and knots in them (*c.f.* Remark 3.36).

Definition 5.1. Suppose that Y is a closed 3-manifold and $z \in Y$ is a basepoint. Let $Y(1)$ be obtained from Y by removing a 3-ball containing z and let δ be a simple closed curve on $\partial Y(1) \cong S^2$. Suppose that $K \subset Y$ is a knot and w is a basepoint on K. Let $Y(K)$ be the knot complement of K and let $\gamma = m \cup (-m)$ consist of two meridians with opposite orientations of K near w. Then $(Y(1), \delta)$ and $(Y(K), \gamma)$ are balanced sutured manifolds. Define

$$\widetilde{\mathbf{H}}(Y, z) := \mathbf{SH}(Y(1), \delta), \text{ and } \mathbf{KH}(Y, K, w) := \mathbf{SH}(Y(K), \gamma).$$

Convention. Different choices of the basepoints give isomorphism vector spaces. Since we only care about the isomorphism class of the vector spaces, we omit the basepoints and simply write $\widetilde{\mathbf{H}}(Y)$ and $\mathbf{KH}(Y, K)$ instead.

To be more specific and consistent with [LY20], in this section, we focus on instanton theory. Based on the discussion in Subsection 2.1, we specify the Floer homology $\mathbf{H}(Y)$ and the cobordism map $\mathbf{H}(W)$ to be $I^{\omega}(Y)$ and $I(W,\nu)$. For a connected closed 3-manifold, the framed instanton Floer homology $I^{\sharp}(Y)$ defined in [KM11] is isomorphic to $\widetilde{\mathbf{H}}(Y)$ when $\mathbf{H}$ is instanton theory. Hence we replace $\widetilde{\mathbf{H}}(Y)$ by $I^{\sharp}(Y)$ throughout this section. Also we replace **SH** and **KH** by **SHI** and **KHI**, respectively. Recall that the definitions of **SHI** and **KHI** *a priori* depend on the choice of a fixed and large genus g of closures. We write

$$\mathbf{SHI}^g \text{ and } \mathbf{KHI}^g$$

explicitly in this section. However, for instanton theory, closures of different genus induce isomorphic groups and we can use closures of genus one to define sutured instanton Floer homology (*c.f.* [KM10b, Section 7]).

In this section, we discuss the canonical $\mathbb{Z}_2$ grading on $\mathbf{KHI}^g$ and the decomposition of $I^{\sharp}$ in Theorem 1.18. For other Floer-type theory, the construction in [LY20, Section 4.3] can be adapted without essential changes and we have a decomposition for $\widetilde{\mathbf{H}}(Y)$ similar to that in Theorem 1.18. The results in this section also apply without essential changes.

5.1. **The case of the unknot.**

In this subsection, we study the model case: the unknot U in S^3. Suppose μ_U and λ_U are the meridian and the longitude of U, respectively. The knot complement is identified with a solid torus $S^1 \times D^2$:

$$\rho : S^3(U) \xrightarrow{\cong} S^1 \times D^2, \tag{5.1}$$

where $\rho(\mu_U) = S^1 \times \{1\}$, and $\rho(\lambda_U) = \{1\} \times \partial D^2$. For co-prime integers x and y, let

$$\gamma_{(x,y)} = \gamma_{x\lambda_U + y\mu_U} \subset \partial S^3(U)$$

be the suture consists of two disjoint simple closed curves representing $\pm(x\lambda_U + y\mu_U)$.

Convention. Note $\gamma_{(x,y)} = \gamma_{(-x,-y)}$. From term (4) in Proposition 2.29, the orientation of the suture does not influence the isomorphism type of formal sutured homology. Hence we do not care about the orientation of the suture and we always assume $y \geqslant 0$.

We describe a closure of the balanced sutured manifold $(S^3(U), \gamma_{(x,y)})$ as follows.

Let Σ be a connected closed surface of genus $g \geqslant 1$. Suppose $Y_\Sigma = S^1 \times \Sigma$ and $\Sigma = \{1\} \times \Sigma$. Pick a non-separating simple closed curve $\alpha \subset \Sigma$ and suppose its complement is $Y_\Sigma(\alpha) = Y_\Sigma \backslash \mathrm{int}(N(\alpha))$. There is a framing on $\partial Y_\Sigma(\alpha)$ induced by the surface Σ. Let μ_α and λ_α be the corresponding meridian and longitude, respectively. Also, suppose $p \in \Sigma$ is a point disjoint from α. According to the discussion in Section 2, we can form a closure $(\bar{Y}, R, \omega)$ of $(S^3(U), \gamma_{(x,y)})$ as follows:

$$\bar{Y} = S^3(U) \underset{\phi}{\cup} Y_\Sigma(\alpha), R = \Sigma, \text{ and } \omega = S^1 \times \{\mathrm{pt}\}, \tag{5.2}$$

where $\phi : \partial S^3(U) \xrightarrow{\cong} \partial Y(\alpha)$ is an orientation reversing diffeomorphism such that

$$\phi(x\lambda_U + y\mu_U) = \lambda_\alpha. \tag{5.3}$$

Note different choices of the preimage of μ_α lead to different closures of $(S^3(U), \gamma_{(x,y)})$. From (5.3), we know that

$$\phi(\lambda_U) = z\lambda_\alpha + y\mu_\alpha,$$

where $z = x$ if $y = 0$, $z = y$ is arbitrary if $y = 1$, and $zx \equiv 1 \pmod{y}$ in other cases. Again different choices of z lead to different closures. From now on, we fix the value of z as follows: $z = x$ of $y = 0$,

$z = 0$ if $y = 1$, and z is the minimal positive integer so that $y|(xz-1)$. Now, composing ϕ with the inverse of the map ρ in (5.1), suppose

$$\widehat{Y} = Y(\alpha) \underset{\phi\circ\rho^{-1}}{\cup} S^1 \times D^2,$$

where $\phi \circ \rho^{-1} : \partial(S^1 \times D^2) \to \partial Y(\alpha)$ is a diffeomorphism such that

$$\phi \circ \rho^{-1}(\{1\} \times \partial D^2) = z\lambda_\alpha + y\mu_\alpha.$$

Hence, $\bar{Y}$ is obtained from Y by performing a y/z surgery and we also write $\bar{Y} = \widehat{Y}_{y/z}$.

Lemma 5.2. *For any suture $\gamma_{(x,y)}$ on $\partial S^3(U)$, we have*

$$\chi(\mathbf{SHI}^g(S^3(U), \gamma_{(x,y)})) = \pm y.$$

Proof. First, we can focus on the closure $(\bar{Y} = \widehat{Y}_{y/z}, R = \Sigma, \omega)$ as in (5.2). We need to compute the Euler characteristic of

$$\mathbf{SHI}^g(S^3(U), \gamma_{(x,y)}) := I^\omega(\widehat{Y}_{y/z}|\Sigma).$$

If $y = 0$, then $x = \pm 1$, but $(S^3(U), \gamma_{(\pm 1,0)})$ are both irreducible and non-taut. By Theorem 2.31, we know that

$$\mathbf{SHI}^g(S^3(U), \gamma_{(1,0)}) = 0.$$

If $y = 1$, then $z = 0$ and $\widehat{Y}_{1/0} = S^1 \times \Sigma$. By Axiom (A1-5), we have

$$\chi(I^\omega(\widehat{Y}_{1/0}|\Sigma)) = -1.$$

If $y > 1$, we have $y > z \geqslant 1$. If $z = 1$, then we have an exact triangle from Axiom (A2)

$$\begin{tikzcd} I^\omega(\widehat{Y}_{y-1}|\Sigma) \arrow[rr] & & I^\omega(\widehat{Y}_y|\Sigma) \arrow[dl, "f"] \\ & I^\omega(\widehat{Y}_{1/0}|\Sigma) \arrow[ul] & \end{tikzcd}$$

where the map f is odd and the rest two are even by Proposition 2.4. Hence we conclude by induction that

$$\chi(I^\omega(\widehat{Y}_y|\Sigma)) = -y.$$

Finally, when $y > z > 1$, suppose the continued fraction of $-y/z$ is

$$-\frac{y}{z} = [a_0, \ldots, a_n] = a_0 - \frac{1}{a_1 - \frac{1}{\cdots - \frac{1}{a_n}}},$$

where $a_n \leqslant -2$. Define

$$-\frac{y'}{z'} = [a_0, \ldots, a_{n-1}], \text{ and } -\frac{y''}{z''} = [a_0, \ldots, a_{n-1} + 1],$$

where $y', y'' \geqslant 0$. From a basic property of continued fraction, we have

$$y = y' + y'', \text{ and } z = z' + z''.$$

From Axiom (A2), there exists an exact triangle

$$\begin{tikzcd} I^\omega(\widehat{Y}_{y''/z''}|\Sigma) \arrow[rr] & & I^\omega(\widehat{Y}_{y/z}|\Sigma) \arrow[dl] \\ & I^\omega(\widehat{Y}_{y'/z'}|\Sigma) \arrow[ul, "f"] & \end{tikzcd}$$

where the map f is odd and the other two are even by Proposition 2.4. □

Remark 5.3. It is worth mentioning that different papers have different normalization for the canonical $\mathbb{Z}_2$ grading. Our choice of normalization in Axiom (A3) is the same as in [KM10a]. In Lidman, Pinzón-Caicedo, and Scaduto's setup [LPCS20], they adapted another normalization and proved $\chi(I^\omega(S^1 \times \Sigma|\Sigma)) = 1$ for Σ of any genus that is at least one.

Corollary 5.4. *Suppose* (Y', R', ω') *is a closure of* $(S^3(U), \gamma_{(x,y)})$*, then*

$$\chi(I(Y|R)) = \pm y.$$

Proof. This corollary follows directly from the fact that canonical maps from $I^\omega(\widehat{Y}_1|\Sigma)$ to $I^{\omega'}(Y'|R')$ is a composition of cobordism maps and hence is homogeneous. □

5.2. **Sutured knot complements.**

Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Any Seifert surface S of K gives rise to a framing on $\partial Y(K)$: the meridian μ can be picked as the meridian of the solid torus $N(K)$, and the longitude λ can be picked as $S \cap \partial Y(K)$. The 'half lives and half dies' fact for 3-manifolds implies that the following map has a 1-dimensional image:

$$\partial_* : H_2(Y(K), \partial Y(K); \mathbb{Q}) \to H_1(\partial Y(K); \mathbb{Q}).$$

Hence any two Seifert surfaces lead to the same framing on $\partial Y(K)$.

Definition 5.5. The framing $(\lambda, -\mu)$ defined as above is called the **canonical framing** of (Y, K). With this canonical framing, let

$$\gamma_{(x,y)} = \gamma_{x\lambda + y\mu} \subset \partial Y(K)$$

be the suture consists of two disjoint simple closed curves representing $\pm(x\lambda + y\mu)$.

Our goal in this subsection is to define a canonical $\mathbb{Z}_2$ grading on $\mathbf{SHI}^g(Y(K), \gamma_{(x,y)})$ for any fixed large enough g. Recall $\underline{\mathrm{SHI}}^g(M, \gamma)$ is the projective transitive system formed by closures of (M, γ) of a fixed genus g. We first assign a $\mathbb{Z}_2$ grading for any closure of $(Y(K), \gamma_{(x,y)})$.

Suppose $(\bar{Y}, R, \omega)$ is a closure of $(Y(K), \gamma_{(x,y)})$. Then we can form a closure $(\bar{Y}_U, R, \omega)$ of $(S^3(U), \gamma_{(x,y)})$ by taking

$$\bar{Y}_U = \bar{Y} \backslash (\mathrm{int}(Y(K))) \underset{\mathrm{id}}{\cup} S^3(U). \tag{5.4}$$

Here id is the diffeomorphism between toroidal boundaries which respect the canonical framings on both boundaries.

Definition 5.6. The **modified** $\mathbb{Z}_2$ **grading** on $I^\omega(\bar{Y}|R)$ is defined as follows.

- If $\chi(I^\omega(\bar{Y}_U|R)) =$ is negative, then it is the same as the canonical $\mathbb{Z}_2$ grading on $I^\omega(\bar{Y}|R)$.
- If $\chi(I^\omega(\bar{Y}_U|R))$ is positive, then it is the same as the canonical $\mathbb{Z}_2$ grading on $I^\omega(\bar{Y}|R)$ with even and odd part switched.

Suppose $(\bar{Y}, R, \omega)$ and $(\bar{Y}', R, \omega)$ are two closures of $(Y(K), \gamma_{(x,y)})$ so that $\bar{Y}'$ is obtained from $\bar{Y}$ by a Dehn surgery along a curve $\beta \subset \bar{Y}$, which is disjoint from $\mathrm{int}(M)$, R and ω. Then there is a map

$$F : I^{\omega}(\bar{Y}|R) \to I^{\omega}(\bar{Y}'|R)$$

associated to the Dehn surgery along $\beta \subset \bar{Y}$. Let $(\bar{Y}_U, R, \omega)$ and $(\bar{Y}'_U, R, \omega)$ be the closures of $(S^3(U), \gamma_{(x,y)})$ constructed as in (5.4). There is also a map

$$F_U : I^{\omega}(\bar{Y}_U|R) \to I^{\omega}(\bar{Y}'_U|R)$$

associated to the same Dehn surgery along $\beta \subset \bar{Y}_U$. Then we have the following.

Lemma 5.7. *The maps F and F_U have the same parity with respect to the canonical $\mathbb{Z}_2$ gradings on corresponding instanton Floer homologies.*

Proof. Note that $H_1(S^3(U); \mathbb{Q}) \cong \mathbb{Q}\langle \mu_U \rangle$ and the map

$$i^U_* : H_1(\partial S^3(U); \mathbb{Q}) \to H_1(S^3(U); \mathbb{Q})$$

induced by the inclusion has a 1-dimensional kernel generated by λ_U. For a null-homologous knot $K \subset Y$, we know that the map

$$i_* : H_1(\partial Y(K); \mathbb{Q}) \to H_1(Y(K); \mathbb{Q})$$

induced by the inclusion has a 1-dimensional kernel generated by the longitude λ of K and has a 1-dimensional image generated by the meridian μ of K. Hence, from the Mayer-Vietoris sequence, we know that there is an injective map

$$j : H_1(\bar{Y}_U; \mathbb{Q}) \hookrightarrow H_1(\bar{Y}; \mathbb{Q}),$$

that sends $[\mu_U]$ to $[\mu]$ and sends every homology class in $\bar{Y}_U \backslash S^3(U) = \bar{Y} \backslash Y(K)$ using the natural map

$$i'_* : H_1(\bar{Y} \backslash Y(K); \mathbb{Q}) \to H_1(\bar{Y}; \mathbb{Q}).$$

Similarly, since $\beta \cap \mathrm{int}(Y(K)) = \varnothing$, we know that there is an injective map

$$j^{\beta} : H_1(\bar{Y}_U(\beta); \mathbb{Q}) \hookrightarrow H_1(\bar{Y}(\beta); \mathbb{Q}),$$

which fits into the following commutative diagram

$$\begin{tikzcd}
H_1(\partial \bar{Y}_U(\beta); \mathbb{Q}) \arrow[r, "\iota^U_*"] \arrow[d, "="] & H_1(\bar{Y}_U(\beta); \mathbb{Q}) \arrow[d, "j^{\beta}"] \\
H_1(\partial \bar{Y}(\beta); \mathbb{Q}) \arrow[r, "\iota_*"] & H_1(\bar{Y}(\beta); \mathbb{Q})
\end{tikzcd}$$

where ι_* and ι^U_* are induced by natural inclusions. Hence we conclude, under the identification $H_1(\partial \bar{Y}_U(\beta); \mathbb{Q}) = H_1(\partial \bar{Y}(\beta); \mathbb{Q})$, two kernels are also identified:

$$\ker(\iota^U_*) = \ker(\iota_*).$$

Since F and F_U are associated to Dehn surgeries along β of the same slopes, we conclude from 2.4 that their parity must be the same. $\square$

Lemma 5.8. *Suppose $(\bar{Y}, R, \omega)$ and $(\bar{Y}', R', \omega')$ are two different closures of $(Y(K), \gamma_{(x,y)})$ so that $g(R) = g(R')$. There is a canonical isomorphism*

$$\Phi : I^{\omega}(\bar{Y}|R) \xrightarrow{\cong} I^{\omega'}(\bar{Y}'|R')$$

as in Definition 2.10. Then with respect to the modified $\mathbb{Z}_2$ grading as in Definition 5.6, the canonical map Φ is homogeneous of degree zero.

Proof. From the definition of Φ and Axiom (A3-3), we know that Φ is always homogeneous. For two closures $(\bar{Y}, R, \omega)$ and $(\bar{Y}', R', \omega')$ of $(Y(K), \gamma_{(x,y)})$, we can form two corresponding closures $(\bar{Y}_U, R, \omega)$ and $(\bar{Y}'_U, R', \omega')$ of $(S^3(U), \gamma_{(x,y)})$ as in (5.2), respectively. There is a canonical map

$$\Phi_U : I^{\omega}(\bar{Y}_U|R) \xrightarrow{\cong} I^{\omega'}(\bar{Y}'_U|R').$$

From Definition 5.6, the modified $\mathbb{Z}_2$ gradings on $I^{\omega}(\bar{Y}|R)$ and $I^{\omega}(\bar{Y}'|R')$ coincide if and only if the canonical $\mathbb{Z}_2$ gradings on $I^{\omega}(\bar{Y}_U|R)$ and $I^{\omega}(\bar{Y}'_U|R')$ coincide. The definition of the modified $\mathbb{Z}_2$ grading automatically makes Φ_U even under the modified $\mathbb{Z}_2$ grading. Hence to show that Φ is even, it suffices to show that Φ and Φ_U have the same parity under the modified $\mathbb{Z}_2$ grading.

From the construction of the canonical maps, there is a sequence of simple closed curves $\beta_1, \ldots, \beta_n$ on R, such that the map Φ is the composition of cobordism maps induced by a diffeomorphism and the sequence of Dehn surgeries. Similarly, the map Φ_U is the composition of the maps induced another diffeomorphism and the same sequence of Dehn surgeries on $\bar{Y}_U$. Since the surgery curves are all on R and disjoint from $\mathrm{int}(Y(K))$, cobordism maps induced by diffeomorphisms are always with even degrees. For Dehn surgeries along β_i, we can apply Lemma 5.7 and then finish the proof. □

Corollary 5.9. *Suppose $(Y(K), \gamma_{(x,y)})$ is the balanced sutured manifold constructed as before and g is a fixed large enough genus of closures, there is a well-defined absolute $\mathbb{Z}_2$ grading on $\mathbf{SHI}^g(Y(K), \gamma_{(x,y)})$. It is called the **canonical** $\mathbb{Z}_2$ grading. In particular, we can define the canonical $\mathbb{Z}_2$ grading on $\mathbf{KHI}^g(Y, K)$.*

Proof. If we fix a genus g, then the fact that $\mathbf{SHI}^g$ and $\mathbf{KHI}^g$ admit well-defined $\mathbb{Z}_2$ gradings follows from above discussions. Now we want to compare the $\mathbb{Z}_2$ grading for closures of different genuses. First we deal with the case of unknot. As in Subsection 5.1, we can constructed a standard closure $(\bar{Y}_{y/z}, \Sigma)$ for $(Y(K), \gamma_{(x,y)})$. Here, the genus of Σ can be arbitrary. To specify the genus, in the proof of this corollary, we temporarily write $(\bar{Y}_{y/z}, \Sigma)$ as $(\bar{Y}^g_{y/z}, \Sigma_g)$. In [BS15], the canonical map

$$\Phi^{g,g+1} : I^{\omega}(\bar{Y}^g_{y/z}|\Sigma_g) \to I^{\omega}(\bar{Y}^{g+1}_{y/z}|\Sigma_{g+1})$$

is constructed as follows: Recall that we have $\alpha \subset \Sigma_g$ being a non-separating simple closed curve, and

$$\bar{Y}^g_{y/z} = Y(K) \cup (S^1 \times \Sigma_g) \backslash N(\alpha).$$

Now pick $\beta \subset \Sigma_g$ be another simple closed curve so that $\alpha \cap \beta = \varnothing$. Take a curve $\theta \subset \Sigma_2$ be non-separating simple closed curve as well. Then we can form $(\bar{Y}^{g+1}_{y/z}, \Sigma_{g+1})$ from $(\bar{Y}^g_{y/z}, \Sigma_g)$ and $(S^1 \times \Sigma_2, \Sigma_2)$ by cutting them open along $S^1 \times \beta$ and $S^1 \times \theta$ respectively, and then glue the two pieces together along toroidal boundaries by the identifications $S^1 = S^1$ and $\beta = \theta$. Then, as in [KM10b], there is a cobordism $W^{g,g+1}$ from $(\bar{Y}^g_{y/z}, \Sigma_g)$ and $(S^1 \times \Sigma_2, \Sigma_2)$ to $(\bar{Y}^{g+1}_{y/z}, \Sigma_{g+1})$, known as the Floer excision cobordism. In the proof of [LPCS20, Proposition 3.6] and [KM10a, Lemma 3.3], the degree of the cobordism induced by cobordisms constructed in the same way as $W^{g,g+1}$ has been computed explicitly. By a similar computation, we see that the degree of canonical map $\Phi^{g,g+1}$ is 0 with respect to the modified $\mathbb{Z}_2$ grading. (Note by the above argument, the modified $\mathbb{Z}_2$

grading for $(\bar{Y}^g_{y/z}, \Sigma_g)$ is the same as the canonical grading.) For any two closures of $(Y(K), \gamma_{(x,y)})$, as in [BS15], the canonical maps are constructed by composing the maps induced by some $W^{g,g+1}$ and canonical maps for closures of the same genuses. Since both types have been shown to be of degree 0, we know that any canonical map is of degree 0 and hence we are done. □

In the rest of this subsection, we deal with bypass exact triangles. Suppose y_3/x_3 is a surgery slope with $y_3 \geqslant 0$. According to Honda [Hon00], there are two basic bypasses on the balanced sutured manifold $(Y(K), \gamma_{(x_3,y_3)})$, whose arcs are depicted as in Figure 16. The sutures involved in the bypass triangles were described explicitly in Honda [Hon00].

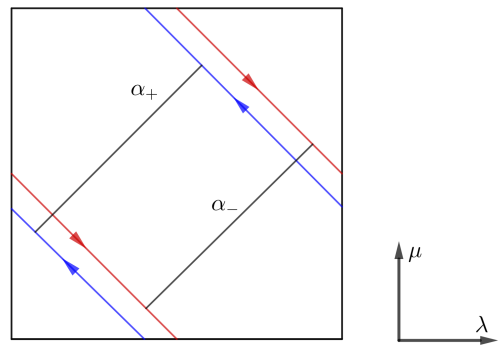

FIGURE 16. Bypass arcs on $\gamma_{(1,-1)}$.

Definition 5.10. For a surgery slope y_1/x_1 with $y_1 \geqslant 0$, suppose its continued fraction is

$$\frac{y_1}{x_1} = [a_0, a_1, \ldots, a_n] = a_0 - \frac{1}{a_1 - \frac{1}{\cdots - \frac{1}{a_n}}},$$

where $a_i \leqslant -2$. If $y_1 > -x_1 > 0$, let

$$\frac{y_2}{x_2} = [a_0, \ldots, a_{n-1}] \text{ and } \frac{y_3}{x_3} = [a_0, \ldots, a_n + 1].$$

If $-x_1 > y_1 > 0$, we do the same thing for $x_1/(-y_1)$. If $y_1 > x_1 > 0$, we do the same thing for $y_1/(-x_1)$. If $x_1 > y_1 > 0$, we do the same thing for $x_1/(-y_1)$. If $y_1/x_1 = 1/0$, then $y_2/x_2 = 0/1$ and $y_3/x_3 = 1/(-1)$. If $y_1/x_1 = 0/1$, let $y_2/x_2 = 1/(-1)$ and $y_3/x_3 = 0/1$. We always require that $y_2 \geqslant 0$ and $y_3 \geqslant 0$.

Remark 5.11. It is straightforward to use induction to verify that for $y_1 > -x_1 > 0$,

$$x_1 = x_2 + x_3, \text{ and } y_1 = y_2 + y_3.$$

Then the bypass exact triangle in Theorem 2.19 becomes the following.

Proposition 5.12. *Suppose $K \subset Y$ is a null-homologous knot, and suppose the surgery slopes y_i/x_i for $i = 1, 2, 3$ are defined as in Definition 5.10. Suppose $\psi^*_{+,*}$ and $\psi^*_{-,*}$ are from two different bypasses, where $*$ means the corresponding slope. Then there are two exact triangles about $\psi^*_{+,*}$ and*

$\psi^*_{-,*}$, *respectively.*

$$\begin{tikzcd}
\mathbf{SHI}^g(-Y(K),-\gamma_{(x_2,y_2)}) \arrow[rr, "\psi^{y_2/x_2}_{\pm,y_3/x_3}"] & & \mathbf{SHI}^g(-Y(K),-\gamma_{(x_3,y_3)}) \arrow[dl, "\psi^{y_3/x_3}_{\pm,y_1/x_1}"] \\
& \mathbf{SHI}^g(-Y(K),-\gamma_{(x_1,y_1)}) \arrow[ul, "\psi^{y_1/x_1}_{\pm,y_2/x_2}"] &
\end{tikzcd}$$

As stated in Proposition 2.20, the bypass maps ψ_1, ψ_2, and ψ_3 are induced by some cobordism maps. Then we have the following.

Lemma 5.13. *Suppose g is a large enough integer and suppose y_i/x_i for $i = 1, 2, 3$ is from Definition 5.10. Suppose further*

$$x_1 = x_2 + x_3, \text{ and } y_1 = y_2 + y_3.$$

With respect to the well-defined $\mathbb{Z}_2$ grading on $\mathbf{SHI}^g$ in Corollary 5.9, the map ψ_2 is odd and the rest two are even. As a consequence,

$$\chi(\mathbf{SHI}^g(-Y(K),-\gamma_{(x_1,y_1)})) = \chi(\mathbf{SHI}^g(-Y(K),-\gamma_{(x_3,y_3)})) + \chi(\mathbf{SHI}^g(-Y(K),-\gamma_{(x_2,y_2)})).$$

Proof. As in Proposition 2.20, we can fix a large enough g so that for $i = 1, 2, 3$, there are closures $(\bar{Y}_i, R, \omega)$ for $(-Y(K), -\gamma_{(x_i,y_i)})$ of genus g, and the bypass maps ψ_1, ψ_2, and ψ_3 have the same $\mathbb{Z}_2$ degrees as the maps induced by Dehn surgeries along three curves ζ_1, ζ_2, and ζ_3 in corresponding closures $\bar{Y}_i \backslash \mathrm{int}(Y(K))$. Since we only care about the $\mathbb{Z}_2$ degrees of maps, in a slight abuse of notation, we do not distinguish the bypass map and the map induced by Dehn surgery.

For $i = 1, 2, 3$, we can form corresponding closures $(\bar{Y}_i^U, R, \omega)$ as in (5.2) so that the curves ζ_1, ζ_2, and ζ_3 remains and suitable surgeries along them induces an exact triangle

$$\begin{tikzcd}
\mathbf{SHI}^g(-S^3(U),-\gamma_{(x_3,y_3)}) \arrow[rr, "\psi^U_3"] & & \mathbf{SHI}^g(-S^3(U),-\gamma_{(x_1,y_1)}) \arrow[dl, "\psi^U_1"] \\
& \mathbf{SHI}^g(-S^3(U),-\gamma_{(x_2,y_2)}) \arrow[ul, "\psi^U_2"] &
\end{tikzcd}$$

As in the proof of Lemma 5.8, with the help of Lemma 5.7, it suffices to check that the maps ψ_1^U, ψ_2^U, and ψ_3^U are odd or even as desired, with respect to the $\mathbb{Z}_2$ grading on $\mathbf{SHI}^g$ from Corollary 5.9.

For the case of the unknot, the argument becomes straightforward: from Definition 5.6 and Lemma 5.2, we know that for any $y > 0$,

$$\chi(\mathbf{SHI}^g(-S^3(U),-\gamma_{(x,y)})) = -y$$

Then the equation $y_1 = y_2 + y_3$ implies that

$$\chi(\mathbf{SHI}^g(-S^3(U),-\gamma_{(x_1,y_1)})) = \chi(\mathbf{SHI}^g(-S^3(U),-\gamma_{(x_3,y_3)})) + \chi(\mathbf{SHI}^g(-S^3(U),-\gamma_{(x_2,y_2)})).$$

Note the maps ψ_i^U for $i = 1, 2, 3$ are coming from a real surgery exact triangle as in Proposition 2.20, while the gradings on $\mathbf{SHI}^g$ could possibly be shifted due to the normalization in Definition 5.6 and the surgery along curves η_1 and η_2 as in Proposition 2.20. Hence they still satisfies the hypothesis of Lemma 2.5. Thus, we conclude that ψ_2^U is odd and the other two are even. Similarly, ψ_2 is odd and the other two are even, and we have

$$\chi(\mathbf{SHI}^g(-Y(K),-\gamma_{(x_1,y_1)})) = \chi(\mathbf{SHI}^g(-Y(K),-\gamma_{(x_3,y_3)})) + \chi(\mathbf{SHI}^g(-Y(K),-\gamma_{(x_2,y_2)})).$$

□

5.3. **Computations and applications.**

Let Y be a closed 3-manifold and let $K \subset Y$ be a null-homologous knot. Suppose S is a minimal genus Seifert surface of K. Its genus is always denoted by $g(S)$, which is distinguished with g, the fixed genus of closures. We refer [LY20, Section 4] for the definitions of sutures $\Gamma_n, \Gamma_n(y/x)$, the admissible surface with stablization S^τ, the bypass maps $\psi^*_{+,*}, \psi^*_{-,*}$, and numbers i^*_{max}, i^*_{min}. To simplify our notation, we write

$$\chi^g_{y/x}(-Y, K, i) = \chi(\mathbf{SHI}^g(-Y(K), -\gamma_{(x,y)}, S^\tau, i)), \tag{5.5}$$

where the Euler characteristic is with respect to the canonical $\mathbb{Z}_2$ grading on $\mathbf{SHI}^g$ as in Corollary 5.9. We write

$$\chi^g_{y/x}(-Y, K) = \sum_{i \in \mathbb{Z}} \chi^g_{y/x}(-Y, K, i) \tag{5.6}$$

When $|x| = 1$, we write y/x as an integer. Also, we write

$$\chi^g_\mu(-Y, K, i) = \chi^g_{1/0}(-Y, K, i)$$

to specify the meridional suture.

Lemma 5.14. *Suppose Y is a closed oriented 3-manifold, and $K \subset Y$ is a null-homologous knot. For $g \in \mathbb{Z}$ large enough and any $i \in \mathbb{Z}$, we have*

$$\chi^g_1(-Y, K, i) = \chi^g_\mu(-Y, K, i), \text{ and } \chi^g_0(-Y, K, i) = 0.$$

Proof. From [LY20, Proposition 4.16], we have the following two bypass exact triangles:

$$\begin{tikzcd}
\mathbf{SHI}^g(-Y(K), -\Gamma_0, S^\tau, i) \arrow[r, "\psi^0_{-,1}"] & \mathbf{SHI}^g(-Y(K), -\Gamma_1, S^\tau, i) \arrow[dl, "\psi^1_{-,\mu}"] \\
\mathbf{SHI}^g(-Y(K), -\Gamma_0, S^\tau, i) \arrow[u, "\psi^\mu_{-,0}"] &
\end{tikzcd}$$

and

$$\begin{tikzcd}
\mathbf{SHI}^g(-Y(K), -\Gamma_0, S^\tau, i+1) \arrow[r, "\psi^0_{+,1}"] & \mathbf{SHI}^g(-Y(K), -\Gamma_1, S^\tau, i) \arrow[dl, "\psi^1_{+,\mu}"] \\
\mathbf{SHI}^g(-Y(K), -\Gamma_0, S^\tau, i) \arrow[u, "\psi^\mu_{+,0}"] &
\end{tikzcd}$$

Hence we obtain the following two equations from Lemma 5.13

$$\chi^g_1(-Y, K, i) = \chi^g_0(-Y, K, i) + \chi^g_\mu(-Y, K, i)$$

$$\chi^g_1(-Y, K, i) = \chi^g_0(-Y, K, i+1) + \chi^g_\mu(-Y, K, i)$$

By Axiom (A1-4), for $i > g(S)$, we have

$$\mathbf{SHI}^g(-Y(K), -\Gamma_0, S^\tau, i) = 0.$$

Hence we conclude by (5.5) and (5.6) that

$$\chi^g_1(-Y, K, g(S)) = \chi^g_\mu(-Y, K, g(S)), \text{ and } \chi^g_0(-Y, K, g(S)) = 0.$$

The lemma follows from the induction on the grading i. □

Lemma 5.15. *Suppose Y is a closed oriented 3-manifold, and $K \subset Y$ is a null-homologous knot. For the suture $\gamma_{(x,y)}$ with $y > 0$, we know that*

$$\chi^g_{y/x}(-Y, K, i) = \sum_{j=0}^{y-1} \chi^g_\mu(-Y, K, i - i^y_{max} + i^\mu_{max} + j).$$

Proof. We only prove the case when $x < 0$. The other case is similar. First, if $x = 1$, then we have a bypass exact triangle (in this case we write $y = n$)

$$\begin{tikzcd}
\mathbf{SHI}^g(-Y(K), -\Gamma_n, S^\tau, i) \arrow[r, "\psi^n_{-,n+1}"] & \mathbf{SHI}^g(-Y(K), -\Gamma_{n+1}, S^\tau, i) \arrow[dl, "\psi^{n+1}_{-,\mu}"] \\
\mathbf{SHI}^g(-Y(K), -\Gamma_\mu, S^\tau, i+1) \arrow[u, "\psi^\mu_{-,n}"] &
\end{tikzcd}$$

Hence, we can apply Lemma 5.13, Lemma 5.14, and the induction to conclude that

$$\chi^g_n(-Y, K, i) = \sum_{j=0}^{n-1} \chi^g_\mu(-Y, K, i - i^n_{max} + i^\mu_{max} + j).$$

If $x > 1$, we can consult to the continued fraction description of y/x and apply an induction in the same spirit as in the proof of Lemma 5.2. □

Corollary 5.16. *Suppose Y is a closed oriented 3-manifold, and $K \subset Y$ is a null-homologous knot. For the suture $\gamma_{(x,y)}$ where $y > 2g(S)$, and for any $i \in \mathbb{Z}$ so that*

$$i^y_{max} - 2g(S) \geqslant i \geqslant i^y_{min} + 2g(S),$$

we know that

$$\chi^g_{y/x}(-Y, K, i) = \chi^g_\mu(-Y, K).$$

Proof. The corollary follows immediately from Lemma 5.15 and the fact that there are only $(2g(S)+1)$ gradings with nontrivial elements for $\mathbf{SHI}^g(-Y(K), -\Gamma_\mu, S^\tau)$. □

Lemma 5.17. *Suppose Y is a closed oriented 3-manifold, and $K \subset Y$ is a null-homologous knot. Then we have*

$$\chi^g_\mu(-Y, K) = \pm\chi(I^\sharp(-Y)).$$

Proof. From Lemma 5.15, we know that for any $n \in \mathbb{Z}_{\geqslant 0}$, we have

$$\chi^g_n(-Y, K) = n \cdot \chi^g_\mu(-Y, K).$$

From [LY20, Lemma 4.11], we know that there is an exact triangle

$$\begin{tikzcd}
\mathbf{SHI}^g(-Y(K), -\Gamma_n) \arrow[rr] & & \mathbf{SHI}^g(-Y(K), -\Gamma_{n+1}) \arrow[dl] \\
& I^\sharp(-Y) \arrow[ul] &
\end{tikzcd}$$

Hence, by Lemma 2.5, we know that there is a proper sign assignment for all n so that

$$\pm\chi^g_n(-Y, K) \pm \chi^g_{n+1}(-Y, K) \pm \chi(I^\sharp(Y)) = 0.$$

Hence the only possibilities are

$$\chi(I^\sharp(Y)) = \pm\chi^g_\mu(-Y, K).$$

□

Proof of Proposition 1.19. It is an immediate corollary following Corollary 5.16, Lemma 5.17 and the definition of the decomposition from [LY20, Section 4.3]. □

For a knot K in S^3, we can actually fix the sign ambiguity coming from different choices of the fixed genus of the closures.

Lemma 5.18. *For any knot $K \subset S^3$ and any positive integer g, we have*

$$\chi_\mu^g(-S^3, K) = -1.$$

Proof. Since we adapt the normalization from Kronheimer and Mrowka [KM10a, Section 2.6], we can directly apply the results from them. In particular, for any knot K, in [KM10a, Section 2.4], a preferred closure $(\bar{Y}_1, \Sigma_1, \omega_1)$ of $(-S^3(K), -\Gamma_\mu)$ with $g(\Sigma_1) = 1$ is chosen. Then they proved that

$$\chi(I^{\omega_1}(\bar{Y}_1|\Sigma_1)) = -\Delta_K(1) = -1.$$

Note this coincide with our choice of modified $\mathbb{Z}_2$ grading: when the Euler characteristic is negative, we do not perform any shifts.

The case $g = 2$ has already been studied in the proof of [KM10a, Lemma 3.3]: another preferred closure $(\bar{Y}_2, \Sigma_2, \omega_2)$ with $g(\Sigma_2) = 2$ for $(-S^3(K), -\Gamma_\mu)$ is constructed, and there is a cobordism W_1 from $\bar{Y}_1 \cup S^1 \times \Sigma_2$ to $\bar{Y}_2$ coming from the Floer's excision theorem. The canonical generator of $I^\omega(S^1 \times \Sigma_2|\Sigma_2)$ is proved to be at the odd grading (*c.f.* [LPCS20, Lemma 3.8], though the normalization of the canonical $\mathbb{Z}_2$ grading is different). Moreover, the degree of the cobordism map W_1 is odd (*c.f.* [LPCS20, Proposition 3.6]).

For general g, it is straightforward to generalize the above construction for $\bar{Y}_1$ and $\bar{Y}_2$ to $\bar{Y}_g$ and $\bar{Y}_{g+1}$. There is a similar cobordism W_g from $\bar{Y}_g \cup S^1 \times \Sigma_2$ to $\bar{Y}_{g+1}$, the degree of which can be computed easily to be odd. Hence by induction we conclude that for all g,

$$\chi_\mu^g(-S^3, K) = -1.$$

□

By Lemma 5.18, we can identify $\chi_\mu^g(-S^3, K)$ for all large enough g, we simply write $\chi_\mu(-S^3, K) = \chi_\mu^g(-S^3, K)$ instead. Applying Lemma 5.15, we know that for any g large enough and $y > 0$,

$$\chi_{y/x}^g(-S^3, K) = -y.$$

Similarly, we simply write $\chi_{y/x}(-S^3, K)$ instead.

Finally, we consider the projectively system $\underline{\mathrm{SHI}}(M, \gamma)$ for a balanced sutured manifold (M, γ) defined in [BS15], which is independent of the choice of the genus of the closures. The isomorphism class of $\underline{\mathrm{SHI}}(M, \gamma)$ and $\mathbf{SHI}^g(M, \gamma)$ are the same. Similar to $\mathbf{SHI}^g(M, \gamma)$, it has a decomposition associated to an admissible surface $S \subset (M, \gamma)$.

Definition 5.19. Suppose (M, γ) is a balanced sutured manifold and S is an admissible surface in (M, γ). For any $i, j \in \mathbb{Z}$, define

$$\underline{\mathrm{SHI}}(M, \gamma, S, i)[j] = \underline{\mathrm{SHI}}(M, \gamma, S, i - j).$$

In [Li19, Section 5], the first author constructed a minus version of the instanton knot homology via the direct system

(5.7)

$$\cdots \to \underline{\mathrm{SHI}}(-S^3(K), \Gamma_n, S^\tau)[g(K) - i_{max}^n] \xrightarrow{\psi_{-,n+1}^n} \underline{\mathrm{SHI}}(-S^3(K), \Gamma_{n+1}, S^\tau)[g(K) - i_{max}^{n+1}] \to \cdots$$

and define $\underline{\mathrm{KHI}}^-(-S^3, K)$ to be the direct limit of (5.7). All $\psi^n_{-,n+1}$ are grading preserving after shifting, so there is a well-defined $\mathbb{Z}$-grading (the Alexander grading) on $\underline{\mathrm{KHI}}^-(-S^3, K)$, which we write as

$$\underline{\mathrm{KHI}}^-(-S^3, K, i).$$

By [Li19, Corollary 2.20], there is a commutative diagram

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-S^3(K), \Gamma_n, S^\tau)[g(K) - i^n_{max}] \arrow[r, "\psi^n_{-,n+1}"] \arrow[d, "\psi^n_{+,n+1}"] & \underline{\mathrm{SHI}}(-S^3(K), \Gamma_{n+1}, S^\tau)[g(K) - i^{n+1}_{max}] \arrow[d, "\psi^{n+1}_{+,n+2}"] \\
\underline{\mathrm{SHI}}(-S^3(K), \Gamma_{n+1}, S^\tau)[g(K) - i^{n+1}_{max}] \arrow[r, "\psi^{n+1}_{-,n+2}"] & \underline{\mathrm{SHI}}(-S^3(K), \Gamma_{n+2}, S^\tau)[g(K) - i^{n+2}_{max}]
\end{tikzcd}$$

Hence the maps $\{\psi^n_{+,n+1}\}$ induces an map U on $\underline{\mathrm{KHI}}^-(-S^3, K)$.

Proof of Proposition 1.21. From Lemma 5.13, the maps $\psi^n_{-,n+1}$ are all even, hence there is a well-defined $\mathbb{Z}_2$ grading on $\underline{\mathrm{KHI}}^-(-S^3, K)$. Again from Lemma 5.13, we know that U is even, *i.e.*, preserving the $\mathbb{Z}_2$ grading on $\underline{\mathrm{KHI}}^-(-S^3, K)$. Finally, we can apply Lemma 5.15 and the fact

$$\chi(\mathbf{KHI}^g(-S^3, K)) = -\Delta_K(t)$$

to conclude the desired formula. Note that by our the normalization, the sign is negative. □

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